Neighborhood Revitalization and Residential Sorting

Matthew Staiger* Giordano Palloni† John Voorheis‡

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Abstract

The HOPE VI Revitalization program sought to transform high-poverty neighborhoods into mixed-income communities. We study its effects on neighborhood poverty rates and subsidized renters' exposure to neighborhood poverty using longitudinal administrative data to compare funded projects with failed applicants. We find that HOPE VI reduced poverty rates in targeted neighborhoods by 8 percentage points within 15 years of the grant, mainly by changing who moved in rather than displacing original residents. These neighborhood-level changes reduced average exposure to neighborhood poverty among subsidized renters city-wide. However, the primary beneficiaries were not original residents, most of whom moved away, but subsidized renters who moved into the revitalized neighborhoods after the intervention. Our estimates imply that revitalizing half of a city's subsidized housing stock reduced subsidized renters' exposure to poverty by 4.6 percentage points.

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*Opportunity Insights, Harvard University. E-mail: mstaiger@g.harvard.edu
†Consumer Financial Protection Bureau. E-mail: giordano.palloni@cfpb.gov
‡U.S. Census Bureau. E-mail: john.l.voorheis@census.gov
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1 Introduction

Subsidized renters in the United States live disproportionately in high-poverty neighborhoods, which can harm residents' health, education, and labor market outcomes (Chyn and Katz, 2021). One potential policy response is to invest directly in disadvantaged neighborhoods. Such place-based policies can improve equity and efficiency when local market failures are present (Kline and Moretti, 2014). But a central concern is that policies that target *places* may not benefit the *people* they are intended to help (Glaeser and Gottlieb, 2008).

To evaluate whether place-based policies benefit people, one must first define which people may be affected. A common approach is to focus on the original residents of targeted places. However, residential churn is often high in low-income neighborhoods—Coulton et al. (2012) find that over half of the residents of low-income neighborhoods move within three years—implying that many of the people who live in a neighborhood after an intervention may not have lived there before. Evaluating place-based policies therefore requires looking not only at the original residents, but also at a broader set of affected groups, such as those who move in after the intervention.

We study these issues in the context of the HOPE VI Revitalization program, one of the largest neighborhood revitalization programs in recent U.S. history. Between 1993 and 2010, the program awarded \$6.3 billion across 262 grants to transform high-poverty neighborhoods into mixed-income communities by demolishing distressed public housing projects and building new subsidized and market-rate housing.[1] HOPE VI sought to reduce concentrated poverty in targeted neighborhoods and enable subsidized renters to live in lower-poverty neighborhoods.

Our analysis addresses three questions: i) Did HOPE VI reduce poverty rates in targeted neighborhoods, and, if so, through what mechanisms? ii) Did these neighborhood-level changes lead the original residents to live in lower-poverty neighborhoods? iii) Did HOPE VI reduce exposure to neighborhood poverty for subsidized renters more broadly, and if so, which groups benefited?[2]

[1]We report HOPE VI expenditures in nominal terms. All other dollar values in this paper are expressed in real 2019 dollars, deflated using the CPI-U.

[2]We use the term subsidized renters to refer primarily to participants of the Public Housing or Housing Choice Vouchers programs.

Our empirical strategy compares neighborhoods that received HOPE VI grants to failed applicants—public housing projects that applied for but never received funding. Although HOPE VI targeted some of the most distressed public housing projects in the country, funding constraints left many similar applicants unfunded, which we use to construct our comparison group. We track individuals and neighborhoods over time using administrative data on income, place of residence, and subsidized housing participation. These data allow us to follow the original residents as they move across neighborhoods and to observe all individuals who move into HOPE VI or failed-applicant neighborhoods after grants were awarded.

We first examine how HOPE VI affected targeted neighborhoods. We find that, 15 years after an award, poverty rates in HOPE VI neighborhoods were 8 percentage points (pp) lower than in comparable failed-applicant neighborhoods. These findings are consistent with Tach and Emory (2017), who use cross-sectional survey data to document similar declines in neighborhood poverty and hypothesize that they may reflect displacement of the original public housing residents. Our individual-level panel data allow us to directly examine the drivers of neighborhood change. We find that HOPE VI reduced poverty rates primarily by changing who moved into the neighborhood rather than by displacing the original residents. The change reflected both a reduction in the stock of public housing units—which led fewer poor households to move in—and the construction of new market-rate units—which attracted higher-income households. In this way, the program achieved one of its primary stated objectives: reducing concentrated poverty in targeted neighborhoods.

Next, we examine whether the poverty reductions in HOPE VI neighborhoods led subsidized renters to live in lower-poverty neighborhoods. We begin by studying subsidized renters who lived in HOPE VI neighborhoods in the year before grant receipt. Five years after the award, 74% of the original residents of HOPE VI public housing projects had moved to a different neighborhood. Because of these high rates of residential mobility, the reduction in poverty rates in HOPE VI neighborhoods is five times larger than the reduction in exposure to neighborhood poverty experienced by the original public housing residents.

Prior work, including two studies commissioned by Congress (Popkin et al., 2009; Buron et al., 2002), also documents high rates of out-migration among original HOPE VI residents, which critics interpret as evidence of forced displacement (Goetz, 2013). However, we find that 56% of original residents of failed-applicant projects also moved within five years, implying that raw migration rates overstate HOPE VI's displacement effects. We also find that nearby residents without subsidized housing were displaced by rising housing prices, but find no evidence of displacement for households with housing vouchers, who were partially shielded from rising rents. Thus, HOPE VI did little to improve original residents' residential outcomes, but large gains for this group were unlikely given the high rates of residential churn in disadvantaged neighborhoods.

We therefore broaden our focus beyond the original residents and examine how HOPE VI affected exposure to neighborhood poverty for all subsidized renters in the city. Our ideal estimand is the effect of HOPE VI on individuals who would have received subsidized housing absent the program. Because we cannot identify this counterfactual population in the data, we instead target a feasible estimand: the difference in exposure to neighborhood poverty between subsidized renters under HOPE VI and subsidized renters in the same cities absent HOPE VI. This estimand consists of two components—the causal effect for those who received subsidized housing after HOPE VI, and a compositional effect from changes in which families receive subsidized housing—and it equals the ideal estimand if HOPE VI did not affect who received subsidized housing.

We use two empirical approaches to estimate effects for all subsidized renters. First, we conduct a city-level analysis comparing cities that applied to revitalize similar shares of their subsidized housing through HOPE VI but were differentially successful in obtaining funding. We find that cities that revitalized more of their subsidized housing experienced larger reductions in exposure to neighborhood poverty for subsidized households. To assess compositional effects, we use the same design to estimate impacts on subsidized renters' characteristics, including income, race, disability status, and family size. We find no evidence that HOPE VI displaced more disadvantaged families out of subsidized housing altogether.

Our second empirical approach directly estimates the causal effect of HOPE VI on exposure to neighborhood poverty among individuals observed receiving subsidized housing after the program.

We partition these individuals into four groups: the original residents of HOPE VI neighborhoods, new residents of HOPE VI neighborhoods, residents who would have moved into HOPE VI neighborhoods absent the intervention, and residents of other neighborhoods affected by spillovers. We estimate group-specific effects using individual-level data to study moves into and out of HOPE VI and failed-applicant neighborhoods. We then calculate aggregate city-level impacts as a weighted average of the group-specific estimates. Our main finding is that cities that revitalized half of their subsidized housing stock—approximately the maximum share affected by HOPE VI—reduced exposure to neighborhood poverty among subsidized renters by 4.6 pp. 59% of this effect is attributable to reductions for the subsidized renters who moved into HOPE VI sites after the program.

Aggregating group-specific estimates yields impacts that are similar to the city-level estimates, suggesting that HOPE VI reduced exposure to neighborhood poverty among subsidized renters primarily through causal effects on subsidized renters rather than through compositional changes in the subsidized population. HOPE VI therefore achieved its goal of enabling subsidized renters to live in lower-poverty neighborhoods.

Our findings point to three lessons for the trade-off between neighborhood revitalization and displacement. First, low-income households in disadvantaged neighborhoods move frequently, so the main beneficiaries of neighborhood-level interventions are likely to be new residents. Unless policymakers can encourage original residents to stay, efforts to maximize a program's impact may be more productive if focused on these newcomers. Second, housing vouchers can help mitigate the displacement effects from rising housing prices. Third, given high residential mobility, efforts to minimize the displacement of the original residents need not conflict with the goal of changing neighborhood composition.

The main contribution of this paper is to estimate the effect of a large place-based intervention on both neighborhoods and individual residential outcomes. Chetty et al. (2026) show that public housing projects revitalized under HOPE VI improve long-run outcomes for children who grow up there. Our findings clarify who benefits from the intervention: primarily families who move in after HOPE VI. This distinction is important because much of the prior work on individual-level impacts focuses on original residents (Kingsley et al., 2003; Jacob, 2004; Chyn, 2018; Haltiwanger et al., 2024). Almagro et al. (2023) interpret neighborhood-level changes through the lens of a structural model and find that HOPE VI reduced welfare for low-income minority renters. Whereas they study impacts on all low-income households, we provide reduced-form estimates of the impact on exposure to neighborhood poverty for subsidized renters—a key input for evaluating the program's overall welfare effects. We also build on research on the neighborhood-level impacts of HOPE VI (Zielenbach and Voith, 2010; Tach and Emory, 2017; Aliprantis and Hartley, 2015; Sandler, 2017; Bruhn, 2018; Blanco, 2023) by using individual-level panel data to decompose neighborhood-level changes into changes in individual outcomes, illuminating the tension between neighborhood revitalization and displacement.

More generally, we contribute to research on urban renewal policies and neighborhood change.[3] Chyn and Katz (2021) note that "a final frontier research area involves the estimation of the impact of place-based policies to improve low-income neighborhoods on the intended beneficiaries—the incumbent (preexisting) adult residents and their children." Our findings confirm the importance of separately estimating impacts on incumbent residents and targeted places, and highlight the value of considering impacts on other beneficiaries—in the case of HOPE VI, the new residents of targeted neighborhoods.

The rest of the paper proceeds as follows. Section 2 provides background on HOPE VI. Section 3 describes the data. Section 4 presents results for neighborhoods and original residents. Section 5 presents results for all households with subsidized housing after HOPE VI. Section 6 concludes.

2 The HOPE VI Revitalization Program

The HOPE VI Revitalization program was created after a report by the National Commission on Severely Distressed Public Housing raised concerns about living conditions in public housing

[3]Papers include Baum-Snow and Marion (2009); Collins and Shester (2013); Busso et al. (2013); Harari et al. (2018); Rojas Ampuero (2022); Blanco and Neri (2023); Gechter and Tsivanidis (2023); Rossi-Hansberg et al. (2010); McKinnish et al. (2010); Guerrieri et al. (2013); Boustan et al. (2019); Diamond and McQuade (2019); Couture et al. (2019); Couture and Handbury (2020); Qiang et al. (2020); Brummet and Reed (2021); Pennington (2021); Su (2022); Lee (2022); Asquith et al. (2023); and Kennedy and Wheeler (2023).

projects (Green and Lane, 1992). The FY1999 Department of Housing and Urban Development (HUD) appropriations bill (Public Law 105-276), which authorized HOPE VI, identified four main goals:

> (1) improving the living environment for public housing residents of severely distressed public housing projects through the demolition, rehabilitation, reconfiguration, or replacement of obsolete public housing projects (or portions thereof); (2) revitalizing sites (including remaining public housing dwelling units) on which such public housing projects are located and contributing to the improvement of the surrounding neighborhood; (3) providing housing that will avoid or decrease the concentration of very low-income families; and (4) building sustainable communities.[4]

In short, the program sought to transform distressed public housing projects into economically integrated neighborhoods where subsidized families would live alongside higher-income neighbors paying market prices for housing.

Most HOPE VI Revitalization funding supported the demolition of distressed public housing projects and the construction of new market-rate and public housing units. The program represented a large neighborhood investment, awarding 262 grants totaling \$6.3 billion from 1993 to 2010, with a mean award of \$24 million. On average, each grant leveraged an additional \$42 million from non-HOPE VI sources, so the total size of the neighborhood-level intervention often exceeded the grant amount (Gress et al., 2016). To put these numbers in perspective, the mean HOPE VI project had 360 occupied units in 1993. Although several distinct programs operated under the broader HOPE VI umbrella, 94 percent of HOPE VI funding went to the Revitalization program, which we refer to as HOPE VI for simplicity.

HOPE VI grants were awarded through a competitive process in which Public Housing Agencies (PHAs)—city-level agencies authorized to administer HUD programs—could submit one application per year. Each year, HUD scored applications based on four factors: need, proposal quality, capacity to undertake development, and amount of non-HOPE VI funding available. HUD then awarded grants to the highest-ranked applicants subject to funding constraints. Because proposals were complex and evaluated by federal officials with limited site-specific knowledge, the scores likely

[4] 42 United States Code §1437v(a).

included noise that led otherwise similar applicants to receive different outcomes. Indeed, HUD's Inspector General found inconsistencies in the scoring process, and HOPE VI officials attributed errors to the short period allowed for grant review (GAO, 2003).

3 Data

We combine survey and administrative data to track neighborhoods and individuals over time. The data have two key strengths: they cover nearly all HOPE VI grants and they contain annual individual-level information on income, residential location, and subsidized housing participation for nearly all individuals in the United States.

Applicants and Awardees. We use publicly available data to identify HOPE VI projects and their neighborhoods. HUD publicly lists all applicants and awardees.[5] We link these records to project-level summary files from HUD and assign each public housing project to one of three categories: i) projects that received a HOPE VI Revitalization grant, ii) projects that applied for but did not receive HOPE VI Revitalization funding, and iii) projects that did not apply for or receive a HOPE VI grant.

We limit our sample to large, non-senior public housing projects with identifiable locations. Table A.1 shows the effect of each sample restriction; after imposing all restrictions, we retain 88% of HOPE VI awardees. We impose the same restrictions on non-HOPE VI projects and additionally require that they be at least one mile from the nearest HOPE VI site, because HOPE VI may have directly affected nearby neighborhoods.[6] Our final sample contains 251 HOPE VI, 166 failed-applicant, and 5,783 non-applicant public housing projects.[7] Figure A.1 maps the locations of each HOPE VI and failed-applicant site.

Neighborhoods. We use HUDUSER's "Picture of Subsidized Households" to identify the Census block group in which each project is located. Block groups are defined by the U.S. Census

[5]The HUD data identify projects using an alphanumeric project ID. We assign project IDs to 99% of Revitalization grantees and 98% of HOPE VI applicants using data from HUDUSER's "Picture of Subsidized Households." We also identify all projects awarded a HOPE VI Demolition grant and drop them from our sample.

[6]Our sample also excludes the 39 demonstration grants awarded between 1993 and 1995.

[7]A single HOPE VI grant or application could affect multiple projects.

Bureau and typically contain about 1,500 residents. Because multiple projects may be located in a single block group and some projects span multiple block groups, we group clusters of nearby public housing projects into neighborhoods defined by the connected sets formed by projects and Census tracts.[8] 90% of neighborhoods contain one or two block groups, and the two largest contain five. For much of our analysis, we collapse the data to these neighborhoods, yielding 125 failed-applicant and 204 HOPE VI neighborhoods.

Before the intervention, HOPE VI neighborhoods contained both public and non-public housing units. The average HOPE VI neighborhood contained 1,059 households in 1993, 45% of whom lived in public housing. Across HOPE VI neighborhoods, the share of households living in public housing ranged from 29 to 60% at the 25th and 75th percentiles, respectively.

Individuals. We use linked survey and administrative data to measure individual outcomes, relying on four main sources. First, HUD administrative data identify who receives subsidized housing and where they live.[9] These data cover several federal rental assistance programs, but 97% of subsidized renters in HOPE VI and failed-applicant neighborhoods participate in public housing, Section 8 vouchers, or project-based Section 8. Second, tax records provide annual measures of income and residential location for all filers. Third, the U.S. Census Bureau's Master Address File–Auxiliary Reference File (MAF-ARF) provides additional address information for individuals not observed in tax records. Fourth, the Decennial Census and American Community Survey provide information on other variables, including income, rent, and homeownership. We link these datasets to construct an individual-level panel with one observation per person-year from 1995 to 2018 for nearly all people living in the United States.[10]

We use adjusted gross income (AGI) to identify poor households. We define individuals as poor if they live in a household where the average AGI over the preceding five years was below

[8] We use tracts to form the connected sets, which allows us to group two projects that are in distinct block groups but in the same tract.

[9] The HUD files we use are the Tenant Rental Assistance Certification System (TRACS) and Public and Indian Housing Information Center (PIC) files.

[10] Although coverage is comprehensive, address data are not available for every individual in every year. We assume that individuals remain at the same address until we observe a new address. For children, we use the address of the parent who most recently claimed them as a dependent.

$15,000.[11] This threshold is approximately the poverty line for a three-person family in 2007. Our measure captures persistent income deprivation (Larrimore et al., 2020). Unless otherwise noted, we use “neighborhood poverty rate” to denote the share of individuals who meet this AGI-based definition of poverty.

Our AGI-based poverty measure closely tracks traditional survey-based measures. Figure A.2 presents a binscatter of the relationship between AGI-based and decennial Census poverty rates at the block-group level in 2000. The measures are closely aligned: the slope is 1.2 (s.e. = 0.002), with an R-squared of 0.69, indicating that our measure captures similar variation in neighborhood poverty to the traditional measure.

We combine the neighborhood and individual data to create two analysis datasets. First, we construct a repeated cross-section summarizing residents of each neighborhood in each year. We use these data to measure neighborhood change over time and to study new residents who move into sample neighborhoods after an award. Second, we construct a panel of original residents that follows individuals who lived in targeted neighborhoods in the year before an award. We use these data to measure the original residents' outcomes and study how this group was affected by HOPE VI. The two datasets provide annual panels tracking neighborhoods and the original residents from 1995 (one year before the first award) to 2018 (eight years after the last award). In some cases, we supplement these data with neighborhood-level information from publicly available summary files from the 1990 and 2000 Decennial Census and annual data from the 2009–2017 ACS, all obtained through IPUMS (Manson et al., 2021).[12]

4 Impacts on Neighborhoods and Original Residents

In this section, we analyze the impact of HOPE VI on two outcomes: i) poverty rates in targeted neighborhoods, and ii) the original residents' exposure to neighborhood poverty. We begin by

[11]For children, we use the AGI of the parent who most recently claimed them as a dependent.

[12]Annual neighborhood-level statistics are from the 5-year ACS estimates, which pool data from the five-year period centered on the year of interest. For example, we use data from the 2015–2019 ACS to estimate neighborhood characteristics in 2017.

defining the target estimands, and then present the empirical strategy and estimates.

4.1 Target Estimands

A PHA (hereafter, "city") submits an application to revitalize a particular neighborhood, and the application is either funded or rejected. Treatment status therefore has two components: T_n indicates whether neighborhood n was targeted by a HOPE VI application, and $D_{c(n)}$ indicates whether city $c(n)$ received the grant.

The outcome space is defined by the triplet $\{N_{it}^d, P_{it}^d, S_{it}^d\}$, where i indexes individuals; $t \in t', t''$ denotes time, with t' before treatment and t'' after treatment; and $d \in 0, 1$ denotes the potential-outcome state corresponding to whether the city received a HOPE VI grant. N_{it}^d is the neighborhood of residence, P_{it}^d is an indicator for income below the poverty line, and S_{it}^d indicates subsidized housing status for individual i at time t. We drop the superscript d to denote observed outcomes; for example, N_{it} is individual i's observed neighborhood of residence at time t.

To simplify the framework, we assume that each city submits at most one HOPE VI application, that there is no migration into or out of the city ($D_{c(N_{ij}^k)} = D_{c(N_{i\ell}^m)}$ for $j, \ell \in t', t''$ and $k, m \in 0, 1$), and that baseline outcomes are unaffected by subsequent treatment ($Y_{it'}^1 = Y_{it'}^0 \equiv Y_{it'}$ for $Y \in N, P, S$). The empirical section below investigates whether violations of these assumptions may bias the estimates.

Impact on Targeted Neighborhoods. Our first target estimand is the average effect on neighborhood poverty rates among neighborhoods that received a HOPE VI grant:

$$\mathbb{E}_n[\bar{P}_{t''}^1(n) - \bar{P}_{t''}^0(n) \mid T_n = 1, D_{c(n)} = 1], \tag{1}$$

where $\bar{P}_t^d(n) \equiv \mathbb{E}_i[P_{it}^d \mid N_{it}^d = n]$ denotes the poverty rate among individuals living in neighborhood n at time t under treatment status d, and $\mathbb{E}_n[\cdot]$ denotes an expectation over neighborhoods, weighting each neighborhood by its baseline population.

To understand mechanisms, we use our longitudinal individual-level data to decompose changes in neighborhood poverty rates into changes in subpopulation sizes defined by three binary variables:

poverty status, subsidized housing receipt, and original-resident status. The neighborhood poverty rate can be expressed as:

$$\bar{P}^d_{t''}(n) = \frac{\sum_{i:N^d_{it''}=n} \sum_{s=0}^{1} \sum_{o=0}^{1} \mathbb{1}\{P^d_{it''} = 1, S^d_{it''} = s, O^d_{it''} = o\}}{\sum_{i:N^d_{it''}=n} \sum_{p=0}^{1} \sum_{s=0}^{1} \sum_{o=0}^{1} \mathbb{1}\{P^d_{it''} = p, S^d_{it''} = s, O^d_{it''} = o\}}, \tag{2}$$

where $O^d_{it''} \equiv \mathbb{1}\{N^d_{it'} = N^d_{it''} = n\}$ indicates whether individual i is an original resident of neighborhood n. We estimate impacts on the size of each subpopulation, $\sum_{i:N^d_{it''}=n} \mathbb{1}\{P^d_{it''} = p, S^d_{it''} = s, O^d_{it''} = o\}$, for all combinations of p, s, and o. This framework allows us to quantify the drivers of neighborhood change: for example, whether poverty rates change because original low-income residents are displaced or because new higher-income residents move in.

In addition to estimating the effect of HOPE VI on poverty rates in targeted neighborhoods, we also examine how the program affected exposure to neighborhood poverty for different groups of individuals. The effect of HOPE VI on individual i's exposure to neighborhood poverty is

$$\Delta_i \equiv \bar{P}^1_{t''}(N^1_{it''}) - \bar{P}^0_{t''}(N^0_{it''}). \tag{3}$$

This effect may reflect changes in where individuals live after treatment, changes in the poverty rates of those neighborhoods, or both.

Impact on Original Subsidized Renters. Our second target estimand is the average effect on exposure to neighborhood poverty for original subsidized renters in HOPE VI neighborhoods:

$$\mathbb{E}_{in}[\Delta_i \mid D_{c(N_{it'})} = 1, T_{N_{it'}} = 1, S_{it'} = 1], \tag{4}$$

where $\mathbb{E}_{in}[\cdot] \equiv \mathbb{E}_n[\mathbb{E}_i[\cdot]]$ denotes an expectation over individuals within neighborhoods and then over neighborhoods.[13] This weighting scheme makes the neighborhood- and individual-level estimates directly comparable.

[13] $\mathbb{E}_{in}[\cdot]$ takes expectations over target neighborhoods, thereby weighting by the baseline population of the target neighborhood. For example, $\mathbb{E}_{in}[\bar{P}^1_{t''}(N^1_{it''}) \mid D_{c(N^1_{it''})} = 1, T_{N_{it'}} = 0, T_{N^1_{it''}} = 1] \equiv \mathbb{E}_n[\mathbb{E}_i[\bar{P}^1_{t''}(N^1_{it''}) \mid N^1_{it''} = n, T_{N_{it'}} = 0] \mid D_{c(n)} = 1, T_n = 1]$.

4.2 Empirical Strategy

The ideal research design to identify the causal effect of HOPE VI on targeted neighborhoods would randomly assign grants across distressed public housing projects. In such an experiment, comparing post-treatment poverty rates in neighborhoods that received a grant to those that did not would identify the causal effect on neighborhood poverty. The same experiment would identify effects on original residents by comparing their post-treatment neighborhood characteristics across treated and control neighborhoods. In practice, HOPE VI grants were not randomly assigned, but we approximate this experiment by comparing HOPE VI grantees to projects that applied for but did not receive funding.

Failed applicants provide a natural comparison group for three reasons. First, funding was scarce: only 29% of applicants in a given year received funding, and 34% never received a grant (see Figure A.3). Second, with average awards exceeding $20 million, PHAs likely could not have replaced HOPE VI funding to carry out planned projects. Third, applying for a HOPE VI grant signals that the PHA viewed the project as a candidate for revitalization, suggesting that failed applicants may be more comparable to awardees—on both observables and, plausibly, unobservables—than non-applicants. Consistent with this, Table A.2 shows that while HOPE VI projects were somewhat more disadvantaged than failed applicants, differences with failed applicants are meaningfully smaller than differences with non-applicants.

Although failed-applicant and HOPE VI neighborhoods differ on average, there are many failed-applicant neighborhoods that are observably similar to HOPE VI neighborhoods. Panel A of Figure A.4 shows substantial overlap in 1990 poverty rates, and Panels B–D show similar overlap for other pre-award characteristics measured in the 1990 Decennial Census.

Given this overlap, we use inverse propensity score weighting (IPW) to adjust for observable differences between the two groups. Let $\pi(X_n) \equiv \Pr(D_{c(n)} = 1 \mid T_n = 1, X_n)$ denote the probability that an applicant neighborhood receives a grant, which is a function of pre-treatment covariates X_n.

For any neighborhood-level outcome Z_n, define the weighted mean as

$$\mathbb{E}_n^w[Z_n \mid D_{c(n)} = d, T_n = 1] \equiv \frac{\mathbb{E}_n[\omega_n^d Z_n \mid D_{c(n)} = d, T_n = 1]}{\mathbb{E}_n[\omega_n^d \mid D_{c(n)} = d, T_n = 1]}, \tag{5}$$

where $d \in \{0, 1\}$ and ω_n^d are the standard IPWs for estimating the average treatment effect on the treated:

$$\omega_n^d = \begin{cases} 1 & \text{if } d = 1 \\ \frac{\pi(X_n)}{1-\pi(X_n)} & \text{if } d = 0. \end{cases} \tag{6}$$

We use $\mathbb{E}_{in}^w[\cdot] \equiv \mathbb{E}_n^w[\mathbb{E}_i[\cdot]]$ to denote expectations over individual-level data using the same IPWs.

In practice, we compare HOPE VI recipients from each grant year to all failed applicants. To estimate propensity scores, we create a stacked dataset with one observation for each HOPE VI neighborhood and 15 observations for each failed-applicant neighborhood, one for each possible grant year from 1996 to 2010. Observations are indexed by neighborhood-by-grant-year pairs (n, g), which assigns each failed-applicant neighborhood a grant year. For simplicity, we refer to these neighborhood-by-grant-year pairs as neighborhoods and denote them using n. For each grant year, we estimate a logistic regression of award receipt on neighborhood population, share Black non-Hispanic, poverty rate, median home value, and public housing project size. This allows selection to vary over time and balances observable characteristics between failed-applicant and HOPE VI neighborhoods within each grant year. In neighborhood-level analyses, we weight observations by the inverse propensity score multiplied by neighborhood population in the year before the award, normalizing failed-applicant weights to equal the total weight of HOPE VI neighborhoods within each grant year.[14]

Propensity score weighting eliminates observable pre-award differences between HOPE VI and failed-applicant neighborhoods. Figure A.5 presents balance tests from regressions of pre-award neighborhood characteristics on an indicator for HOPE VI grant receipt. Panel A weights

[14]For results based on individual-level data, we multiply the inverse propensity score weights by baseline neighborhood population divided by the number of individuals in the analysis sample. Thus, the sum of weights across individuals within a project equals the weight used in the neighborhood-level analysis. This weighting scheme makes results comparable across specifications because all analyses weight by the same measure: baseline neighborhood population.

by population and shows some pre-grant differences between HOPE VI and failed-applicant neighborhoods. Panel B uses IPWs and shows that these weights eliminate observable differences between the two groups.[15]

Because our empirical strategy compares HOPE VI and failed-applicant neighborhoods, the following assumption is central to our empirical analysis.

Assumption 1: Conditional Random Assignment of HOPE VI Grants. Among applicant neighborhoods, HOPE VI grant receipt is as good as randomly assigned conditional on pre-treatment covariates X_n. This assumption implies mean independence of potential outcomes for subpopulations defined by potential neighborhood residence and potential subsidized housing status. Specifically, let $\mathcal{A}$ denote a set of individuals defined by some combination of $N_{it}^d = n$ and S_{it}^d, for $t \in \{t', t''\}$ and $d \in \{0, 1\}$. Then:

$$\mathbb{E}_n^w[\mathbb{E}_i[\bar{P}_j^k(N_{i\ell}^m) \mid \mathcal{A}] \mid D_{c(n)} = 1, T_n = 1] = \mathbb{E}_n^w[\mathbb{E}_i[\bar{P}_j^k(N_{i\ell}^m) \mid \mathcal{A}] \mid D_{c(n)} = 0, T_n = 1], \quad (7)$$

for $j \in \{t', t''\}$, $k \in \{0, 1\}$, $\ell \in \{t', t''\}$, and $m \in \{0, 1\}$. The following paragraphs discuss two special cases that clarify the content of this assumption.

For outcomes that are only observed after treatment, we estimate effects using cross-sectional comparisons between HOPE VI and failed-applicant neighborhoods. Specifically, we estimate

$$y_n = \alpha + \beta D_n + \lambda_g + \epsilon_n \quad (8)$$

where y_n is the outcome for neighborhood n, typically averaged over 10 to 15 years after the award; λ_g is a grant-year fixed effect; and D_n is an indicator for receiving a HOPE VI grant. We estimate this specification by weighted least squares (WLS) using inverse propensity score weights and clustering standard errors by neighborhood.[16] When available, we include pre-treatment covariates

[15]Figure A.6 presents analogous results for contiguous Census block groups. Propensity score weighting eliminates observable pre-award differences for these nearby Census blocks for all but one characteristic.

[16]As discussed above, we use n to denote a neighborhood-by-grant-year pair. Because failed-applicant neighborhoods appear multiple times, once for each grant year, we cluster standard errors at the neighborhood level rather than the neighborhood-by-grant-year level.

to improve precision. The estimation sample is the stacked dataset used to estimate the propensity score, comparing HOPE VI recipients from each grant year to all failed applicants.

The cross-sectional comparison in Equation 8 identifies the target estimands under Assumption 1. In the special case of Assumption 1 in which $j = \ell = t''$, $k = m = 0$, and $\mathcal{A} = \{N^0_{it''} = n\}$, Equation 7 becomes

$$\mathbb{E}^w_n[\bar{P}^0_{t''}(n) \mid D_{c(n)} = 1, T_n = 1] = \mathbb{E}^w_n[\bar{P}^0_{t''}(n) \mid D_{c(n)} = 0, T_n = 1]. \tag{9}$$

In other words, absent treatment, the average poverty rate in neighborhoods that received a HOPE VI grant would have been the same as in failed applicant neighborhoods. This allows us to estimate the first target estimand in Equation 1 by comparing poverty rates in HOPE VI and failed-applicant neighborhoods. See Appendix B for details.

In addition to implying mean independence of potential poverty rates in targeted neighborhoods, Assumption 1 also applies to subpopulations defined by baseline residence and subsidized-housing status. When $j = \ell = t''$, $k = m = 0$, and $\mathcal{A} = \{N_{it'} = n, S_{it'} = 1\}$, Equation 7 becomes

$$\begin{aligned}\mathbb{E}^w_{in}[\bar{P}^0_{t''}(N^0_{it''}) \mid & D_{c(N_{it'})} = 1, T_{N_{it'}} = 1, S_{it'} = 1] \\ &= \mathbb{E}^w_{in}[\bar{P}^0_{t''}(N^0_{it''}) \mid D_{c(N_{it'})} = 0, T_{N_{it'}} = 1, S_{it'} = 1].\end{aligned} \tag{10}$$

In other words, absent treatment, original subsidized residents of HOPE VI neighborhoods would have experienced the same post-period exposure to neighborhood poverty as original subsidized residents of failed-applicant neighborhoods, on average. This allows us to estimate the second target estimand in Equation 4 by comparing the post-period neighborhood poverty rates for original residents of HOPE VI projects to those for original residents of the failed-applicant projects. See Appendix B for details.

For outcomes observed both before and after treatment, we estimate effects using a stacked difference-in-differences estimator. To address concerns from staggered treatment timing, we follow Cengiz et al. (2019) and stack the data to create a panel with a clean "event" for each grant year. For each grant year from 1996 to 2010, the panel includes all failed applicants and the awardees from that year. We then stack the grant-year datasets and estimate

$$y_{nt} = \sum_{j \neq -1} \beta^j D_{nt}^j + \lambda_{gt} + \delta_n + \epsilon_{nt}, \tag{11}$$

where δ_n is a neighborhood fixed effect, λ_{gt} is a grant-year-by-relative-year fixed effect, and D_{nt}^j is a treatment indicator equal to one if neighborhood n received a HOPE VI grant and year t is j years after the award. We estimate this specification by WLS using IPWs and cluster standard errors at the neighborhood level. The coefficients β^j trace out treatment effects over time relative to the year before the award.

The difference-in-differences estimator in Equation 11 identifies treatment effects under a weaker version of Assumption 1 requiring only *parallel trends* in potential outcomes. For example, the parallel-trends assumption needed to identify the first target estimand is

$$\mathbb{E}_n^w[\bar{P}_{t''}^0(n) - \bar{P}_{t'}^0(n) \mid D_{c(n)} = 1, T_n = 1] = \mathbb{E}_n^w[\bar{P}_{t''}^0(n) - \bar{P}_{t'}^0(n) \mid D_{c(n)} = 0, T_n = 1]. \tag{12}$$

We assess the plausibility of this assumption by examining estimated pre-treatment effects and evaluating the sensitivity of the results to alternative comparison groups, estimators, and data sources.

An implicit assumption in our empirical analyses is that potential outcomes do not depend on the treatment status of other projects. This assumption could be violated if HOPE VI reduced poverty in surrounding neighborhoods and failed applicants were located in these nearby neighborhoods. Several contextual details help mitigate this concern. First, 71% of failed-applicant neighborhoods are in PHAs where no project received a HOPE VI grant. Second, we drop failed-applicant neighborhoods within one mile of a HOPE VI site. Third, 40% of HOPE VI neighborhoods are the only such project in their PHA, and another 40% are in PHAs with two or three other sites. Even in PHAs with multiple sites, 85% of project pairs are more than one mile apart, and 98% are more than half a mile apart. Finally, only 3% of individuals who ever live in a HOPE VI or failed-applicant neighborhood subsequently appear in another target neighborhood in our sample. Together, this suggests that physical proximity and residential mobility between target neighborhoods are limited, reducing the risk of bias from spatial spillovers.

4.3 Estimated Impacts on Targeted Neighborhoods

HOPE VI led to a large, persistent reduction in neighborhood poverty rates. Figure 1(A) presents estimates of β^j from Equation 11 and shows that the program reduced the share of residents who were poor by 8 pp 15 years after the award. Figure A.7 explores these results in more detail, showing that 10 to 15 years after the award, HOPE VI reduced the share of households in all income categories below \$15,000 and increased the share with incomes above \$20,000, with the largest increases observed for households with AGI between \$20,000 and \$100,000.

To characterize the timing of the intervention, Figure A.8 presents effects on neighborhood population. HOPE VI led to a 26-log-point decline in population five years after the award, but this effect dissipated over the next five years. The results suggest the following timeline: (*short-run*) in the five years following the award, residents moved out and housing units were demolished; (*medium-run*) over the next five years, residents moved back as housing units were rebuilt; and (*long-run*) around ten years after the award, construction was largely complete and populations had fully recovered.[17]

Table A.3 assesses the robustness of the results to alternative estimators, comparison groups, and data sources. We regress the change in block-group poverty rates from 1990 to 2017 on an indicator for receiving a HOPE VI grant and baseline covariates.[18] The outcome and covariates are measured using publicly available data from the 2017 ACS or 1990 Decennial Census. Column 1 of Panel A uses the failed-applicant comparison group and shows similar effects with the alternative estimator and data: HOPE VI reduced neighborhood poverty rates by 7.7 pp 7–21 years after the award.[19] Columns 2 and 3 compare HOPE VI sites to block groups containing projects that never

[17]Figure A.9 presents effects on the physical housing stock, showing a reduction in total housing units in the first five years and construction of new units over the subsequent five years.

[18]Dube et al. (2025) describe this estimator, which, like the stacked event-study estimator, accounts for issues arising from staggered treatment adoption and heterogeneous effects.

[19]As discussed in Section 3, we classify households with AGI below \$15,000 as poor. Figure A.10 presents estimates

applied for HOPE VI funding and to block groups without public housing projects, respectively; both specifications show similarly large impacts on neighborhood poverty rates. Columns 4 and 5 use the same comparison groups but limit the sample to block groups within one to five miles of a HOPE VI site and include fixed effects for the closest HOPE VI site. These estimates again show large impacts when comparing HOPE VI sites to observably similar nearby neighborhoods. The specifications in Panel B do not control for baseline differences and yield substantially larger estimates. Taken together, Table A.3 shows similar effects across alternative comparison groups, provided that we control for observable pre-award differences between HOPE VI and comparison neighborhoods.[20]

We use the decomposition in Equation 2 to examine the drivers of neighborhood change. Figure 1(B) presents estimates from Equation 8, where the outcome is the average number of people in each group defined by poverty, original-resident status, and subsidy status in the neighborhood 10–15 years after the award.[21] Neighborhood change was driven by two factors: i) more higher-income families moved into market-rate units after the award, and ii) fewer poor families with subsidized housing moved in. Importantly, neighborhood-level changes primarily reflect changes in who moves in, not displacement of original residents. As we discuss in the next section, high residential churn limits displacement effects for original residents.

The program reduced poverty rates in targeted neighborhoods but reduced subsidized renters' access to those neighborhoods. Figure 1(B) implies that the program reduced the size of the population with subsidized housing by 20%.[22] Figure A.14 presents estimates from Equation 11 and shows a long-run decline in the subsidized population of a similar magnitude. One displacement-

from Equation 11 using survey-based neighborhood poverty rates as the outcome. We impute missing values between Census surveys by assuming poverty rates do not change unless observed data indicate otherwise. The survey-based measure yields similar results: HOPE VI reduced poverty rates by 11 pp 15 years after the award.

[20]Figure A.11 shows that estimating Equation 11 with baseline-population weights yields results similar to our main IPW-weighted results. This is because pre-grant differences in observable characteristics are modest and failed-applicant neighborhoods were relatively stable throughout the study period, while revitalized neighborhoods changed dramatically. See Figures A.12 and A.13 and Appendix C for details.

[21]We use Equation 8 for this analysis because we estimate impacts on the size and characteristics of the population that moved in after the award, and this population is not defined before the award. Table A.4 presents these same estimates.

[22]We arrive at these calculations by taking the sum of the treatment effects divided by the sample mean size of the populations: -0.20=(-43-168-20-2)/(108+696+52+335). See Columns 7 through 14 of Table A.4.

related concern is that reductions in neighborhood poverty may be driven by an influx of childless adults (McKinnish et al., 2010). This does not appear to be the case: we find similar reductions in poverty rates among households with and without children (see Figure A.15).

HOPE VI reduced poverty rates by attracting residents who would have had higher incomes regardless of where they lived, rather than by increasing new residents' incomes. For each neighborhood and year, we calculate the poverty rate among individuals who move into the neighborhood using their income in the previous, current, or subsequent year.[23] Figure 2 presents estimates from Equation 11, using these three poverty measures as outcomes. The red circles use income in the year of the move, while the blue squares and green diamonds use income in the year before and after the move, respectively. The three sets of estimates are statistically indistinguishable. Thus, higher-income residents who moved into revitalized sites already had higher incomes before moving.

Figure A.16 shows how HOPE VI affected neighborhoods' racial and ethnic composition. Many HOPE VI projects were predominantly Black non-Hispanic, which explains why the Black non-Hispanic population share declined in the short run as public housing projects were demolished. By 15 years after the award, the Black non-Hispanic share had largely recovered, and we cannot reject that there is no difference between HOPE VI and failed applicant neighborhoods in the share of residents who were Black non-Hispanic.

HOPE VI increased housing prices in treated neighborhoods. ACS data 10–15 years after the award show that rents, home values, and monthly mortgage payments rose by more than 10%, with similar estimates after controlling for housing-unit characteristics and restricting to units that existed prior to the intervention. These patterns suggest that the program increased neighborhood desirability rather than simply improving the physical housing stock. Appendix C discusses these results, which are presented in Table A.5.

Finally, we investigate whether HOPE VI had spillover effects on surrounding neighborhoods. We measure changes in poverty rates for all block groups within a 3-mile radius of each target

[23] In contrast to our main poverty measure, these outcomes are based on the individual's income in a single year rather than a five-year average.

neighborhood and group surrounding block groups into half-mile bins based on the distance between their centroid and the target neighborhood centroid. We then regress the change in poverty rates from the year before the award to 15 years after the award on the interaction between the HOPE VI indicator and distance from the targeted neighborhood indicators. Figure 3 presents the estimates, which show limited evidence of spatial spillovers on nearby neighborhoods.[24] Together, these results indicate that the effects were highly local and faded quickly with distance from the targeted neighborhoods.

4.4 Estimated Impacts on the Original Residents

We begin our analysis of original residents by examining migration responses. Figure 4(A) plots the share of the original residents of HOPE VI neighborhoods—i.e., residents of the neighborhood in the year before the award—who had moved away 0–15 years after an award, disaggregated by baseline housing subsidy type. Within five years of an award, over 60% of the original residents had moved, and migration rates were high for households with and without subsidized housing. These high migration rates overstate displacement effects, as out-migration rates were similarly high in failed-applicant neighborhoods (see Figure A.18).

To directly assess displacement, Figure 4(B) presents effects on the share of the original residents who had left the target neighborhood by a given year, based on Equation 11. Displacement effects for the original public housing residents peak at 18 pp five years after the award but fade to 6 pp after 15 years, as most original residents of failed-applicant sites had also moved away by then. Original residents without subsidized housing were also 5 pp more likely to have moved within five years of an award, and this difference persisted 15 years after the award. One explanation is that rising local housing costs—documented in Table A.5—priced these residents out of the neighborhood. Consistent with this hypothesis, we find no displacement effect for the original residents with access to other forms of subsidized housing—primarily vouchers—which would have limited their exposure to rent increases.

[24]Figure A.17 presents similar results using Equation 11 to estimate impacts on poverty rates in contiguous Census block groups.

Because of these high migration rates, HOPE VI's effect on the original residents' exposure to neighborhood poverty was much smaller than its effect on targeted neighborhoods. The red triangles in Figure 5(A) present the estimated effects for the original residents—i.e., all residents of the neighborhood in the year before the award. HOPE VI reduced the poverty rate of the neighborhoods where original residents lived 15 years after an award by 1.5 pp. For comparison, the blue circles reproduce the estimates from Figure 1(A) for poverty rates in targeted neighborhoods. The reduction in targeted-neighborhood poverty rates is 5.4 times as large as the reduction in the original residents' exposure to neighborhood poverty. Figure 5(B) presents estimates of the second estimand in Equation 4 for subgroups of original residents defined by baseline housing subsidy type. Regardless of their baseline subsidy type—non-subsidized housing, public housing, or other subsidized housing—all original residents experienced similar reductions in exposure to neighborhood poverty.[25]

We also find no evidence that HOPE VI affected the income of adult original residents. Figure A.20 presents estimates from Equation 11, where the outcome is the original residents' poverty rate rather than exposure to neighborhood poverty. For all periods, the estimates are economically small and statistically insignificant.

We conclude that HOPE VI reduced poverty rates in targeted neighborhoods by attracting higher-income households into market-rate units. The program had a limited effect on the original residents' exposure to neighborhood poverty because of high residential mobility among this group.

5 Impact of HOPE VI on All Subsidized Renters

In this section, we broaden the analysis beyond the original residents and examine how HOPE VI affected exposure to neighborhood poverty for all subsidized renters in the city.

5.1 Target Estimand

Building on the framework in Section 4.1, our ideal estimand is the average effect on exposure to neighborhood poverty among all individuals in the city who would have received subsidized

[25] Figure A.19 presents estimates for original residents without subsidized housing by baseline poverty status. We find no evidence of worsening neighborhood conditions for poor residents without subsidized housing.

housing absent HOPE VI:

$$\tau^{\text{ideal}} = \mathbb{E}_i[\Delta_i \mid D_{c(N^0_{it''})} = 1, S^0_{it''} = 1]. \tag{13}$$

This estimand is difficult to estimate because we do not observe who would have received subsidized housing absent HOPE VI. We therefore define a feasible target estimand and assess how it differs from the ideal estimand.

Impact on All Subsidized Households. The feasible target estimand compares exposure to neighborhood poverty among subsidized households under HOPE VI to exposure among subsidized households in those cities absent HOPE VI:

$$\tau^{\text{feas}} = \mathbb{E}_i[\bar{P}^1_{t''}(N^1_{it''}) \mid D_{c(N^1_{it''})} = 1, S^1_{it''} = 1] - \mathbb{E}_i[\bar{P}^0_{t''}(N^0_{it''}) \mid D_{c(N^0_{it''})} = 1, S^0_{it''} = 1]. \tag{14}$$

This feasible estimand can be decomposed into two components:

$$\tau^{\text{feas}} = \tau^{\text{comp}} + \tau^{\text{caus}}, \tag{15}$$

$$\tau^{\text{comp}} = \mathbb{E}_i[\bar{P}^0_{t''}(N^0_{it''}) \mid D_{c(N^1_{it''})} = 1, S^1_{it''} = 1] - \mathbb{E}_i[\bar{P}^0_{t''}(N^0_{it''}) \mid D_{c(N^0_{it''})} = 1, S^0_{it''} = 1], \tag{16}$$

$$\tau^{\text{caus}} = \mathbb{E}_i[\Delta_i \mid D_{c(N^1_{it''})} = 1, S^1_{it''} = 1]. \tag{17}$$

The first term, τ^{comp}, captures compositional changes in who receives housing subsidies. The second term, τ^{caus}, captures the causal effect on exposure to neighborhood poverty among those with subsidized housing after HOPE VI. If HOPE VI does not affect who receives subsidized housing ($S^0_{it''} = S^1_{it''}$), then $\tau^{\text{comp}} = 0$ and $\tau^{\text{feas}} = \tau^{\text{caus}} = \tau^{\text{ideal}}$.

Although HOPE VI reduced poverty rates in targeted neighborhoods, these neighborhood-level changes need not have led subsidized renters to live in lower-poverty neighborhoods. The program could have displaced subsidized renters to other high-poverty neighborhoods, potentially increasing both their exposure to neighborhood poverty and that of other residents in destination neighborhoods.

We formalize this point in Appendix D, where we interpret HOPE VI through a model of residential sorting (Schelling, 1971, 2006; Glaeser, 2008).

We organize the empirical analysis in three steps. First, Section 5.2 uses variation across PHAs to estimate τ^{feas} and quantify the potential role of τ^{comp}. Second, Section 5.3 builds on the empirical design in Section 4 to estimate τ^{caus}. Finally, Section 5.4 compares these estimates to assess whether they provide a consistent account of HOPE VI's aggregate effects.

5.2 Effect on All Subsidized Renters and Their Composition

We estimate τ^{feas} by comparing PHAs that applied to revitalize similar shares of their subsidized housing through HOPE VI but were differentially successful in obtaining funding. To facilitate this comparison, we construct a PHA-level dataset and estimate:[26]

$$y_c = \alpha + \lambda A_c + \delta H_c + \beta D_c + \phi X_c + \theta A_c \times X_c + \epsilon_c, \tag{18}$$

where c denotes the PHA, A_c is the share of all subsidized units in the PHA that were targeted by a successful or unsuccessful HOPE VI grant application (i.e., the number of public housing units in projects that applied for funding divided by the total number of public and voucher units in the PHA), H_c is the share of all subsidized units in the PHA that were targeted by a successful HOPE VI grant application, D_c equals one if the PHA received a HOPE VI grant and allows for level differences between PHAs that did and did not receive funding,[27] and X_c is the average 1990 poverty rate of HOPE VI and failed-applicant sites.[28] The sample consists of PHAs that either received HOPE VI funding or applied but received none. The main coefficient of interest is δ, which captures the effect of revitalizing a larger share of subsidized units.

[26]This PHA-level analysis uses data from HUDUSER's "Picture of Subsidized Households" and focuses on changes between 1997, the earliest year in which all necessary variables are available, and 2018.

[27]Among PHAs where HOPE VI applications targeted a small share of the overall subsidized housing stock (A_c is close to zero), those that win HOPE VI funding have similar outcomes to those that do not. Because of this, we find similar results when we estimate a model without including D_c as a covariate.

[28]We control for baseline poverty rates because they are a key potential confounder. We estimate that HOPE VI reduced poverty rates in targeted neighborhoods by 7.7 pp when regressing block-group poverty rates in 2017 on the HOPE VI indicator and a vector of 15 baseline covariates (see Column 1 of Table A.3). If we estimate the same regression controlling only for baseline poverty rates, the estimate is 6.9 pp. With no covariates, the estimate falls to 3.5 pp. Following Feigenberg et al. (2025), we include the interaction between A_c and X_c.

Table 1 presents estimates from Equation 18. The outcome in Column 1 is the change from 1997 to 2018 in the average poverty rate of the Census tracts where subsidized households lived.[29] The estimates imply that revitalizing half of subsidized units in a city—approximately the maximum observed in the sample—reduced exposure to neighborhood poverty among subsidized renters by 9 pp, on average: $\hat{\tau}^{\text{feas}}\big|_{H_c=0.5} = -0.09 = -0.183 \times 0.5$.

A key question is whether reductions in exposure to neighborhood poverty reflect causal impacts on individuals (τ^{caus}) or displacement of the neediest families from subsidized housing altogether (τ^{comp}). To assess the displacement channel, Column 2 of Table 1 shows that HOPE VI reduced the share of subsidized housing units that were public housing. This effect could reflect both a reduction in public housing units and an expansion of the voucher program, as HUD allowed PHAs to apply for additional housing vouchers to offset units removed through HOPE VI redevelopment.[30] Column 3 shows no effect on the aggregate number of subsidized units. In other words, HOPE VI shifted the type of subsidy from public housing to vouchers but does not appear to have reduced the total number of subsidized units available in the city.

Although HOPE VI did not reduce the aggregate number of subsidized housing units, the shift from public housing to vouchers may raise displacement concerns, as existing research suggests that voucher recipients, particularly those that are more disadvantaged, often have difficulty finding landlords willing to accept vouchers (Ellen, 2020). To assess changes in the composition of subsidized renters, Columns 4–8 of Table 1 present estimated effects on characteristics likely associated with difficulty using a voucher: household income below \$10,000, Black household head, number of people per housing unit, households with children headed by a single parent, and households below age 62 in which the household head, spouse, or co-head has a disability. We find no evidence that HOPE VI displaced any of these groups out of subsidized housing altogether. This suggests that HOPE VI did not change the composition of subsidized households, consistent with

[29]To see the connection to the target estimand, note that the outcome for the treated sample in column 1 is $\mathbb{E}_i[\bar{P}_{t''}(N_{it''}) \mid D_{c(N_{it''})} = 1, S_{it''} = 1] - \mathbb{E}_i[\bar{P}_{t'}(N_{it'}) \mid D_{c(N_{it'})} = 1, S_{it'} = 1]$.

[30]Typically, additional housing vouchers were awarded to families who previously occupied demolished public housing units. After these families exited the voucher program, however, the voucher could be reissued to another eligible household on the waiting list.

$\tau^{\text{comp}} = 0$.

Column 9 of Table 1 shows no effect on county-level poverty rates, implying that HOPE VI altered the spatial distribution of poverty within cities but did not change aggregate poverty rates, at least for the current generation of adults. The outcome in Column 10 is the component of changes in exposure to neighborhood poverty (the outcome in Column 1) attributable to changes in poverty rates in targeted sites.[31] The estimates imply that reductions in neighborhood poverty in revitalized sites explain 75% of the total reduction in exposure to neighborhood poverty (0.75 = 0.138/0.183). Thus, the gains among subsidized renters appear concentrated among families who live in HOPE VI neighborhoods after the intervention. We examine this directly in the next section.

We also use Equation 18 to estimate impacts on exposure to neighborhood poverty for all other households in the county. We estimate changes in exposure to neighborhood poverty from 1990 to 2017 for household groups defined by income. Table A.6 presents the results. We find no evidence of impacts on the broader population. For example, Column 1 implies that revitalizing half of subsidized units is associated with a 0.2 pp increase in exposure to neighborhood poverty ($0.002 = 0.004 \times 0.5$) for households with annual incomes below \$10,000. We find similarly small and statistically insignificant impacts for all eight income groups. These analyses have limited power to detect effects on the broader population, as expected given that subsidized households represent a small share of the overall population.

5.3 Impacts on Subgroups in Subsidized Housing After HOPE VI

This section estimates τ^{caus}, the average causal effect of HOPE VI on exposure to neighborhood poverty among individuals who receive subsidized housing after the intervention. Rather than estimate this aggregate effect directly, we decompose it into effects for conceptually distinct subgroups and then aggregate the subgroup-specific effects. Specifically, we partition individuals who receive subsidized housing in HOPE VI cities after treatment, $\{D_{c(N^1_{it''})} = 1, S^1_{it''} = 1\}$, into

[31] We can write the average poverty rate for subsidized households in a given PHA at time t as, $p_t = \omega_t p^r_t + (1-\omega_t)p^n_t$, where ω_t is the share of units in targeted neighborhoods and p^r and p^n denote the average poverty rates in targeted and other neighborhoods, respectively. Then we can write the change as, $p_{t+1} - p_t = \omega_t(p^r_{t+1} - p^r_t) + p^r_{t+1}(\omega_{t+1} - \omega_t) + (1-\omega_{t+1})p^n_{t+1} - (1-\omega_t)p^n_t$. The outcome in Column 1 of Table 1 is $p_{t+1} - p_t$ and the outcome in Column 10 is $\omega_t(p^r_{t+1} - p^r_t)$.

four mutually exclusive, exhaustive groups:

1. *Original Residents Who Received Subsidized Housing After.* Those who lived in targeted neighborhoods before the award and received subsidized housing after:

$$\mathcal{G}_1 = \left\{ D_{c(N^1_{it''})} = 1, S^1_{it''} = 1, T_{N_{it'}} = 1 \right\}. \tag{19}$$

2. *New Residents of HOPE VI Neighborhoods.* Subsidized renters who moved into HOPE VI neighborhoods after the award:

$$\mathcal{G}_2 = \{D_{c(N^1_{it''})} = 1, S^1_{it''} = 1, T_{N_{it'}} = 0, T_{N^1_{it''}} = 1\}. \tag{20}$$

3. *Residents Who Would Have Moved in Absent HOPE VI.* Subsidized renters who, absent HOPE VI, would have moved into a treated neighborhood but instead reside elsewhere:

$$\mathcal{G}_3 = \{D_{c(N^1_{it''})} = 1, S^1_{it''} = 1, T_{N_{it'}} = 0, T_{N^1_{it''}} = 0, T_{N^0_{it''}} = 1\}. \tag{21}$$

4. *Residents of Other Neighborhoods.* Subsidized renters who reside in other neighborhoods, potentially affected by spillovers:

$$\mathcal{G}_4 = \{D_{c(N^1_{it''})} = 1, S^1_{it''} = 1, T_{N_{it'}} = 0, T_{N^1_{it''}} = 0, T_{N^0_{it''}} = 0\}. \tag{22}$$

Because these four groups partition the set of individuals who receive subsidized housing after the program, the weighted sum of group-specific effects equals the causal component of the feasible estimand:

$$\tau^{\text{caus}} = \sum_{g=1}^{4} \omega_g \mathbb{E}_i \left[\Delta_i \mid i \in \mathcal{G}_g \right], \tag{23}$$

where ω_g is the share of individuals in $\mathcal{G}_g$. We therefore estimate the following group-specific treatment effects.

Impacts on Subgroups in Subsidized Housing After HOPE VI. The target estimand is the average effect on exposure to neighborhood poverty for each of the four subgroups observed in subsidized housing after HOPE VI:[32]

$$\bar{\Delta}_g \equiv \mathbb{E}_{in} \left[\Delta_i \mid i \in \mathcal{G}_g \right] \text{ for } g \in [1, 4]. \tag{24}$$

[32]We use the same weighting scheme as in Section 4 so that the neighborhood- and individual-level estimates are directly comparable.

5.3.1 Effect on Original Residents Who Received Subsidized Housing After

We estimate effects for original residents of HOPE VI neighborhoods who received subsidized housing after HOPE VI by comparing them to analogous residents of failed-applicant neighborhoods. Identification relies on the following assumption.

Assumption 2. Among original residents of failed-applicant neighborhoods, the average potential outcomes for those who would have subsidized housing in the post-period under the treatment regime is the same for those who would have subsidized housing in the post-period under the non-treatment regime. Specifically,

$$\begin{aligned}\mathbb{E}^{w}_{in}[\bar{P}^{0}_{t''}(N^{0}_{it''}) \mid D_{c(N_{it'})} = 0, T_{N_{it'}} = 1, S^{0}_{it''} = 1] \\ = \mathbb{E}^{w}_{in}[\bar{P}^{0}_{t''}(N^{0}_{it''}) \mid D_{c(N_{it'})} = 0, T_{N_{it'}} = 1, S^{1}_{it''} = 1].\end{aligned} \tag{25}$$

This assumption holds if HOPE VI does not affect which original residents appear in subsidized housing in the post-treatment period—i.e., if $S^{0}_{it''} = S^{1}_{it''}$.

Two pieces of evidence support Assumption 2. First, Figure A.21(A) shows no effect on the likelihood that original residents appear in subsidized housing 10–15 years after the award.[33] The remaining concern is compositional: the program could have changed which types of households remained in subsidized housing. We assess this concern by comparing pre-program observable characteristics for residents who remain in subsidized housing in HOPE VI and failed-applicant neighborhoods, i.e., under treatment and control. The two groups are nearly identical in income, the characteristic we expect to be most strongly related to potential outcomes for neighborhood poverty—i.e., $\mathbb{E}[P_{it'} \mid D(N_{it'}) = 1, S^{1}_{it''} = 1] = \mathbb{E}[P_{it'} \mid C(N_{it'}) = 1, S^{0}_{it''} = 1]$. Specifically, among original residents who remain in subsidized housing ten years after the award, the difference in pre-award income between HOPE VI and failed-applicant neighborhoods is \$45 (s.e. = \$396), statistically indistinguishable from zero and economically small.

[33]Figure A.21(B) presents results for original residents who had subsidized housing before the award. The probability of living in public housing declined, but this reduction was fully offset by a corresponding increase in voucher receipt.

Assumptions 1 and 2 imply the following identification result:

$$\begin{aligned}\bar{\Delta}_1 =& \mathbb{E}^w_{in}[\bar{P}_{t''}(N_{it''}) \mid D_{c(N_{it'})} = 1, T_{N_{it'}} = 1, S_{it''} = 1] \\ &- \mathbb{E}^w_{in}[\bar{P}_{t''}(N_{it''}) \mid D_{c(N_{it'})} = 0, T_{N_{it'}} = 1, S_{it''} = 1].\end{aligned} \tag{26}$$

See Appendix B for details.

The green squares in Figure 5(A) present the empirical analog of Equation 26, re-estimating our main event-study specification in Equation 11 on the subsample of all original residents (regardless of original subsidy status) who later appear in subsidized housing ten years after the award. The results imply that HOPE VI reduced exposure to neighborhood poverty for original residents by 2.0 pp 10–15 years after the award: $\hat{\bar{\Delta}}_1 = -0.020$.

5.3.2 Effect on New Residents of HOPE VI Neighborhoods

A reduction in poverty rates in revitalized neighborhoods does not, by itself, imply a reduction in exposure to neighborhood poverty for new residents. The program could have attracted new residents who would have chosen lower-poverty neighborhoods elsewhere absent the intervention. Formally, we can decompose the treatment effect for new residents into two components:

$$\bar{\Delta}_2 = \underbrace{\mathbb{E}^w_n[\bar{P}^1_{t''}(n) - \bar{P}^0_{t''}(n) \mid D_{c(n)} = 1, T_n = 1]}_{\text{Effect on Neighborhoods}} - \underbrace{\mathbb{E}^w_{in}[\bar{P}^0_{t''}(N^0_{it''}) - \bar{P}^0_{t''}(N^1_{it''}) \mid i \in \mathcal{G}_2]}_{\text{Sorting Effect}}. \tag{27}$$

The effect on new residents' exposure to neighborhood poverty depends not only on how the program affected poverty rates in targeted neighborhoods, but also on where these individuals would have lived absent the intervention. If they would have lived in lower-poverty areas absent the program, the sorting effect could offset the neighborhood-level poverty reduction.

To assess empirically the importance of the sorting effect, we compare subsidized renters who move into HOPE VI and failed-applicant neighborhoods after an award. We estimate differences in the poverty rates of their *origin neighborhoods*—the neighborhoods where they lived before moving—and use these origins as proxies for where individuals would have lived absent the intervention. If newcomers to revitalized neighborhoods came from lower-poverty areas than those entering failed-applicant neighborhoods, this might suggest that the program attracted individuals who would otherwise have lived in lower-poverty neighborhoods elsewhere. Assumption 3 formalizes this intuition.

Assumption 3. There is a linear relationship between the poverty rate in the origin neighborhood and non-treated counterfactual,

$$\bar{P}^0_{t''}(N^0_{it''}) = \alpha + \frac{1}{\theta}\bar{P}_{t'}(N_{it'}) + \epsilon_i \text{ with } \theta > 0, \tag{28}$$

where ϵ_i is an idiosyncratic error term.

Assumptions 1 and 3 yield the following identification result:

$$\begin{aligned} \theta\bar{\Delta}_2 =& \mathbb{E}^w_{in}[\bar{P}_{t'}(N_{it'}) \mid D_{c(N_{it''})} = 1, T_{N_{it''}} = 1, T_{N_{it'}} = 0, S_{it''} = 1] \\ & - \mathbb{E}^w_{in}[\bar{P}_{t'}(N_{it'}) \mid D_{c(N_{it''})} = 0, T_{N_{it''}} = 1, T_{N_{it'}} = 0, S_{it''} = 1]. \end{aligned} \tag{29}$$

In other words, the sorting effect is proportional to the difference in origin-neighborhood poverty rates between subsidized renters who move into HOPE VI and failed-applicant neighborhoods in the post-period. See Appendix B for details.

To estimate this difference, we construct a sample of subsidized households who move into a failed applicant or HOPE VI neighborhood between 5 years and 15 years after an award and estimate Equation 11 on individual-level panel data where the outcome is the poverty rate in the *origin* neighborhood (i.e., the neighborhood the individual lived in before moving to the HOPE VI or failed-applicant neighborhood).

We find that subsidized renters who moved into revitalized neighborhoods came from neighborhoods with poverty rates similar to those of subsidized renters who moved into failed-applicant neighborhoods. The blue circles in Figure 6 show that origin-neighborhood poverty rates for new residents of revitalized neighborhoods were never statistically different and were only 0.8 pp lower 15 years after the award. For comparison, the red squares show effects on destination-neighborhood poverty rates, replicating Figure 1(A): families who moved into HOPE VI neighborhoods after the award moved into lower-poverty neighborhoods than those who moved into failed-applicant neighborhoods. These results suggest that sorting effects are small relative to effects on poverty rates in targeted neighborhoods.

To further quantify the importance of sorting, we estimate the effect on destination-neighborhood poverty rates using a modified version of Equation 11 that controls for origin-neighborhood poverty rates, individual income before the move, and a quadratic in age. The green diamonds in Figure 6 display these estimates. The results are nearly identical: 15 years after the award, the adjusted estimate is only 0.06 pp, or 1%, smaller than the unadjusted estimate. If the covariates fully capture differences in outside housing options for subsidized families moving into HOPE VI and failed-applicant neighborhoods, the adjusted estimates represent the causal effect of HOPE VI on new residents' exposure to neighborhood poverty.

New residents of HOPE VI neighborhoods could also differ along other dimensions. We explore this possibility by controlling for a richer set of characteristics. Panel A of Table A.7 presents estimates from regressions of outcomes on the HOPE VI indicator and interactions between award year and move year, estimated on individuals who move in after the award. The outcomes in Columns 1 and 2 are the poverty rates in the prior and current neighborhoods, respectively. These columns replicate Figure 6, but summarize average differences rather than present effects for each year relative to the award. Columns 3–5 add increasingly rich covariate sets: Column 3 adds the baseline covariates used to produce the green series in Figure 6; Column 4 additionally controls for contemporaneous income; and Column 5 adds 14 characteristics from the HUD data, including race, ethnicity, sex, disability status, and income from various sources. These additional covariates do not materially affect the estimates, suggesting little sorting along these observable dimensions.

We view the estimates in Figure 6 as strong evidence that the reduction in poverty in targeted neighborhoods translated into lower exposure to neighborhood poverty for new subsidized households. Specifically, the estimates suggest that HOPE VI reduced new residents' exposure to neighborhood poverty by 7.6 pp 15 years after the award: $\hat{\hat{\Delta}}_2 = -0.076$.

Figure A.22 presents the effect of HOPE VI on out-migration. We estimate Equation 11, where the outcome is the share of current residents who move to another neighborhood the following year. For subsidized renters, out-migration rates spike four years after the award but decline in the long run: subsidized renters living in HOPE VI neighborhoods 10–15 years after the award are 3 pp less likely to move than those in failed-applicant neighborhoods. Consistent with the previous analysis, this pattern suggests that new residents stay longer because they prefer the lower-poverty, revitalized neighborhoods. For non-subsidized households, we find much smaller short-run displacement effects and no evidence of long-run differences in out-migration rates.

5.3.3 Effect on Residents Who Would Have Moved in Absent HOPE VI

HOPE VI reduced the subsidized housing population in targeted neighborhoods by approximately 20% in the long run (see Figure A.14). However, as we show in Section 5.4, the overall citywide supply of subsidized housing remained unchanged because of an offsetting expansion in housing vouchers. This implies that some subsidized renters who might otherwise have moved into HOPE VI neighborhoods instead lived elsewhere. While we cannot directly identify these individuals, we approximate their counterfactual outcomes using origin-neighborhood poverty rates among those who moved into failed-applicant neighborhoods after an award. Our analysis relies on the following assumption.

Assumption 4. This assumption has two components. First, for individuals who would have moved into a HOPE VI neighborhood absent the award, the poverty rate of their post-award neighborhood is equal, on average, to the poverty rate of their origin neighborhood:

$$\mathbb{E}_{in}[\bar{P}^1_{t''}(N^1_{it''}) \mid i \in \mathcal{G}_3] = \mathbb{E}_{in}[\bar{P}^0_{t''}(N_{it'}) \mid i \in \mathcal{G}_3], \tag{30}$$

Second, among all subsidized renters who move into failed-applicant neighborhoods, the poverty rate of the origin neighborhood is the same as for those who would move into those neighborhoods only absent treatment. Specifically,

$$\begin{aligned} &\mathbb{E}^w_{in}[\bar{P}^0_{t''}(N_{it'}) \mid D_{c(N_{it''})} = 0, T_{N_{it'}} = 0, T_{N_{it''}} = 1, S_{it''} = 1] \\ &\qquad = \mathbb{E}^w_{in}[\bar{P}^0_{t''}(N_{it'}) \mid D_{c(N^0_{it''})} = 0, T_{N_{it'}} = 0, T_{N^0_{it''}} = 1, T_{N^1_{it''}} = 0, S^1_{it''} = 1], \end{aligned} \tag{31}$$

Assumption 4 holds if individuals who would have moved into targeted neighborhoods absent HOPE VI but do not do so under treatment remain in their origin neighborhoods, treatment does not change the poverty rate of those origin neighborhoods, and these individuals are not systematically different from those who move into targeted neighborhoods regardless of treatment.

Under Assumptions 1 and 4, we obtain

$$\bar{\Delta}_3 = \mathbb{E}^{w}_{in}[\bar{P}_{t''}(N_{it'}) - \bar{P}_{t''}(N_{it''}) \mid D_{c(N_{it''})} = 0, T_{N_{it'}} = 0, T_{N_{it''}} = 1, S_{it''} = 1], \tag{32}$$

where the right-hand side is the difference in average poverty rates between the origin and destination neighborhoods of subsidized renters who move into failed-applicant neighborhoods after the award. See Appendix B for details.

To estimate this, we identify all subsidized renters who moved into failed-applicant neighborhoods 10–15 years after the award. For these individuals, we compute the difference between the poverty rates in their origin and destination neighborhoods. On average, the destination neighborhood had a poverty rate 14 pp higher than the origin neighborhood, implying that subsidized renters who move into failed-applicant neighborhoods experienced large increases in neighborhood poverty. Under Assumptions 1 and 4, this implies that subsidized renters who would have moved into HOPE VI neighborhoods absent the intervention instead experienced a 14 pp reduction in exposure to neighborhood poverty: $\hat{\bar{\Delta}}_3 = -0.14$.

We argue that Assumption 4 is plausible because voucher recipients typically rent units in neighborhoods with poverty rates similar to those of their origin neighborhoods (Feins and Patterson, 2005). The magnitude of this estimate is also consistent with the extreme poverty of targeted neighborhoods before treatment. In 1990, 86% of these neighborhoods were in the top decile of the county-level poverty distribution. The 25th and 50th percentiles of the targeted-neighborhood distribution corresponded to the 95th and 98th percentiles of the broader county distribution, respectively. For households diverted away from these neighborhoods, it would have been difficult to relocate elsewhere without reducing exposure to neighborhood poverty.

5.3.4 Spillover Effects on Residents of Other Neighborhoods

If HOPE VI reduced poverty rates in targeted neighborhoods by changing where poor adults lived rather than by changing individual incomes, then those reductions must be offset by increases in poverty elsewhere. If poor households were displaced into neighborhoods with many subsidized renters, these increases could offset the gains experienced by families in HOPE VI neighborhoods. We estimate an upper bound for this "migration spillover" effect: the effect attributable to individuals who would have lived in HOPE VI neighborhoods absent the award but instead live elsewhere. Our approach uses Figure 1(B) to estimate the number of displaced poor households and then allocate these households in a way that is consistent with observed migration flows and maximizes the potential increase in subsidized renters' average exposure to neighborhood poverty. The following assumption allows us to recover this upper bound.

Assumption 5. This assumption has four components. First, individuals who would have moved into targeted sites absent treatment, instead stay in their origin neighborhoods under treatment:

$$N^1_{it''} = N_{it'} \text{ for } i : T_{N_{it'}} = 0, T_{N^1_{it''}} = 0, T_{N^0_{it''}} = 1. \tag{33}$$

Second, HOPE VI does not impact the size of neighborhoods that are not directly targeted by the program:

$$\sum_i \mathbb{1}\{N^1_{it''} = n\} = \sum_i \mathbb{1}\{N^0_{it''} = n\} \text{ for } T_n = 0. \tag{34}$$

Third, individuals who would have moved into targeted sites absent treatment only displace the people they replace and do not lead to a broader change in neighborhood composition. Specifically, the set of individuals who live in a non-targeted neighborhood under treatment can be partitioned into those who would also live there absent treatment and those who would have moved into targeted sites absent treatment:

$$\{N^1_{it''} = n\} = \{N^1_{it''} = n, T_{N^0_{it''}} = 1\} \cup \{N^1_{it''} = n, N^0_{it''} = n\} \text{ for } T_n = 0. \tag{35}$$

Fourth, HOPE VI has no effect on the income of individuals who live in the same non-targeted neighborhood under both treatment and control:

$$P^0_{it''} = P^1_{it''} \text{ for } \{N^1_{it''} = n, N^0_{it''} = n, T_n = 0\}. \tag{36}$$

Assumption 5 implies that the spillover effect can be bounded as follows:

$$\begin{aligned} \bar{P}^1_{t''}(n) - \bar{P}^0_{t''}(n) \leq & \frac{\sum_i \mathbb{1}\{T_{N_{it'}} = 1, N^1_{it''} = n\}}{\sum_i \mathbb{1}\{N^1_{it''} = n\}} \\ & + \frac{\sum_i \mathbb{1}\{N_{it'} = n, T_{N^0_{it''}} = 1\}}{\sum_i \mathbb{1}\{N^0_{it''} = n\}}, \end{aligned} \tag{37}$$

for all n such that $T_n = 0$. See Appendix B for details.

Assumption 5 implies that HOPE VI affects poverty rates in non-targeted neighborhoods only because residents displaced from HOPE VI sites move to those neighborhoods. While we cannot identify the specific individuals who were displaced, the number of displaced households must be no greater than the total number of individuals who move between targeted and non-targeted neighborhoods. Further assuming that displaced households replace non-poor incumbents allows us to construct an upper bound for the spillover effects. The two terms on the right-hand side of Equation 37 represent spillover effects from two groups of displaced households: i) original residents who would have remained in HOPE VI neighborhoods absent treatment, and ii) individuals who lived in a non-targeted neighborhood before the award and would have moved into a HOPE VI neighborhood absent treatment. The first term is observed in the HOPE VI sample, and the second term is observed in the failed-applicant sample.

We estimate the two components in Equation 37 separately. For each component and target neighborhood, we implement the following procedure:

1. Estimate the size of the displaced population.
2. Estimate the maximum number of poor individuals displaced into each neighborhood.
3. Rank destination neighborhoods by the number of subsidized renters.

4. Calculate a counterfactual poverty rate in the destination neighborhood with the most subsidized renters by removing displaced poor individuals from this neighborhood and assuming they are replaced by non-poor individuals. If the displaced population exceeds the number of people who moved to this neighborhood, remove the remaining displaced households from the second-ranked neighborhood, and so on.

5. Summarize spillover effects from displaced original residents by taking the weighted average of the increase in neighborhood poverty rates among destination neighborhoods with non-zero changes in poverty, weighting by the size of the subsidized population in each destination neighborhood.

This methodology yields an estimate of migration spillover effects for each target neighborhood. Table A.8 provides a formal description of each step. We summarize these effects by taking a weighted average across projects, using baseline populations in targeted neighborhoods as weights.

We start with the results for the displaced original residents: $\frac{\sum_i \mathbb{1}\{N^1_{it''}=n, D(N_{it'})=1\}}{\sum_i \mathbb{1}\{N_{it''}=n\}}$. For each HOPE VI neighborhood, we implement Steps 1–5 above. In Step 1, we estimate the displaced population as the number of poor original residents who remain in the neighborhood 10 years after the award multiplied by 0.35, based on Column 3 of Table A.4.[34] In Step 2, we identify all poor original HOPE VI residents who lived in a different neighborhood 10 years after the award and count the number who moved to each destination neighborhood. Migration patterns were dispersed: 97% of flows between HOPE VI and destination neighborhoods consisted of fewer than 10 people. In Step 3, we find that original residents tended to move to neighborhoods with fewer subsidized renters: the average HOPE VI neighborhood contains 5.6 times more subsidized renters than the average destination neighborhood. Step 4 estimates how much lower poverty rates would be in destination neighborhoods absent the displaced households, and Step 5 averages across neighborhoods potentially affected by each HOPE VI. Taking the weighted average across all HOPE VI projects implies that migration spillovers from displaced original residents could have increased poverty rates by at most 0.30 pp for subsidized renters.

We follow an analogous approach to estimate spillover effects attributable to households who would have moved in after an award: $\frac{\sum_i \mathbb{1}\{N_{it'}=n, D(N^0_{it''})=1\}}{\sum_i \mathbb{1}\{N_{it''}=n\}}$. Because we cannot identify who

[34]Column 3 of Table A.4 shows that Revitalization reduced the number of poor original residents who remained in Revitalization neighborhoods by 26% (0.26 = 74/285). Thus, this population would have been 1.35 times larger absent the program ($1.35 = 1/(1 - 0.26)$), and the displaced population is $0.35 \times p = 1.35 \times p - p$, where p is the number of remaining poor original residents.

would have moved into HOPE VI neighborhoods absent treatment, we study migration flows into failed-applicant neighborhoods. For each failed-applicant neighborhood, we proceed as follows. In Step 1, we estimate the displaced population as the number of new poor residents who lived in the neighborhood 10 years after the award multiplied by 0.15, based on Column 4 of Table A.4 ($0.15 = 195/1,290$). In Step 2, we identify all new, poor residents and count the number who moved from each origin neighborhood to the failed-applicant neighborhood. Again, migration patterns are dispersed: 95% of flows between origin and failed-applicant neighborhoods consisted of fewer than 10 people. In Step 3, we show that new residents came from neighborhoods with far fewer subsidized renters: the average failed-applicant neighborhood contains 31 times more subsidized renters than the average origin neighborhood. After the weighted aggregation in Steps 4 and 5, we find that migration spillovers from the displacement of households who would have moved in absent an award could have increased poverty rates by at most 0.19 pp for subsidized renters.

Column 6 of Table A.4 shows that the program also increased the number of non-poor households who moved in after the award by 23% ($0.23 = 199/879$). Poverty rates could have risen in the neighborhoods from which these households moved if they were replaced by poor households. Following an analogous methodology, we find that displacement of non-poor households who would have moved in absent the award could have increased poverty rates by 0.27 pp in the neighborhoods where these households would have lived absent the award. Column 5 of Table A.4 shows that the program reduced the number of non-poor original residents, suggesting that spillovers from this group would reduce, rather than increase, exposure to neighborhood poverty.

Taken together, the results suggest that households displaced by HOPE VI were dispersed across many neighborhoods with fewer subsidized households than HOPE VI neighborhoods. As a result, spillover-induced increases in poverty rates elsewhere in the city were small and had little effect on subsidized renters' exposure to neighborhood poverty, especially relative to the reduction in poverty rates in HOPE VI neighborhoods. Summing the displacement effects discussed above yields $\hat{\hat{\Delta}}_4 \leq 0.0076 = 0.0030 + 0.0019 + 0.0027$.

5.4 A Comparison of Aggregate Effects

Equation 23 writes the average causal effect of HOPE VI among individuals who receive subsidized housing after the intervention, τ^{caus}, as the weighted average of the group-specific effects presented in Sections 5.3.1–5.3.4. Following the PHA-level analysis in Equation 18, we express this aggregate effect as a function of H_c, the share of a city's subsidized units targeted by HOPE VI, using weights $\omega_g(H_c)$.

We construct the group-specific weights as follows. The first group, $\mathcal{G}_1$, consists of all original residents of the HOPE VI neighborhoods (regardless of baseline subsidy type) who have subsidized housing in the post-period. Because 45% of original residents of the targeted neighborhoods live in the HOPE VI public housing projects, the size of the full neighborhood population is $H_c/0.45$. Furthermore, we find that 27.1% of these individuals resided in subsidized housing 15 years after the award, implying that $\omega_1(H_c) = 0.271 \times H_c/0.45$. For $\mathcal{G}_2$, Table A.4 shows that HOPE VI reduced the number of subsidized units in revitalized neighborhoods by 20%, and that 90% of residents of HOPE VI neighborhoods after the intervention were new residents rather than original residents, implying that $\omega_2(H_c) = (1 - 0.2) \times 0.9 \times H_c$.[35] For $\mathcal{G}_3$, the 20% reduction in subsidized units in targeted neighborhoods combined with the null effect on the total number of subsidized units in the city (see Table 1), implies that $\omega_3(H_c) = 0.2 \times H_c$. Finally, we assume that $\omega_4(H_c) = 1 - \omega_1(H_c) - \omega_2(H_c) - \omega_3(H_c)$, which provides an upper bound on spillover effects because some subsidized renters may not have been affected by migration spillovers.

Figure 7(A) presents the decomposition in Equation 23 for a PHA that revitalizes half of its subsidized housing units, approximately the maximum share.[36] The height of the first four bars represents the group-specific estimate, $\omega_g(0.5) \times \hat{\hat{\Delta}}_g$, and the bar on the right presents the sum of these effects. Our estimates imply that PHAs that revitalized half of their subsidized units reduced

[35]We arrive at the latter number using the estimates in Columns 7, 9, 11, and 13 of Table A.4. We estimate the size of the population in HOPE VI neighborhoods as the failed-applicant average plus the coefficient estimate: $0.9 = (696 - 168 + 335 - 2)/(108 - 43 + 696 - 168 + 52 - 20 + 335 - 2)$.

[36]Among PHAs that received HOPE VI funding, the average share of units affected by HOPE VI was 17%, and the 5th and 95th percentiles were 4% and 44%, respectively. Thus, some PHAs revitalized close to half of their subsidized housing units.

the average neighborhood poverty rate for all subsidized renters by 4.6 pp.[37] Reductions in exposure to neighborhood poverty for new subsidized residents of HOPE VI neighborhoods explain 59% of the aggregate effect (0.59=0.027/0.046).

The solid black line in Figure 7(B) displays the estimate of τ^{caus}, constructed by aggregating the group-specific estimates, as a function of the share of units targeted by HOPE VI. The dashed line presents estimates of τ^{feas} from the PHA-level analysis in Equation 18, i.e., a visualization of Column 1 of Table 1.[38] The two empirical approaches yield similar estimates, consistent with $\tau^{\text{feas}} \approx \tau^{\text{caus}}$ and $\tau^{\text{comp}} \approx 0$.

We conclude that HOPE VI reduced exposure to neighborhood poverty among subsidized renters primarily by causing new residents of revitalized HOPE VI neighborhoods to live in lower-poverty areas, not by displacing disadvantaged households out of subsidized housing.

6 Conclusion

Addressing the harms of residential segregation is a central goal of housing policy. Yet place-based efforts to improve disadvantaged neighborhoods often raise concerns about displacing the populations they are intended to help. This paper uses individual-level administrative data to study how neighborhood transformation affects individual residential outcomes in the context of a large place-based intervention: the HOPE VI Revitalization program. We find that HOPE VI achieved its intended neighborhood-level goal and led to a large, persistent reduction in poverty rates in targeted neighborhoods. These changes led subsidized renters to live in lower-poverty neighborhoods, on average. However, the primary beneficiaries were not original residents, most of whom moved away, but subsidized renters who moved into the revitalized neighborhoods after the intervention.

The program required a large upfront investment, but produced durable changes in neighborhood conditions. Combining our estimates with estimates of households' willingness to pay for neighbor-

[37]These estimates do not include the spatial spillovers described in Figure A.17. If subsidized households were more likely to live near HOPE VI neighborhoods than elsewhere in the city, these spatial spillovers would reduce exposure to neighborhood poverty. In that case, our estimates understate the program's effect.

[38]We estimate a more flexible specification by replacing the linear terms for S_h and H_c with indicators for the deciles of these two variables. The diamond markers in Figure 7(B) depict the estimates from this more flexible specification.

hood amenities from Galiani et al. (2015), suggests that the present discounted value of cumulative benefits to subsidized renters from reduced exposure to neighborhood poverty exceeds program costs over horizons of at least 16 years. Appendix E provides details. This calculation considers only the direct financial costs and benefits from reduced exposure to neighborhood poverty for subsidized renters. While these are key inputs to any cost-benefit analysis, the program could have imposed other costs—such as displacement costs due to rising housing prices studied by Almagro et al. (2023)—and produced other benefits—such as the long-run gains for children documented by Chetty et al. (2026)—that our analysis does not capture. Nonetheless, the estimates suggest that the program generated gains for subsidized renters that were substantial even relative to its large upfront costs.

Our results show how place-based investments in low-income neighborhoods can improve neighborhood conditions for low-income households.[39] However, high rates of residential mobility imply that place-based policies may be ineffective at targeting specific individuals, such as the original neighborhood residents, while still effectively targeting broader populations of interest, such as subsidized renters. Providing housing vouchers to original residents offers one way to attenuate price-driven displacement of low-income residents after place-based investments that meaningfully change neighborhoods.

[39] HOPE VI projects were more disadvantaged than public housing projects on average, so our results may not generalize to all public housing projects.

References

Aliprantis, Dionissi and Daniel Hartley, “Blowing it up and knocking it down: The local and city-wide effects of demolishing high concentration public housing on crime,” *Journal of Urban Economics*, 2015, *88*, 67–81.

Almagro, Milena, Eric Chyn, and Bryan A Stuart, “Urban Renewal and Inequality: Evidence from Chicago’s Public Housing Demolitions,” Technical Report, National Bureau of Economic Research 2023.

Ampuero, Fernanda Catalina Rojas, “Sent away: The long-term effects of slum clearance on children and families.” PhD dissertation, UCLA 2022.

Asquith, Brian J, Evan Mast, and Davin Reed, “Local effects of large new apartment buildings in low-income areas,” *The Review of Economics and Statistics*, 2023, *105* (2), 359–375.

Baum-Snow, Nathaniel and Justin Marion, “The effects of low income housing tax credit developments on neighborhoods,” *Journal of Public Economics*, 2009, *93* (5-6), 654–666.

Blanco, Hector, “Pecuniary effects of public housing demolitions: Evidence from Chicago,” *Regional Science and Urban Economics*, 2023, *98*, 103847.

_ **and Lorenzo Neri**, “Knocking it down and mixing it up: The impact of public housing regenerations,” *IZA Discussion Paper*, 2023.

Boustan, Leah Platt, Robert A Margo, Matthew M Miller, James M Reeves, and Justin P Steil, “Does Condominium Development Lead to Gentrification?,” *National Bureau of Economic Research*, 2019.

Bruhn, Jesse, “Crime and public housing: A general equilibrium analysis,” *SSRN 3064909*, 2018.

Brummet, Quentin and Davin Reed, “The effects of gentrification on incumbent residents,” *ERN: Urban Economics & Public Policy (Topic)*, 2021.

Buron, Larry, Susan J. Popkin, Diane K. Levy, and David Price, “The HOPE VI Resident Tracking Study: A Snapshot of the Current Living Situation of Original Residents from Eight Sites,” Technical Report, Abt Associates Inc., Cambridge, MA 2002.

Busso, Matias, Jesse Gregory, and Patrick Kline, “Assessing the incidence and efficiency of a prominent place based policy,” *American Economic Review*, 2013, *103* (2), 897–947.

Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer, “The effect of minimum wages on low-wage jobs,” *The Quarterly Journal of Economics*, 2019, *134* (3), 1405–1454.

Chetty, Raj, Rebecca Diamond, Thomas B Foster, Lawrence F Katz, Sonya Porter, Matthew Staiger, and Laura Tach, “Creating High-Opportunity Neighborhoods: Evidence from the HOPE VI Program,” Technical Report, National Bureau of Economic Research 2026.

Chyn, Eric, “Moved to opportunity: The long-run effects of public housing demolition on children,” *American Economic Review*, 2018, *108* (10), 3028–56.

_ **and Lawrence F Katz**, “Neighborhoods matter: Assessing the evidence for place effects,” *Journal of Economic Perspectives*, 2021, *35* (4), 197–222.

Collins, William J and Katharine L Shester, “Slum clearance and urban renewal in the United States,” *American Economic Journal: Applied Economics*, 2013, *5* (1), 239–273.

Coulton, Claudia, Brett Theodos, and Margery A Turner, “Residential mobility and neighborhood change: Real neighborhoods under the microscope,” *Cityscape*, 2012, pp. 55–89.

Couture, Victor and Jessie Handbury, “Urban revival in America,” *Journal of Urban Economics*, 2020, *119*, 103267.

— **, Cecile Gaubert, Jessie Handbury, and Erik Hurst**, "Income growth and the distributional effects of urban spatial sorting," Technical Report, National Bureau of Economic Research 2019.

Diamond, Rebecca and Tim McQuade, "Who wants affordable housing in their backyard? An equilibrium analysis of low-income property development," *Journal of Political Economy*, 2019, *127* (3), 1063–1117.

Dube, Arindrajit, Daniele Girardi, Oscar Jorda, and Alan M Taylor, "A Local Projections Approach to Difference-in-Differences," *Journal of Applied Econometrics*, 2025.

Ellen, Ingrid G., "What do we know about housing choice vouchers?," *Regional Science and Urban Economics*, 2020, *80.*

Feigenberg, Benjamin, Ben Ost, and Javaeria A Qureshi, "Omitted variable bias in interacted models: A cautionary tale," *Review of Economics and Statistics*, 2025, pp. 1–15.

Feins, Judith D and Rhiannon Patterson, "Geographic mobility in the housing choice voucher program: A study of families entering the program, 1995—2002," *Cityscape*, 2005, pp. 21–47.

Galiani, Sebastian, Alvin Murphy, and Juan Pantano, "Estimating Neighborhood Choice Models: Lessons from a Housing Assistance Experiment," *American Economic Review*, 2015, *105* (11), 3385–3415.

GAO, "Public Housing: HUD's Oversight of HOPE VI Sites Needs to Be More Consistent," Technical Report GAO-03-555, U.S. Government Accountability Office, Washington, D.C. May 2003.

Gechter, Michael and Nick Tsivanidis, "Spatial spillovers from high-rise developments: Evidence from the Mumbai Mills," *Unpublished manuscript*, 2023.

Glaeser, Edward L, *Cities, agglomeration, and spatial equilibrium*, OUP Oxford, 2008.

— **and Joshua D Gottlieb**, "The economics of place-making policies," Technical Report, National Bureau of Economic Research 2008.

Goetz, Edward G, "The audacity of HOPE VI: Discourse and the dismantling of public housing," *Cities*, 2013, *35*, 342–348.

Green, Bill and Vincent Lane, *The final report of the National Commission on Severely Distressed Public Housing*, A report to the Congress and the Secretary of Housing and Urban Development, 1992.

Gress, Taryn, Seungjong Cho, and Mark Joseph, "HOPE VI data compilation and analysis," *National Initiative on Mixed-Income Communities, Case Western Reserve University*, 2016.

Guerrieri, Veronica, Daniel Hartley, and Erik Hurst, "Endogenous gentrification and housing price dynamics," *Journal of Public Economics*, 2013, *100*, 45–60.

Haltiwanger, John C, Mark J Kutzbach, Giordano Palloni, Henry O Pollakowski, Matthew Staiger, and Daniel H Weinberg, "The children of HOPE VI demolitions: National evidence on labor market outcomes," *Journal of Public Economics*, 2024, *239*, 105188.

Harari, Mariaflavia, Maisy Wong et al., "Slum upgrading and long-run urban development: Evidence from Indonesia," in "2018 Meeting Papers. Society for Economic Dynamics" 2018.

Jacob, Brian A, "Public housing, housing vouchers, and student achievement: Evidence from public housing demolitions in Chicago," *American Economic Review*, 2004, *94* (1), 233–258.

Kennedy, Patrick and Harrison Wheeler, "High-End Housing and Gentrification: Evidence from a San Francisco Lottery," Technical Report 2023.

Kingsley, Thomas G, Jennifer Johnson, and Kathryn LS Pettit, "Patterns of Section 8 relocation in the HOPE VI program," *Journal of Urban Affairs*, 2003, *25* (4), 427–447.

Kline, Patrick and Enrico Moretti, "People, Places, and Public Policy: Some Simple Welfare Economics of Local Economic Development Programs," *The Annual Review of Economics*, 2014, *6*, 629–662.

Larrimore, Jeff, Jacob Mortenson, and David Splinter, "Presence and persistence of poverty in US tax data," Technical Report, National Bureau of Economic Research 2020.

Lee, Sun Kyoung, "When Cities Grow: Urban Planning and Segregation in the Prewar US," *University of Michigan, mimeo*, 2022.

Manson, Steven, Jonathan Schroeder, David Van Riper, Tracy Kugler, and Steven Ruggles, "IPUMS National Historical Geographic Information System: Version 16.0 [dataset]," *Minneapolis, MN: IPUMS. 2021. http://doi.org/10.18128/D050.V16.0*, 2021.

McKinnish, Terra, Randall Walsh, and T Kirk White, "Who gentrifies low-income neighborhoods?," *Journal of urban economics*, 2010, *67* (2), 180–193.

Pennington, Kate, "Does building new housing cause displacement?: the supply and demand effects of construction in San Francisco," *Working paper*, 2021.

Popkin, Susan J, Diane K Levy, and Larry Buron, "Has HOPE VI transformed residents' lives? New evidence from the HOPE VI panel study," *Housing Studies*, 2009, *24* (4), 477–502.

Qiang, Ashley J, Christopher Timmins, and Wen Wang, "Displacement and the consequences of gentrification," Technical Report, Working paper 2020.

Rossi-Hansberg, Esteban, Pierre-Daniel Sarte, and Raymond Owens III, "Housing externalities," *Journal of political Economy*, 2010, *118* (3), 485–535.

Sandler, Danielle H, "Externalities of public housing: The effect of public housing demolitions on local crime," *Regional Science and Urban Economics*, 2017, *62*, 24–35.

Schelling, Thomas C, "Dynamic models of segregation," *Journal of mathematical sociology*, 1971, *1* (2), 143–186.

_ , *Micromotives and macrobehavior*, WW Norton & Company, 2006.

Su, Yichen, "The rising value of time and the origin of urban gentrification," *American Economic Journal: Economic Policy*, 2022, *14* (1), 402–439.

Tach, Laura and Allison Dwyer Emory, "Public housing redevelopment, neighborhood change, and the restructuring of urban inequality," *American journal of sociology*, 2017, *123* (3), 686–739.

Zielenbach, Sean and Richard Voith, "HOPE VI and neighborhood economic development: The importance of local market dynamics," *Cityscape*, 2010, pp. 99–131.

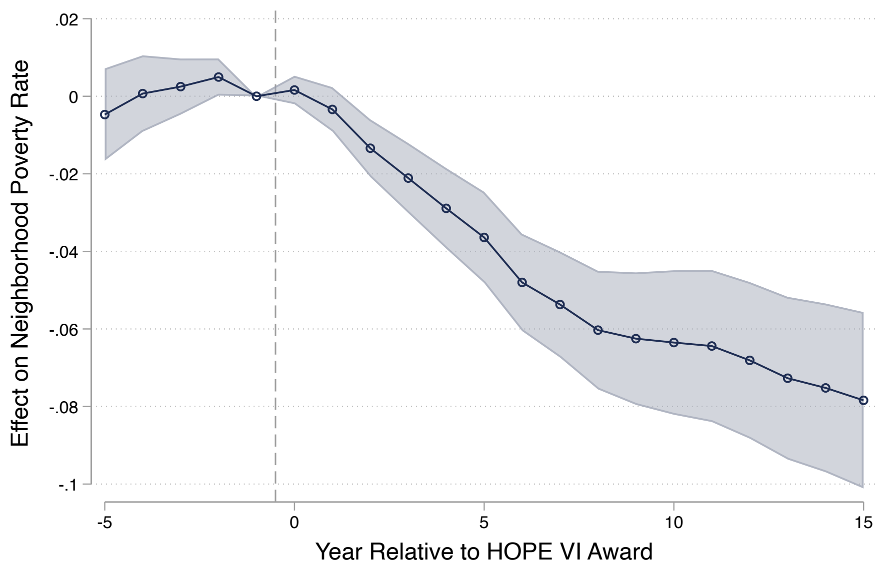
Table 1: Analysis of Public Housing Authorities, Changes Between 1997 and 2018

				Characteristics of Subsidized Renters						
	Subsidized Renters' Exposure to Neighbor-hood Poverty	Share of Subsidized Units Public Housing	Log No. of Voucher and Public Housing Units	Share with Income Below $10k	Share Black	Average Person Per Housing Unit	Share Single Parents	Share Disabled	County-Level Poverty Rate	Poverty Exposure from Targeted Neighbor-hoods
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Share Subsidized Units Targeted	-0.183	-0.512	0.345	0.124	0.167	0.151	0.181	-0.139	0.036	-0.138
by HOPE VI Award	(0.078)	(0.202)	(0.283)	(0.136)	(0.120)	(0.312)	(0.131)	(0.114)	(0.032)	(0.028)
	[0.019]	[0.012]	[0.224]	[0.363]	[0.166]	[0.629]	[0.169]	[0.226]	[0.256]	[0.000]
Received Any HOPE VI Funding	-0.029	-0.046	0.036	-0.030	-0.104	-0.008	-0.049	0.029	-0.007	0.002
	(0.015)	(0.040)	(0.056)	(0.027)	(0.023)	(0.060)	(0.026)	(0.022)	(0.006)	(0.006)
	[0.063]	[0.253]	[0.525]	[0.275]	[0.000]	[0.893]	[0.065]	[0.184]	[0.282]	[0.698]
Share Subsidized Units Targeted	0.245	-0.095	-0.155	0.133	-0.188	1.304	0.571	0.135	0.057	0.264
by HOPE VI Application	(0.141)	(0.369)	(0.517)	(0.260)	(0.219)	(0.554)	(0.251)	(0.203)	(0.058)	(0.051)
	[0.083]	[0.797]	[0.764]	[0.608]	[0.393]	[0.020]	[0.024]	[0.508]	[0.329]	[0.000]
Observations	193	194	194	169	173	178	162	177	194	193

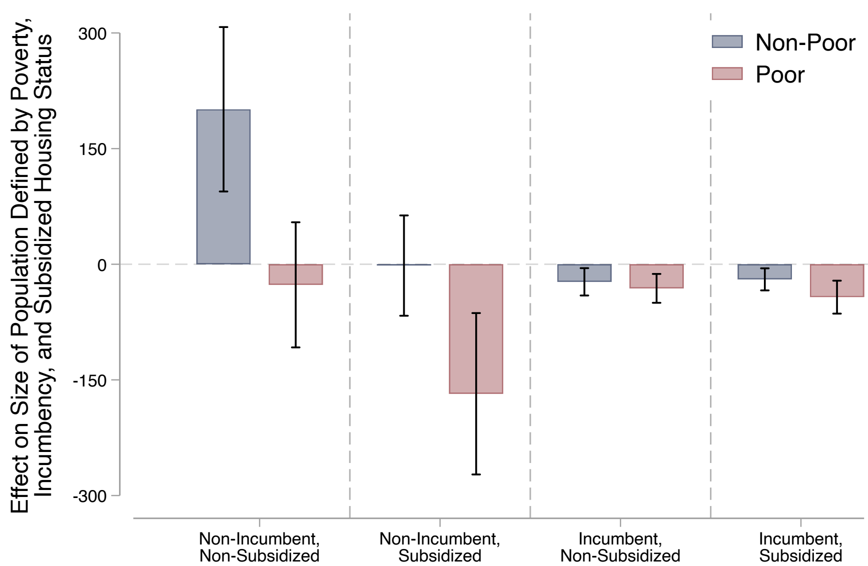
Notes: Each column presents estimates from a separate PHA-level regression. The dependent variables measure changes between 1997 and 2018. The regressions include the share of all subsidized units in the PHA that were targeted by a successful or unsuccessful HOPE VI Revitalization application; the share of all subsidized units in the PHA that were targeted by a successful HOPE VI Revitalization application; an indicator for whether the PHA received any HOPE VI Revitalization funding; the average 1990 poverty rate of the neighborhoods targeted by successful and unsuccessful HOPE VI applications; and the interaction between the share of subsidized units targeted by a successful or unsuccessful application and the average 1990 poverty rate of targeted neighborhoods. The outcome in Column 1 is the average poverty rate of the Census tracts in which subsidized renters reside. The outcome in Column 2 is the share of subsidized units that are public housing. The outcome in Column 3 is the log number of households in public or voucher housing. The outcomes in Columns 4 to 8 are characteristics of subsidized renters, including: having a household income below $10,000, being Black, number of people per housing unit, share of households with children with single parent, and share of households below age 62 in which either the household head or spouse (or co-head) has a disability. The outcome in Column 9 is the county-level poverty rate and the outcome in Column 10 is the component of column 1 attributable to changes in poverty rates in the targeted neighborhoods. Each observation represents a PHA. The sample includes PHAs that either received HOPE VI Revitalization funding or applied for HOPE VI Revitalization funding but received none. Each regression is estimated by weighted least squares, using the total number of subsidized units in 1997 as weights. Standard errors are presented in parentheses and p-values are presented in brackets.
Source: Authors' calculations based on the 1997 and 2018 vintages of HUD User's Picture of Subsidized Households.

Figure 1: Effect of HOPE VI Revitalization on Neighborhood Poverty

(A) Effect on Poverty Rates in Targeted Neighborhoods

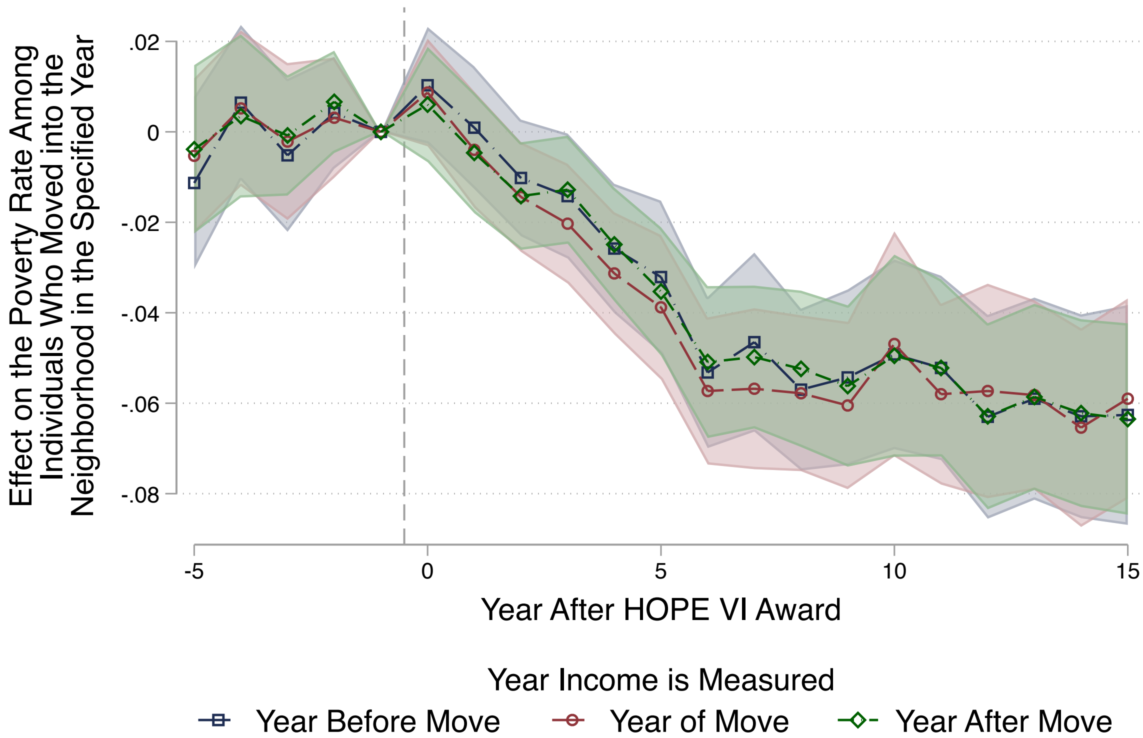


(B) Effect on Size of Subpopulations



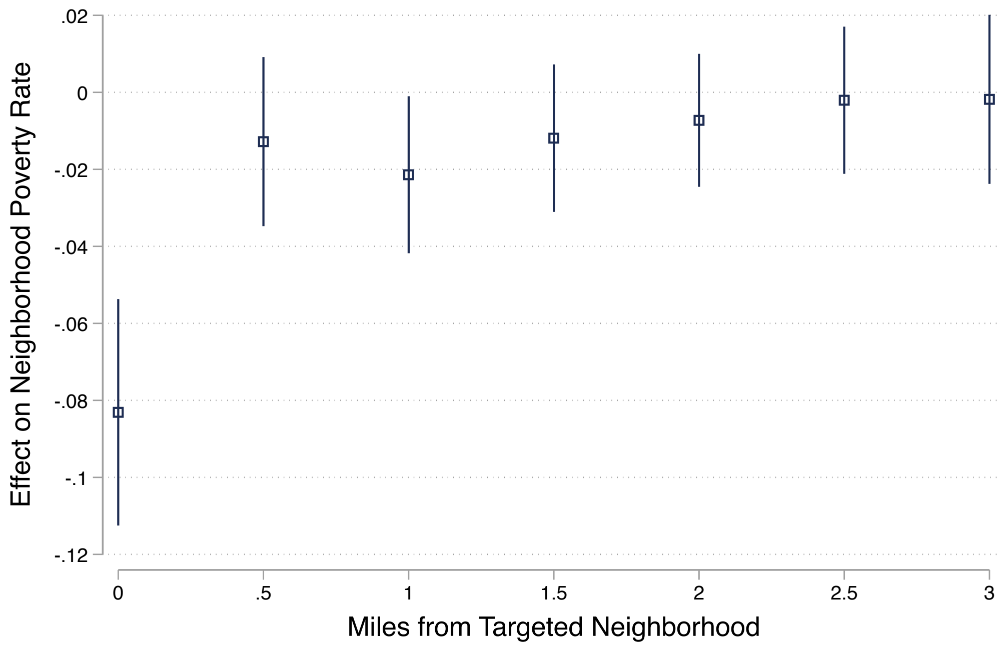
Notes: Panel A plots estimates from the stacked difference-in-differences specification described in Equation 11. The outcome is the share of current residents who are poor. The shaded regions denote the 95% confidence intervals. Panel B plots estimates from the cross-sectional estimator described in Equation 8. Each bar represents an estimate from a separate regression in which an outcome variable is regressed on the HOPE VI indicator, controlling for grant-year fixed effects as well as the number of poor and non-poor households who lived in the neighborhood in the year before the grant. The outcome variable is the size of the population defined by poverty, incumbency, and subsidized housing status. The vertical bars denote the 95% confidence intervals.

Figure 2: Effect of HOPE VI Revitalization on the Poverty Rate of New Residents



Notes: This figure plots estimates from the stacked difference-in-differences specification described in Equation 11. We limit the sample to the population of individuals who move in during the year specified by the horizontal axis. The three series represent estimates from three separate regressions in which we measure the poverty rate of this population using their income in the previous, current, or subsequent year. The shaded regions denote the 95% confidence intervals.

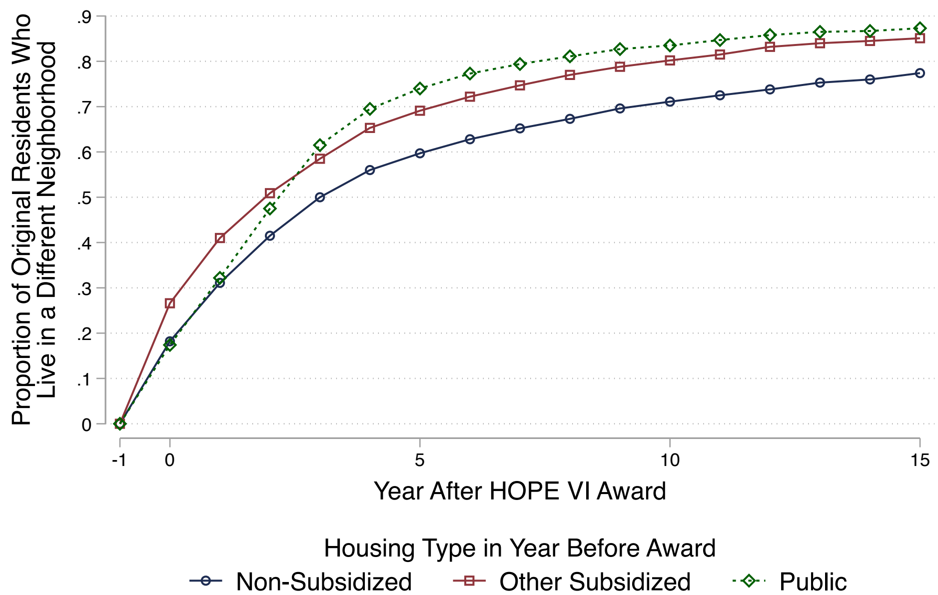
Figure 3: Effect of HOPE VI Revitalization on Poverty Rates in Nearby Neighborhoods



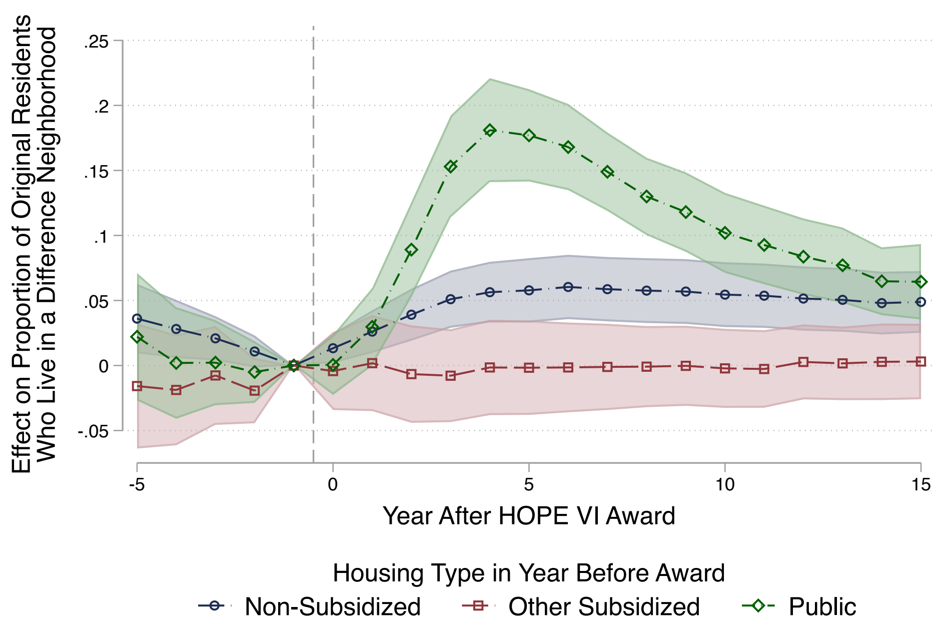
Notes: The sample includes HOPE VI and failed applicant neighborhoods as well as all block groups within three miles of these targeted neighborhoods. We estimate a regression where the outcome variable is the change in neighborhood poverty rate between one year before and fifteen years after the award. The independent variables in the specification include the interaction between the Revitalization indicator and the distance category (grouped into .5 mile increments) and the block group poverty rate in the year before the award. The regression is estimated via weighted least squares, weighting by the inverse propensity score weights. Standard errors are clustered at the level of the targeted neighborhood and the vertical spikes denote the 95% confidence intervals.

Figure 4: Residential Mobility of the Original Residents of Targeted Neighborhoods

(A) Rates of Residential Mobility in HOPE VI Revitalization Neighborhoods



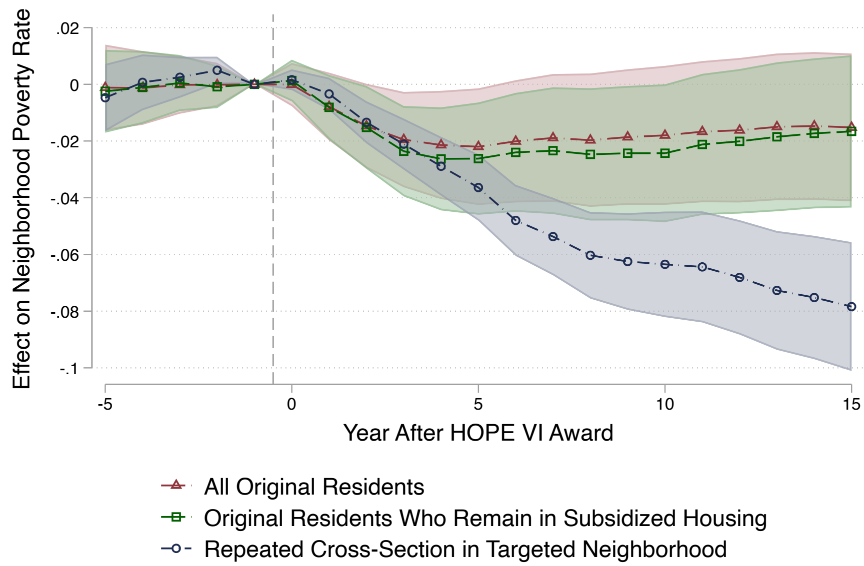
(B) Effect of the HOPE VI Revitalization Program on Residential Mobility



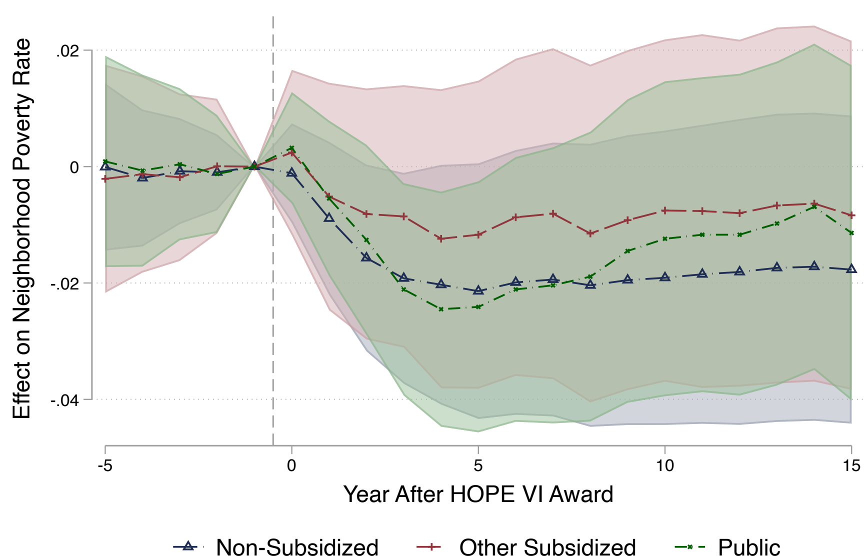
Notes: Panel A plots the share of original residents of HOPE VI Revitalization neighborhoods who live in a different census block group by each year after the award. Panel B plots estimates from the stacked difference-in-differences specification described in Equation 11. The outcome is the share of original residents who live in a different census block group. Shaded regions denote 95% confidence intervals. All results are presented separately by original housing type: non-subsidized, other subsidized, or public housing.

Figure 5: Effect on Exposure to Neighborhood Poverty Among Original Residents

(A) All Original Residents

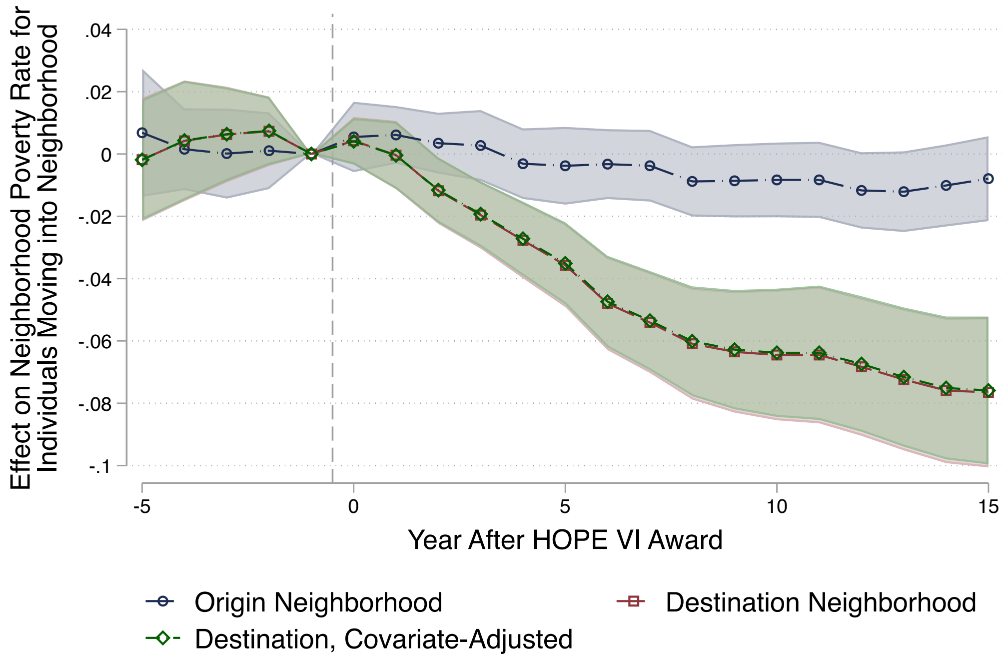


(B) By Housing Subsidy Status in the Year Before the HOPE VI Award



Notes: This figure plots estimates from the stacked difference-in-differences specification described in Equation 11. In Panel A, the red triangles present the estimated effect on exposure to neighborhood poverty among all original residents of targeted neighborhoods. The green squares repeat the analysis but restrict the sample to original residents who were in subsidized housing 10 years after the award. The blue circles reproduce the estimated effect on poverty rates in targeted neighborhoods from Figure 1(A). Panel B presents estimates analogous to the red triangles in Panel A, estimated separately by whether original residents lived in public housing, other subsidized housing, or non-subsidized housing in the year before the award. Shaded regions denote 95% confidence intervals.

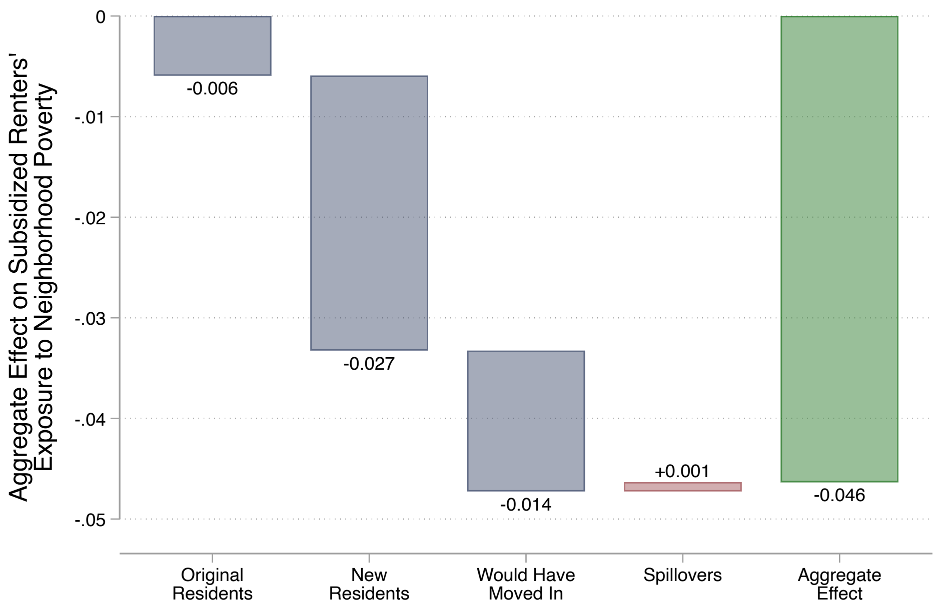
Figure 6: Effect on Exposure to Neighborhood Poverty Among New Subsidized Renters



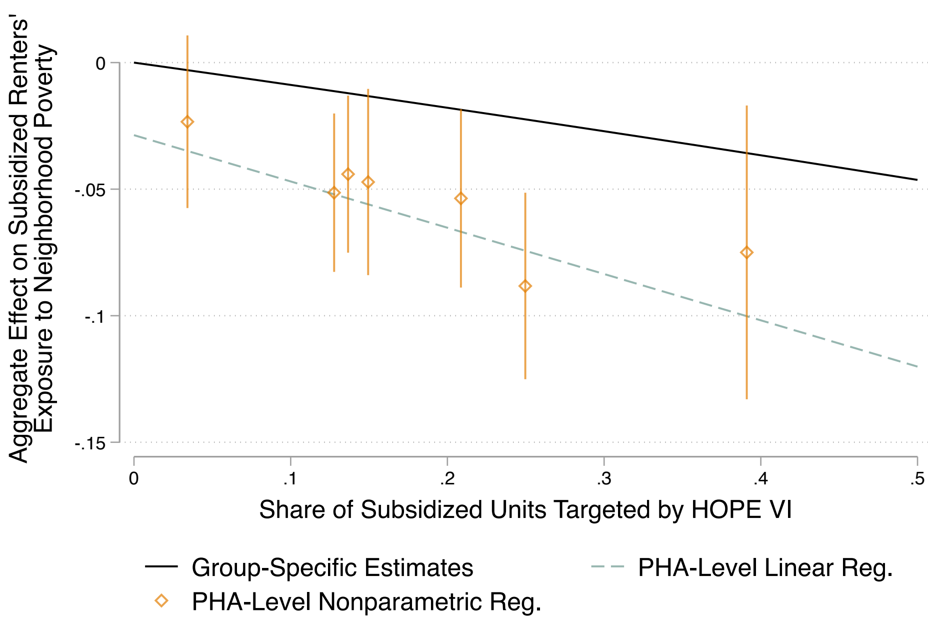
Notes: This figure plots estimates from the stacked difference-in-differences specification described in Equation 11, estimated on individual-level panel data. The sample includes subsidized renters who move into a HOPE VI Revitalization or failed-applicant neighborhood in the year specified by the horizontal axis. For the blue circles, the outcome is the neighborhood poverty rate in the origin neighborhood. For the red squares and green diamonds, the outcome is the poverty rate in the destination neighborhood. The specification underlying the green diamonds additionally controls for the poverty rate in the origin neighborhood, the individual's income before the move, and a quadratic in age. Shaded regions denote 95% confidence intervals.

Figure 7: Aggregate Effect on Subsidized Renters' Exposure to Neighborhood Poverty

(A) Aggregation of Group-Specific Estimates



(B) Comparison to PHA-Level Analysis



Notes: The height of the first four bars in Panel A represents estimates of $\omega_g(0.5)\mathbb{E}[\Delta_i \mid \mathcal{G}_g]$ from Equation 23 for a PHA that revitalized half of its subsidized housing units. The bar on the right presents the aggregate effect. Panel B compares aggregate effects based on the group-specific estimates to those obtained from the PHA-level regression described in Equation 18. The figure presents estimated effects as a function of the share of subsidized units targeted by HOPE VI.

Appendix A Additional Tables and Figures

Table A.1: Sample Restrictions

	Non-Applicant		Failed Applicant		HOPE VI	
	Count	Percent	Count	Percent	Count	Percent
Full sample	16,300	100%	244	100%	286	100%
Located in the 50 states or DC	15,932	98%	242	99%	286	100%
Not in indian housing	13,233	81%	242	99%	286	100%
In picture of subsidized housing	13,233	81%	233	95%	272	95%
Not scattered site or vacant	12,618	77%	220	90%	271	95%
At least 25 units in 1993	8,913	55%	210	86%	266	93%
Not senior housing	6,946	43%	204	84%	262	92%
Non-missing location	6,084	37%	198	81%	251	88%
At least one mile from Revitalization	5,783	35%	166	68%	251	88%

Notes: The column headers define the sample (the three samples are mutually exclusive). Each row presents the count of projects that remain after imposing the sample restriction as well as the percent of the full sample.
Source: Authors' calculations based on project-level summary files from HUD and the publicly available list of awardees of and applicants to HOPE VI Revitalization funding.

Table A.2: Baseline Characteristics

	Non-Applicant		Failed Applicant		HOPE VI	
	mean	s.d.	mean	s.d.	mean	s.d.
A. Project						
Occupied units	430	960	293	200	667	756
Percent minority	61.6	38.9	77.7	33.0	77.9	35.6
Percent majority income earned	21.1	15.2	19.9	11.2	14.1	10.5
Average household size	2.3	1.1	2.7	1.0	2.4	1.1
Percent 62 or older	26.4	21.4	15.0	11.8	13.1	11.7
Percent with disability	12.9	11.1	10.9	7.5	10.0	8.9
Percent single parent	35.5	25.0	47.4	23.4	40.3	27.2
Percent with female head	69.0	25.7	75.4	24.6	71.4	31.5
Percent crowded housing	6.8	8.5	7.4	6.8	8.0	7.1
Average rent	273	145	248	123	189	130
Average income (thousands)	13.1	6.9	11.6	5.3	8.9	5.9
Observations	5,783		166		251	
B. Census block group						
Percent Black	31.9	33.8	48.3	36.2	72.0	32.0
Median rent	606	260	501	207	451	222
Median home value (thousands)	119	103	111	113	109	107
Percent with public assistance	19.2	13.0	31.3	16.8	38.2	17.2
Percent unemployed	12.1	8.2	19.4	12.1	24.0	14.1
Percent below poverty	30.8	17.2	45.0	19.0	55.0	20.6
Number of people (thousands)	1.5	0.6	1.6	0.6	1.9	0.8
Observations	5,391		178		308	

Notes: The columns define one of three mutually exclusive samples including non-HOPE VI, failed applicant, and HOPE VI Revitalization neighborhoods. Each row presents statistics for a different variable. In panel A the variables are measured at the project level in 1993. In panel B the variables are measured at the block group level in 1990. In panels A and B we weight by the population of the project and block group, respectively.
Source: Authors' calculations based on project-level summary files from HUD and summary files from the 1990 Decennial Census.

Table A.3: Effect on Neighborhood Poverty Using Alternative Estimator, Data, and Comparison Groups

	Change in Block Group Poverty Rate Between 1990 and 2017				
	(1)	(2)	(3)	(4)	(5)
A. With Covariates					
HOPE VI Revitalization	-0.077	-0.118	-0.093	-0.101	-0.066
	(0.016)	(0.008)	(0.005)	(0.012)	(0.006)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Covariates	baseline	baseline	baseline	baseline+fe	baseline+fe
Comparison group	failed applicants	non-applicants	non-public housing	non-applicants near HOPE VI	non-public housing near HOPE VI
Observations	485	5,694	209,685	1,130	25,512
B. Without Covariates					
Revitalization	-0.132	-0.202	-0.222	-0.191	-0.228
	(0.020)	(0.008)	(0.005)	(0.012)	(0.006)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Covariates	none	none	none	fe	fe
Comparison group	failed applicants	non-applicants	non-public housing	non-applicants near HOPE VI	non-public housing near HOPE VI
Observations	485	5,694	209,722	1,130	25,522

Notes: Each column presents estimates from a separate regression in which the change in Census block group poverty rate between 1990 and 2017 is regressed on an indicator for receiving a HOPE VI Revitalization grant. The comparison groups in columns 1-5 are block groups that contain failed applicant projects, projects that did not apply for HOPE VI funding, no public housing projects, projects that did not apply for HOPE VI funding and are located within 1-5 miles of a HOPE VI site, and no public housing projects and are located within 1-5 miles of a HOPE VI site. The vector of baseline covariates include the following characteristics of the block group measured in 1990: poverty rate, share of residents that are White, share of residents that are Black, median rent, median home value, share of households receiving public assistance income, share of residents with a college degree, unemployment rate, share of residents that are new renters or owners, share of units that are owner-occupied, share of housing built before 1960, total population, total housing, and share of units vacant. Columns 4 and 5 include fixed effects (fe) for the closest HOPE VI site. Each regression is estimated via weighted least squares, using block group population in 1990 as the weights. Standard errors are presented in parentheses and p-values in brackets.

Source: Authors' calculations based on summary files of Decennial Census survey and the ACS.

Table A.4: Effect of HOPE VI Revitalization on Size of Sub-Populations

	By Income		By Income and Tenure				By Income, Tenure, and Subsidized Housing							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
HOPE VI	-269	156	-74	-195	-42	199	-43	-31	-168	-27	-20	-23	-2	201
Revitalization	(74)	(65)	(15)	(71)	(10)	(65)	(11)	(10)	(53)	(41)	(7)	(9)	(33)	(54)
	[0.000]	[0.017]	[0.000]	[0.006]	[0.000]	[0.002]	[0.000]	[0.001]	[0.002]	[0.520]	[0.008]	[0.012]	[0.958]	[0.000]
Mean	1580	1100	285	1290	224	879	108	177	696	595	52	172	335	544
Poor	yes	no	yes	yes	no	no	yes	yes	yes	yes	no	no	no	no
Original resident			yes	no	yes	no	yes	yes	no	no	yes	yes	no	no
Subsidized housing							yes	no	yes	no	yes	no	yes	no
Observations	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000

Notes: Each column presents estimates from a separate regression in which an outcome variable is regressed on an indicator for the HOPE VI Revitalization award. The outcome variable differs across columns and is the size of the population defined by poverty, tenure, and subsidized housing status. Regressions are estimated via weighted least squares using the inverse propensity score. All regressions control for the year of first award as well as the number of poor and non-poor households who lived in the neighborhood in the year before the award. Standard errors are clustered at the neighborhood and are presented in parentheses and p-values are presented in brackets.

Table A.5: Effect of HOPE VI Revitalization on Housing Prices and Home Ownership

	Log Rent			Log Home Value			Log Monthly Mortgage			Owner	Renter
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
HOPE VI Revitalization	0.249	0.192	0.136	0.246	0.136	0.161	0.125	0.044	0.101	-0.014	0.014
	(0.062)	(0.061)	(0.044)	(0.074)	(0.067)	(0.069)	(0.052)	(0.048)	(0.044)	(0.023)	(0.023)
	[0.000]	[0.002]	[0.002]	[0.001]	[0.042]	[0.021]	[0.018]	[0.357]	[0.021]	[0.537]	[0.537]
Housing Characteristics		X	X		X	X		X	X		
Exclude New Housing			X			X			X		
Mean	6.560	6.560	6.730	11.700	11.700	11.700	6.760	6.760	6.690	0.348	0.652
Observations	30,000	31,000	12,000	33,000	32,000	23,000	18,000	18,000	12,000	2,000	2,000

Notes: Each column presents estimates from a separate regression in which an outcome variable is regressed on an indicator for the HOPE VI Revitalization award. Regressions are estimated via weighted least squares using the inverse propensity score. All regressions control for the year of first award and the calendar year as well as the vacancy rate, median home value, neighborhood poverty rate, and share of population that is Black in the year before the award. When indicated some regressions further control for housing characteristics (building type, year built, persons per room) and limit to housing units built before the award. In columns 1-9 the unit of observation is the housing unit. In columns 10 and 11 the data are collapsed to the project level. Standard errors are clustered at the neighborhood and are presented in parentheses and p-values are presented in brackets. We also report the mean for the failed applicant neighborhoods.
Source: Authors' calculations based on linked survey and administrative data.

Table A.6: Analysis of Public Housing Authorities, Changes Between 1997 and 2018 for all Households

	Change in Average Census Tract Poverty Rate Between 1990 and 2017, for Households in Specified Income Category							
	0-10k	10k-20k	20k-30k	30k-40k	40k-50k	50k-75k	75k-100k	100k+
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Share Subsidized Units Targeted	0.004	0.017	0.042	0.026	0.028	0.025	0.017	0.025
by HOPE VI Award	(0.043)	(0.037)	(0.032)	(0.029)	(0.028)	(0.025)	(0.022)	(0.020)
	[0.934]	[0.648]	[0.193]	[0.373]	[0.319]	[0.323]	[0.430]	[0.228]
Received Any HOPE VI Funding	-0.011	-0.004	-0.005	-0.006	-0.004	-0.001	0.000	0.000
	(0.009)	(0.007)	(0.006)	(0.006)	(0.005)	(0.005)	(0.004)	(0.004)
	[0.207]	[0.580]	[0.440]	[0.320]	[0.477]	[0.788]	[0.961]	[0.933]
Share Subsidized Units Targeted	0.118	0.072	0.026	0.044	0.024	0.018	-0.013	-0.048
by HOPE VI Application	(0.079)	(0.067)	(0.059)	(0.054)	(0.050)	(0.046)	(0.040)	(0.037)
	[0.134]	[0.280]	[0.657]	[0.411]	[0.639]	[0.698]	[0.749]	[0.198]
Observations	194	194	194	194	194	194	194	194

Notes: Each column presents estimates from a separate PHA-level regression. The outcome variable is the change in the exposure to census tract poverty between 1990 and 2017 for households who live in the county and have a household income in the range defined by the column header. The regressions include the share of all subsidized units in the PHA that were targeted by a successful or unsuccessful HOPE VI Revitalization application; the share of all subsidized units in the PHA that were targeted by a successful HOPE VI Revitalization application; an indicator for whether the PHA received any HOPE VI Revitalization funding; the average 1990 poverty rate of the neighborhoods targeted by successful and unsuccessful HOPE VI applications; and the interaction between the share of subsidized units targeted by a successful or unsuccessful application and the average 1990 poverty rate of targeted neighborhoods. Each observation represents a PHA. The sample includes PHAs that either received HOPE VI Revitalization funding or applied for HOPE VI Revitalization funding but received none. Each regression is estimated by weighted least squares, using the total number of subsidized units in 1997 as weights. Standard errors are presented in parentheses and p-values are presented in brackets.
Source: Authors' calculations based on the 1997 and 2018 vintages of HUD User's Picture of Subsidized Households.

Table A.7: Effect on Exposure to Poverty for New Subsidized Renters, Additional Covariates

	Poverty Rate In Prior Neighborhood	Poverty Rate In Current Neighborhood			
	(1)	(2)	(3)	(4)	(5)
A. Moved In After Award					
HOPE VI Revitalization	-0.009	-0.072	-0.069	-0.068	-0.068
	(0.007)	(0.015)	(0.014)	(0.014)	(0.014)
	[0.203]	[0.000]	[0.000]	[0.000]	[0.000]
R-squared	0.016	0.089	0.154	0.159	0.181
Observations (Thousands)	1,082	1,082	1,082	1,082	1,082
B. Moved In Before Award					
HOPE VI Revitalization	-0.002	-0.003	-0.001	-0.002	-0.001
	(0.010)	(0.019)	(0.015)	(0.016)	(0.015)
	[0.874]	[0.888]	[0.935]	[0.894]	[0.931]
R-squared	0.102	0.187	0.364	0.347	0.377
Observations (Thousands)	1,362	1,362	1,256	1,075	1,256

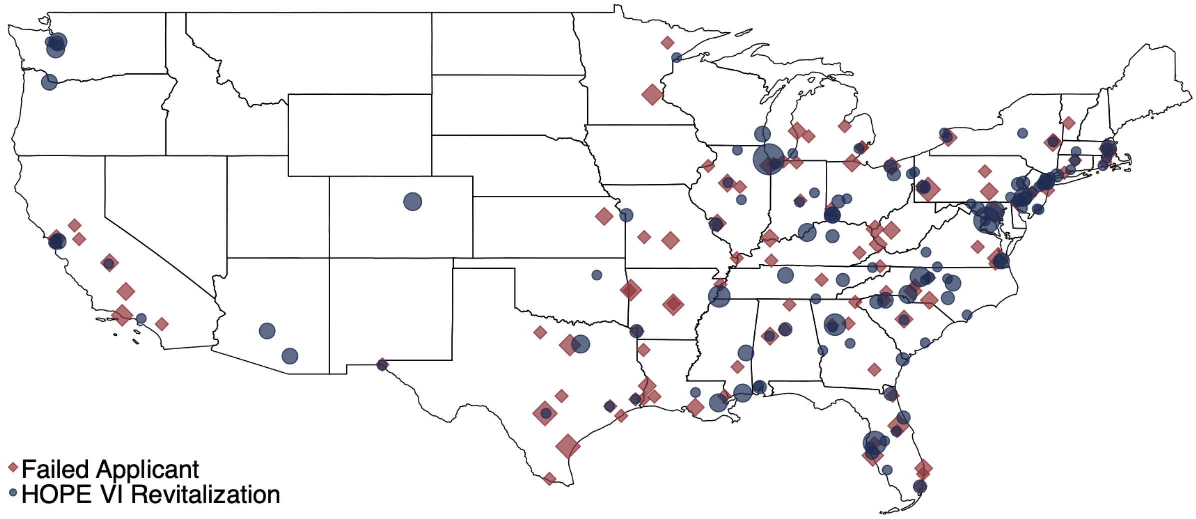
Notes: The sample in panels A and B includes the set of individuals who move into a HOPE VI Revitalization or failed applicant neighborhood after or before the award, respectively. Each column presents an estimate from a separate regression of an outcome variable on the HOPE VI Revitalization indicator, controlling for the interaction between the award year and years since the award. The outcome variable in Column 1 is the poverty rate of the neighborhood that the individual lived in prior to moving into the target neighborhood. The outcome variable in Columns 2-5 is the poverty rate of the target neighborhood. Columns 3-5 add additional covariates. Column 3 adds controls for the poverty rate in the prior neighborhood, own income in the year before moving into the target neighborhood, and a quadratic in age. Column 4 adds a control for income in the current year. Column 5 adds controls for a vector of characteristics from the HUD administrative data, including ethnicity, race, sex, disability status, household head disability status, household size, type of subsidized housing, a flag for hardship, hard to house, separate measures for the amount of income from various sources (welfare, wages, pension, other, and unknown).

Table A.8: Algorithm for Constructing the Spillovers Estimator

Step	**Description**	**Formal Notation**
1	Estimate the size of the displaced population.	Let Γ_0 denote the size of the displaced population.
2	Estimate the maximum size of the population of poor individuals displaced into each neighborhood.	Let $\Gamma(n)$ denote the maximum size of the population of poor individuals displaced into neighborhood n.
3	Rank destination neighborhoods by the number of subsidized renters.	Let $s(n) = \sum_i \mathbb{1}\{N_{it''} = n, S_{it''} = 1\}$ be the number of subsidized renters in n and let $r(\cdot)$ be a ranking function that ranks neighborhoods by $s(n)$ in descending order such that $r(j) < r(k)$ if and only if $s(j) > s(k)$ for all neighborhoods j and k.
4	Calculate a counterfactual poverty rate in the destination neighborhood with the most subsidized renters by removing the displaced poor individuals from this neighborhood and assuming that they are replaced by non-poor individuals. If the size of the displaced population exceeds the number of people who moved to this neighborhood, then remove the remaining displaced households from the second-ranked neighborhood, and so on.	Iterate sequentially for $k = \{1, 2, 3, ...\}$. Select neighborhood j such that $r(j) = k$ and define $d_j = min\{\Gamma(j), \tilde{\Gamma}_k\}$, where $\tilde{\Gamma}_k = \Gamma_{k-1}$. Define $\delta_j = (\sum_i \mathbb{1}\{N_{it''} = j, P_{it''} = 1\} - d_j)/\sum_i \mathbb{1}\{N_{it''} = j\}$. Stop iterating when $d_j = 0$.
5	Summarize the spillover effects from the displaced original residents by taking the weighted average of the increase in neighborhood poverty rates among the set of destination neighborhoods that had a non-zero change in poverty, weighting by the size of the subsidized population in the destination neighborhood.	Let $\mathcal{J} = \{j : d_j > 0\}$ denote the set of neighborhoods that had a non-zero change in poverty rate and let $\pi(j) = \sum_i \mathbb{1}\{N_{it''} = j, S_{it''}\}/\sum_i \mathbb{1}\{N_{it''} \in \mathcal{J}, S_{it''}\}$. Then we summarize the spillover effects as $\sum_{j\in\mathcal{J}} \pi(j)\delta_j$.

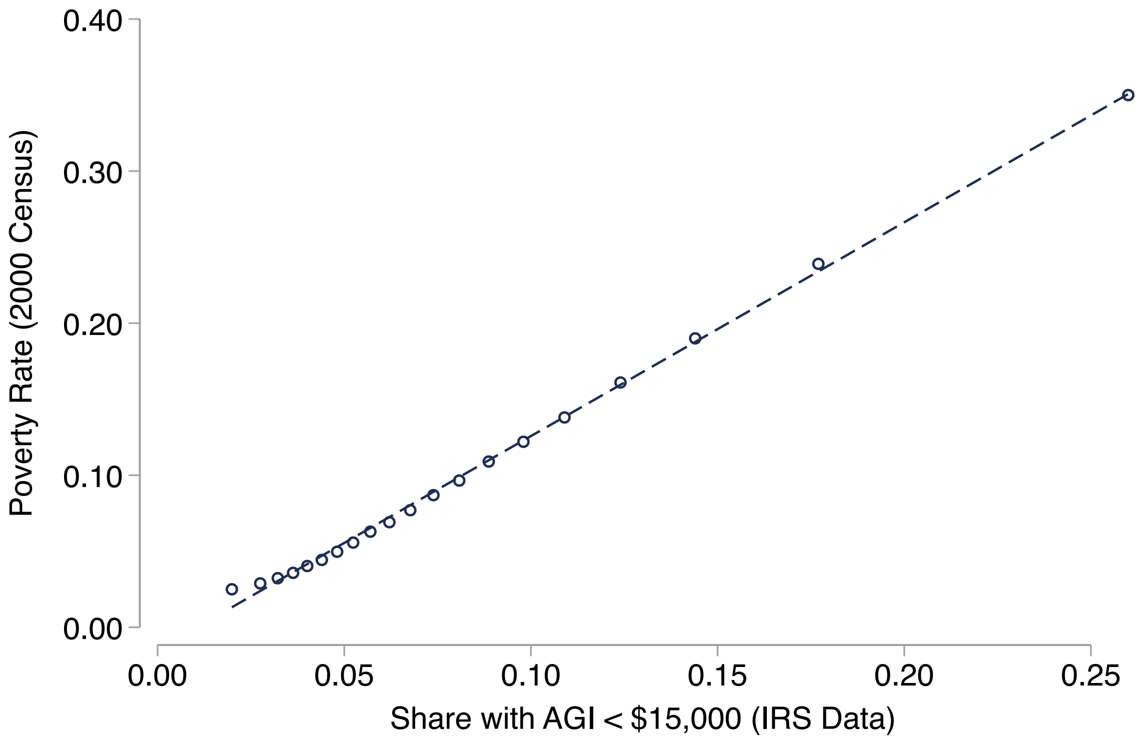
Notes: This table provides a step-by-step guide to the construction of the spillovers estimator, corresponding to the discussion in 5.3.4. We implement this procedure separately for each target neighborhood but do not index by target neighborhood to simplify exposition.

Figure A.1: Location of HOPE VI Revitalization Applicants and Awardees



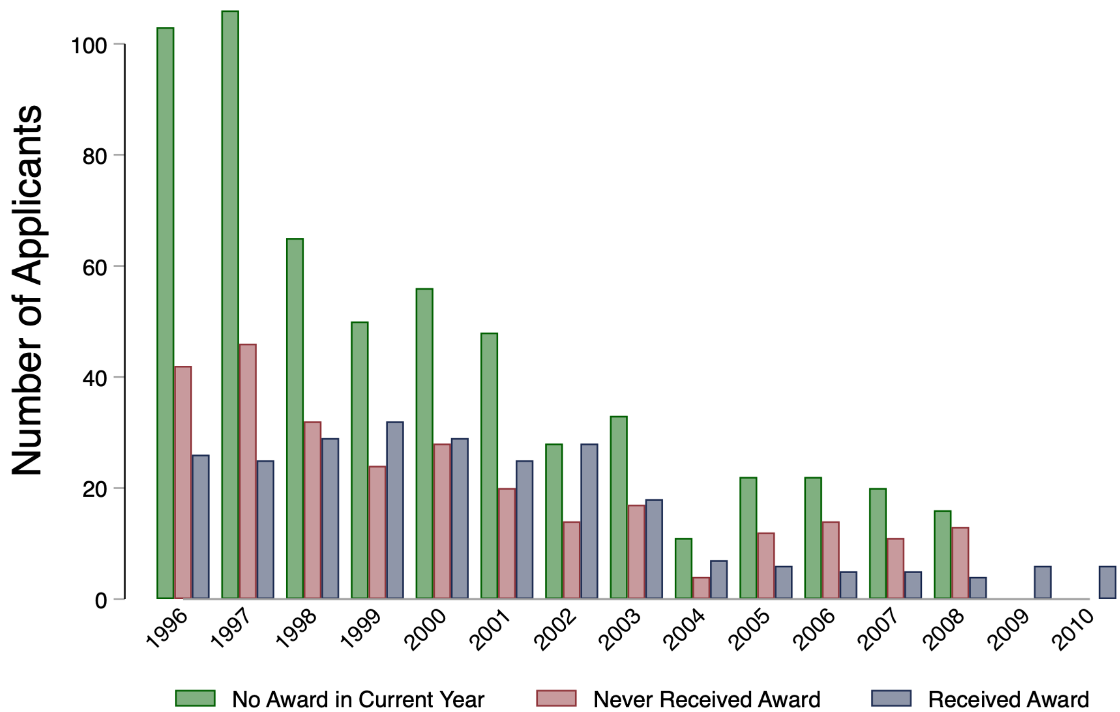
Notes: The blue circles and red diamonds represent the location of HOPE VI Revitalization and failed applicant sites, respectively. The size of the markers is proportional to the number of units in the project.
Source: Authors' calculations based on project-level summary files from HUD.

Figure A.2: Alignment of IRS and Census Income Measures Across Neighborhoods



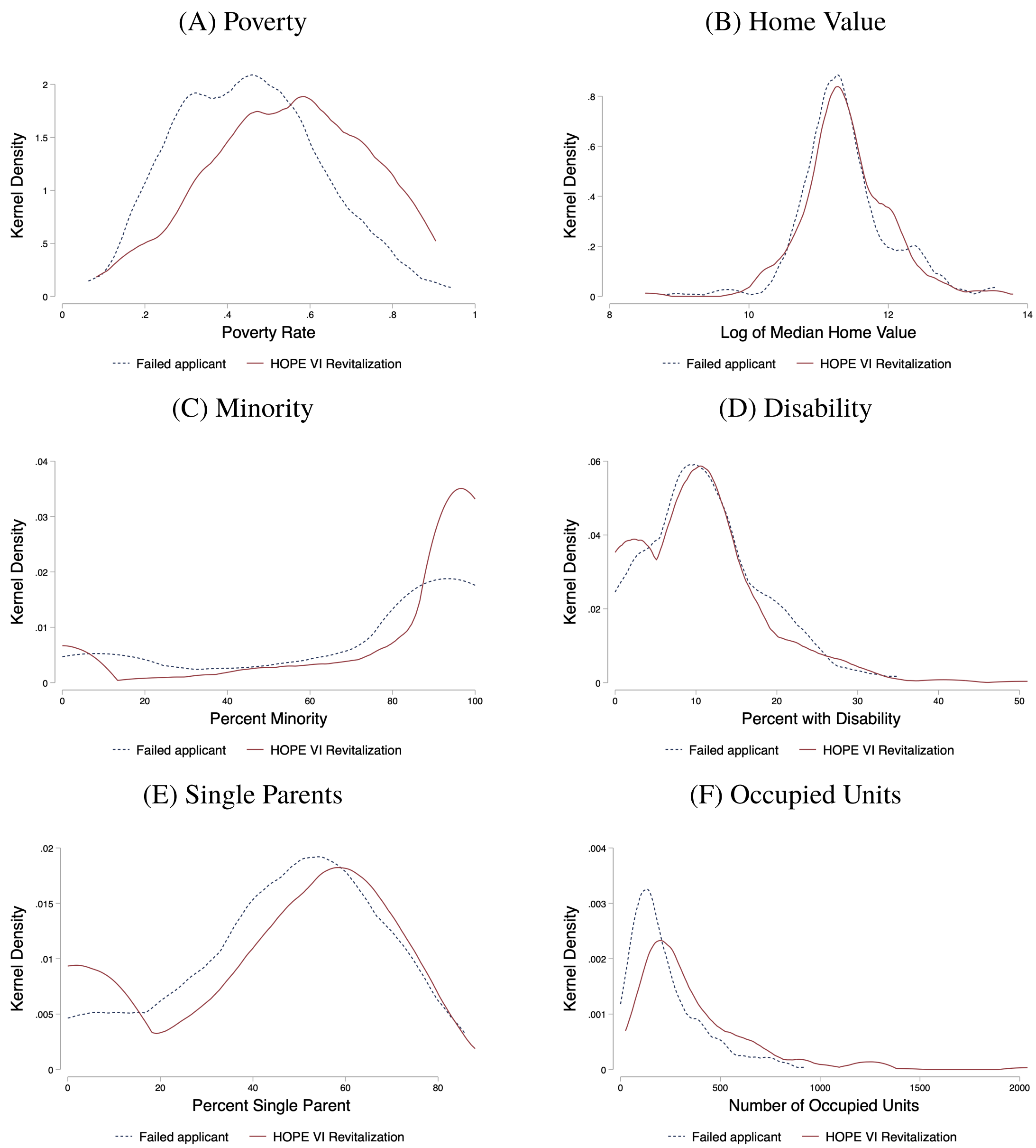
Notes: We measure household poverty using responses from the 2000 Decennial Census and identify whether individuals had adjusted gross income (AGI) below $15,000 using IRS data. We compute the average of each measure at the census block group level based on residents living there in 2000. Block groups are grouped into ventiles based on the share of residents with AGI below $15,000, and we plot the average poverty rate and AGI-based measure by ventile.
Source: Authors' calculations based on linked administrative data.

Figure A.3: Number of Awardees and Failed Applicants by Year



Notes: The figure presents the number of public housing projects per grant year for projects that applied for and: (1) did not receive funding in that year, (2) did not receive funding in any year, and (3) received a HOPE VI Revitalization grant in that year. If a project received more than one HOPE VI grant, then we report the earliest year. Projects not awarded a grant in one year could apply in a subsequent year. We do not observe application data for 2009 and 2010.
Source: Authors' calculations based on the publicly available list of awardees of and applicants to HOPE VI funding.

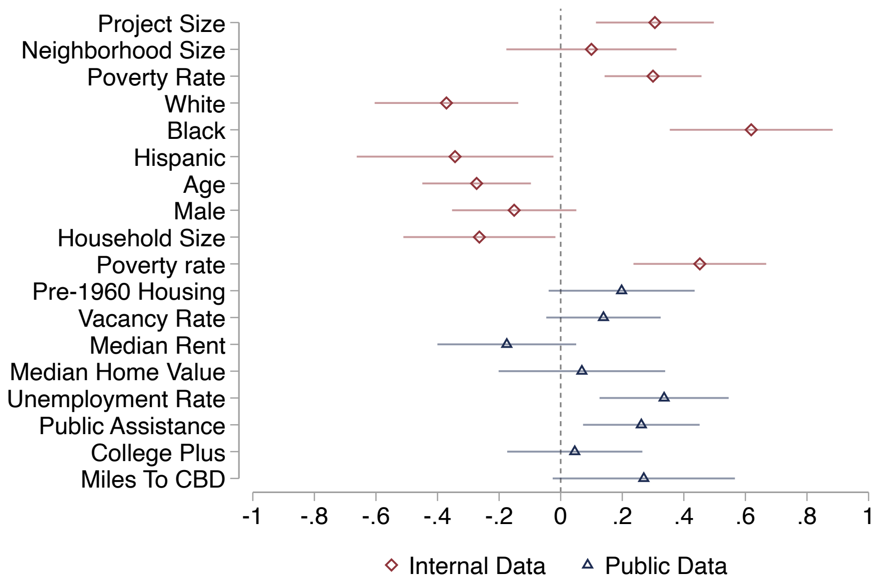
Figure A.4: Distribution of Baseline Characteristics



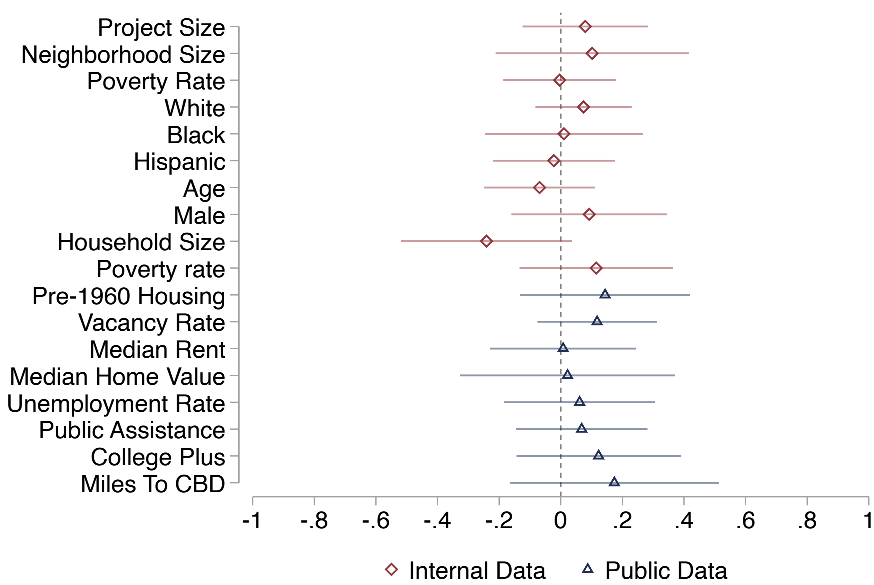
Notes: Each figure presents a kernel density plot of characteristics measured prior to the award. Panels A and B summarize characteristics of the block group and are based on data from the 1990 Decennial Census. Panels C-F summarize characteristics of the projects as measured in the 1993 vintage of HUD User's Picture of Subsidized Households. Distributions are presented separately by treatment status.
Source: Authors' calculations based on summary files of 1990 Decennial Census survey and project-level summary files from HUD.

Figure A.5: Balance of Baseline Characteristics

(A) Population Weights



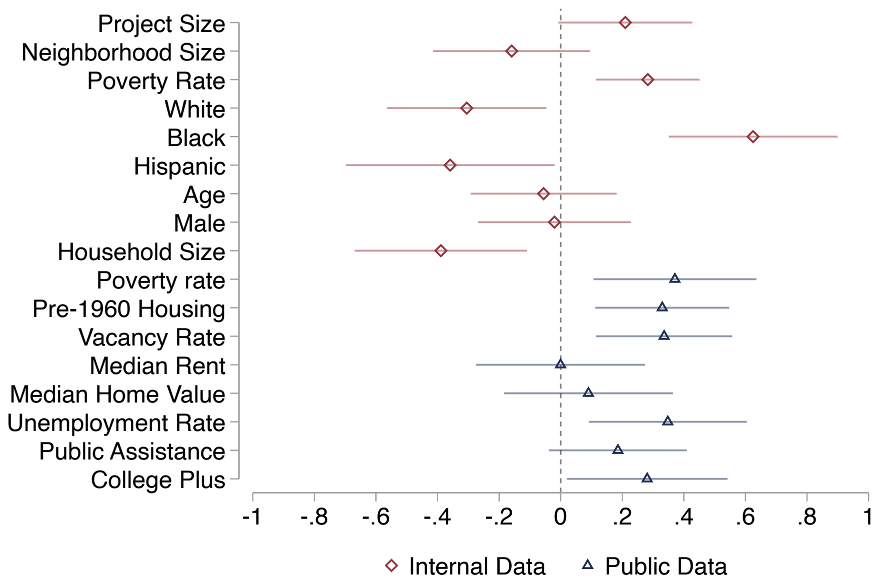
(B) Inverse Propensity Score Weights



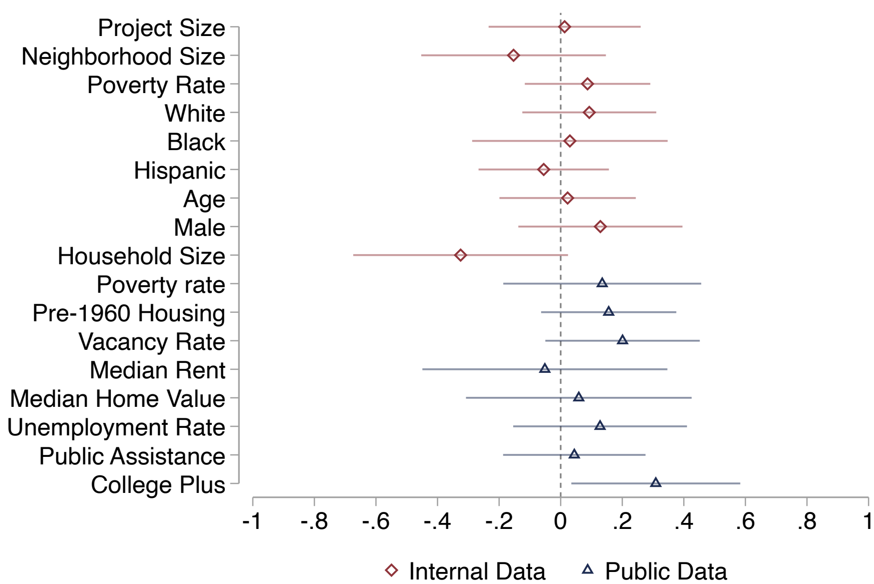
Notes: Each point plots the coefficient from a separate regression of a pre-award neighborhood characteristic (standardized by mean and standard deviation) on an indicator for whether the neighborhood received a HOPE VI Revitalization grant. The sample includes HOPE VI and failed applicant neighborhoods. Panel A weights by neighborhood population; Panel B uses inverse propensity score weights. Red diamonds represent estimates using internal data; blue triangles use publicly available data. Standard errors are clustered at the neighborhood level, and 95 percent confidence intervals are shown with horizontal bars.
Source: Authors' calculations based on linked administrative data.

Figure A.6: Balance of Baseline Characteristics in Surrounding Neighborhoods

(A) Population Weights

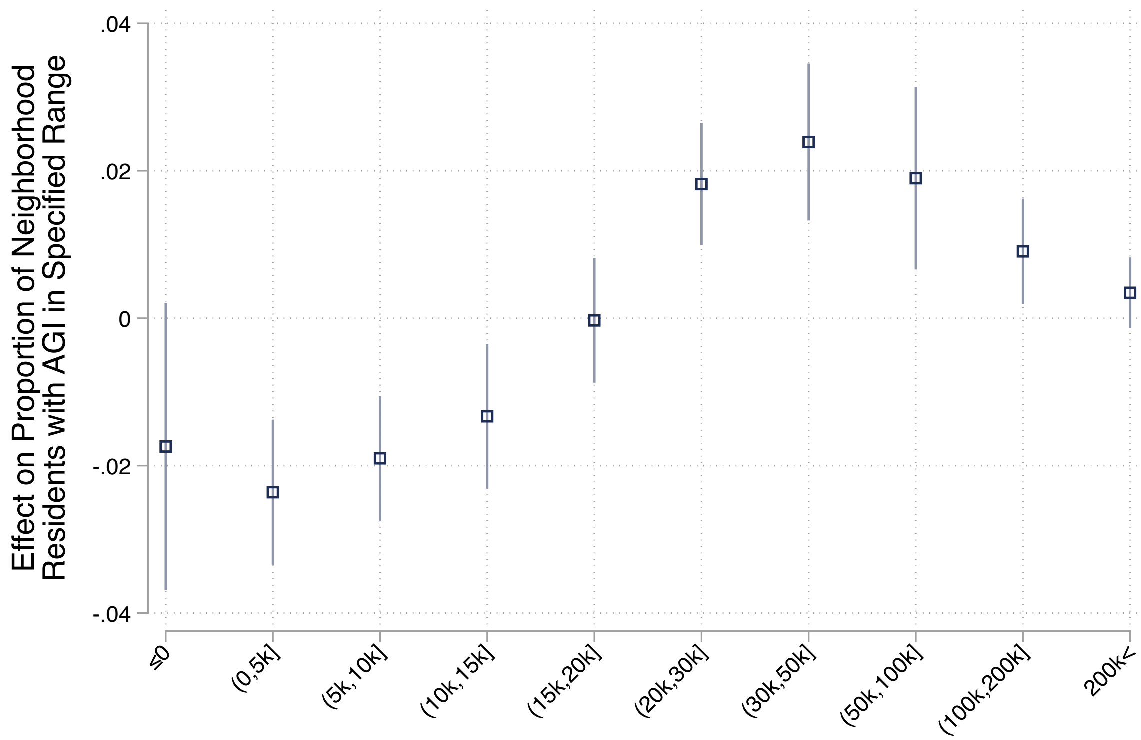


(B) Inverse Propensity Score Weights



Notes: Each point plots the coefficient from a separate regression of a pre-award neighborhood characteristic (standardized by mean and standard deviation) on an indicator for whether the neighborhood received a HOPE VI Revitalization grant. The sample includes the contiguous Census block groups to the HOPE VI and failed applicant neighborhoods. Panel A weights by neighborhood population; Panel B uses inverse propensity score weights. Red diamonds represent estimates using internal data; blue triangles use publicly available data. Standard errors are clustered at the neighborhood level, and 95 percent confidence intervals are shown with horizontal bars.
Source: Authors' calculations based on linked administrative data.

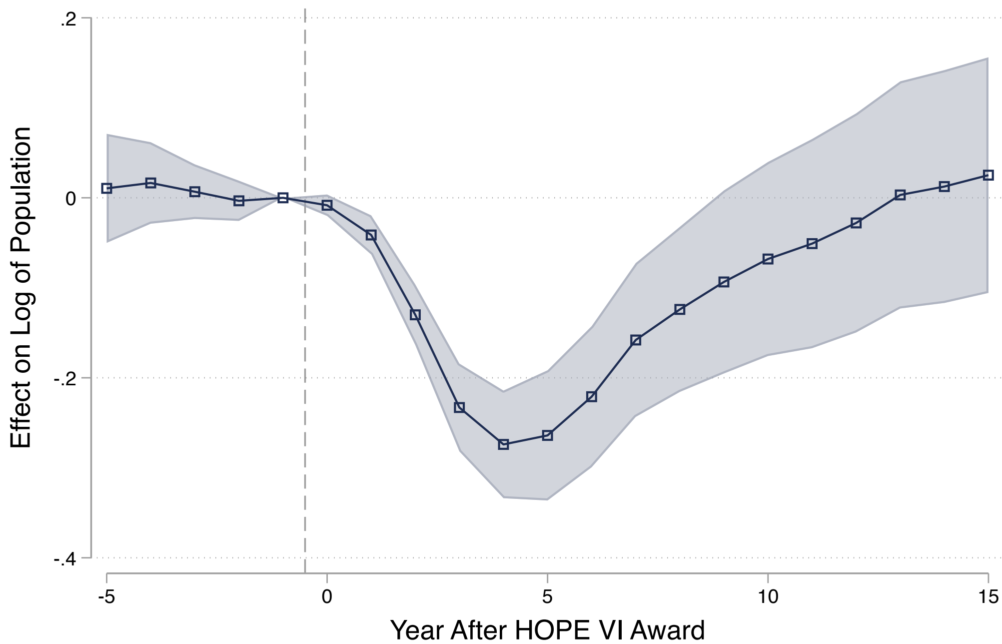
Figure A.7: Effect of HOPE VI Revitalization on Neighborhood Income Distribution



Notes: This figure plots estimates from Equation 8. Each point represents an estimate from a separate regression in which the outcome variable is the proportion of individuals living in the neighborhood 10 to 15 years after the award whose AGI falls within the range defined by the horizontal axis. The vertical bars denote the 95 percent confidence intervals.

Source: Authors' calculations based on linked administrative data.

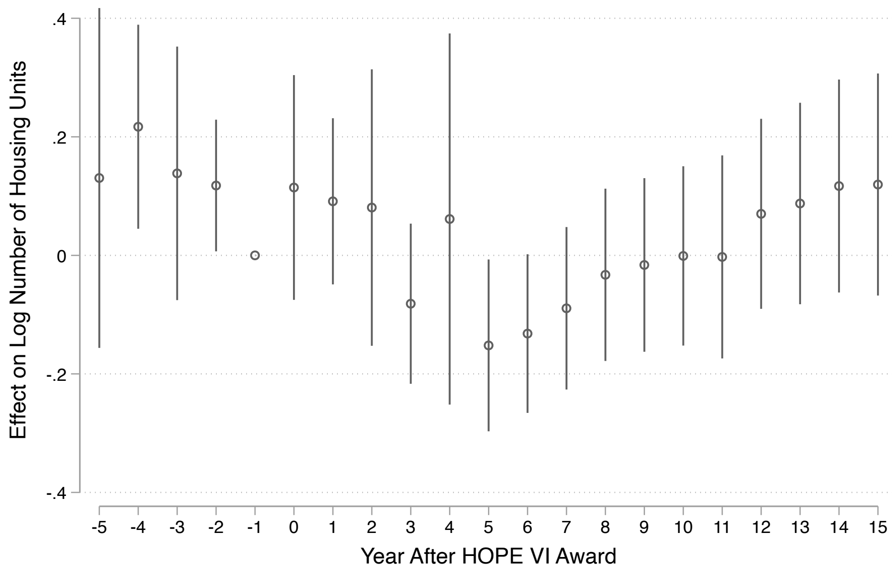
Figure A.8: Effect of HOPE VI Revitalization on Neighborhood Population Size



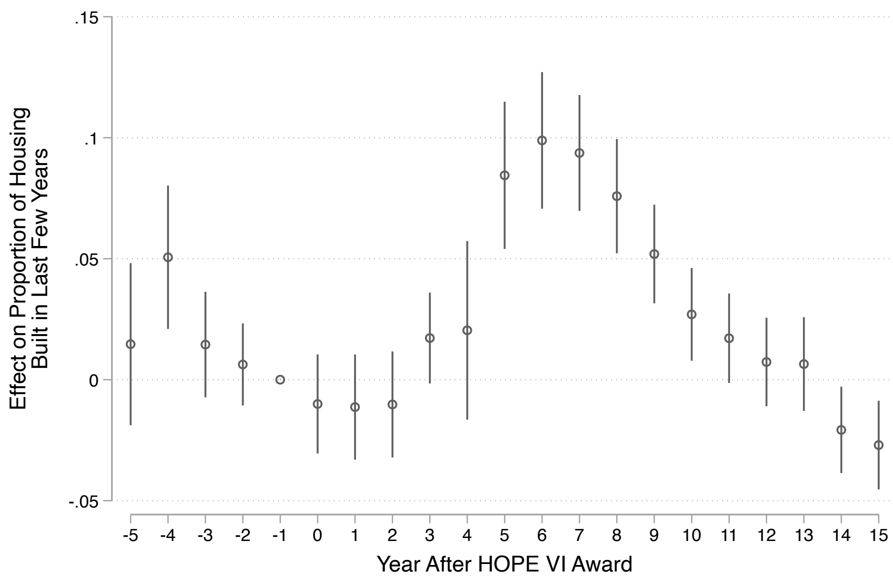
Notes: This figure plots estimates from the stacked difference-in-differences specification described in Equation 11. The outcome variable is the log of the total number of people living in the neighborhood. The shaded regions denote the 95 percent confidence intervals.
Source: Authors' calculations based on linked administrative data.

Figure A.9: Effect of HOPE VI Revitalization on Housing Stock

(A) Total Housing

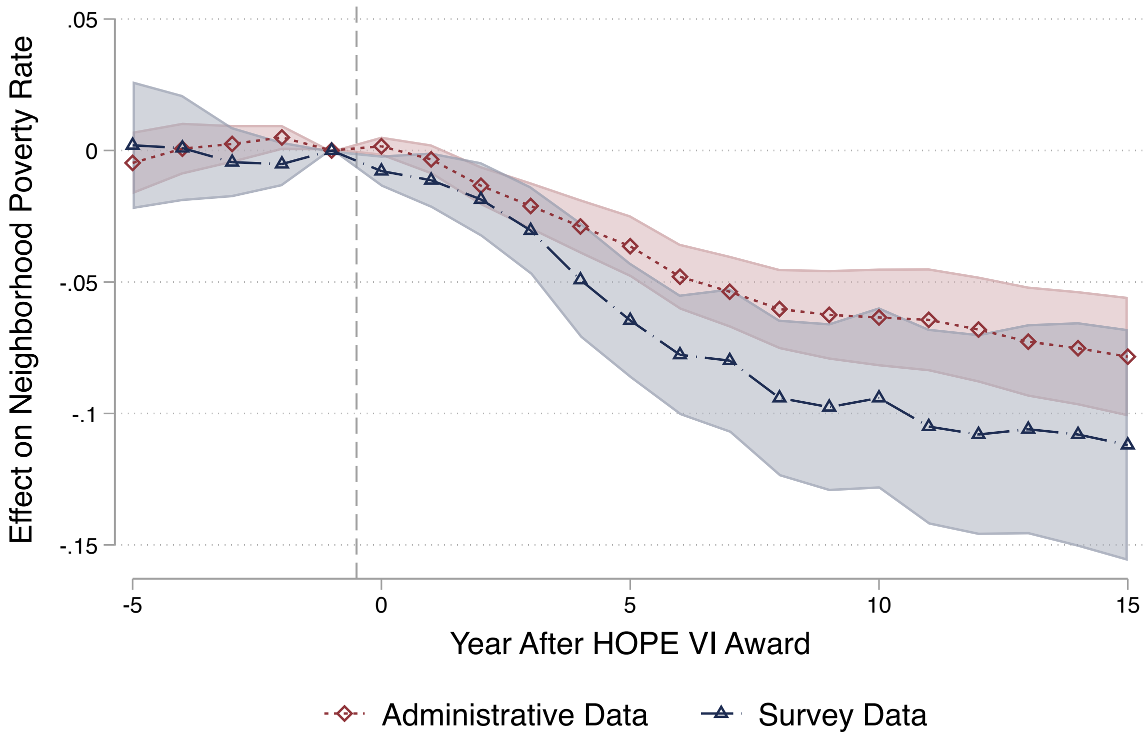


(B) New Housing



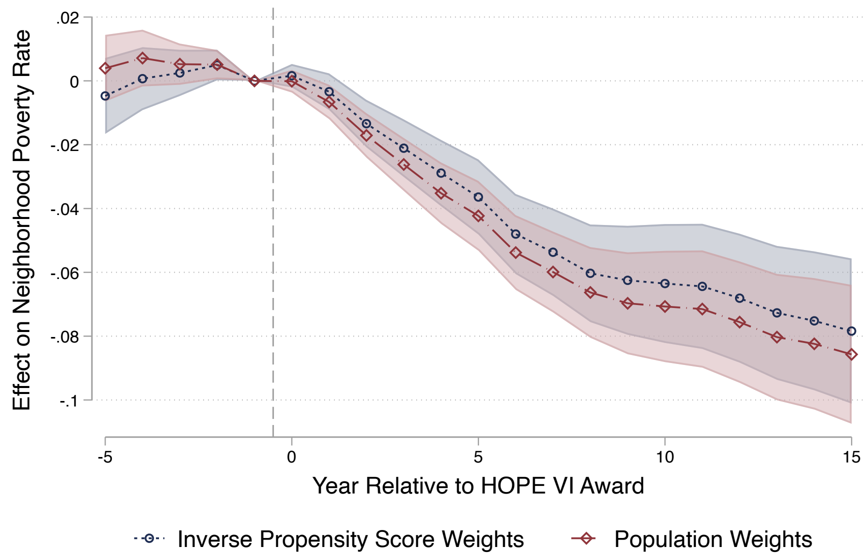
Notes: This figure plots estimates from the stacked difference-in-differences specification described in Equation 11. The outcomes are the total housing stock and the proportion of housing built within the past few years. The vertical bars denote the 95 percent confidence intervals.
Source: Authors' calculations based on summary files of Decennial Census surveys and the ACS.

Figure A.10: Effect of HOPE VI Revitalization on Alternative Measures of Poverty Rates



Notes: This figure plots estimates from the specification described in Equation 11. The outcome variable is either the share of the population with an AGI below $15,000 (based on the administrative data) or the share of the population whose income falls below the poverty line (based on survey data). The shaded region denotes the 95 percent confidence intervals.
Source: Authors' calculations based on linked administrative data.

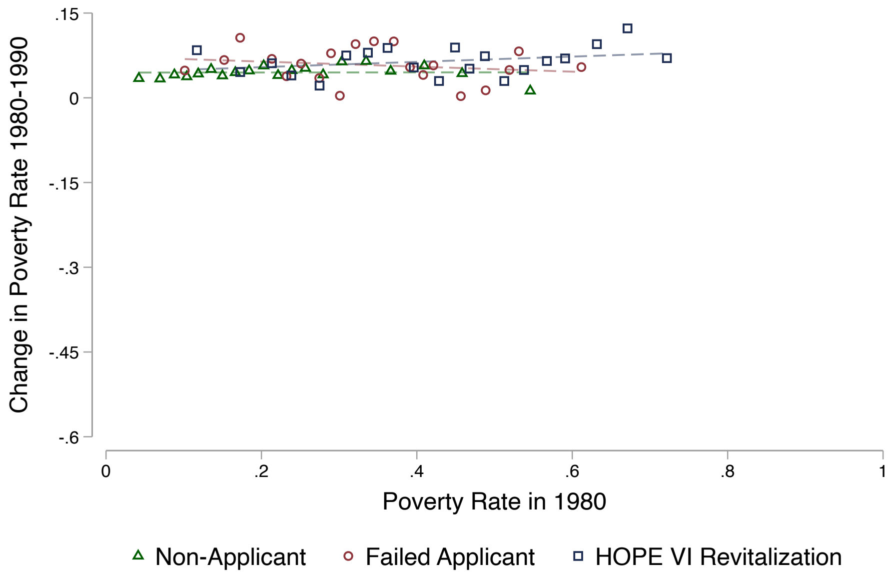
Figure A.11: Effect of HOPE VI Revitalization Using Alternative Weights



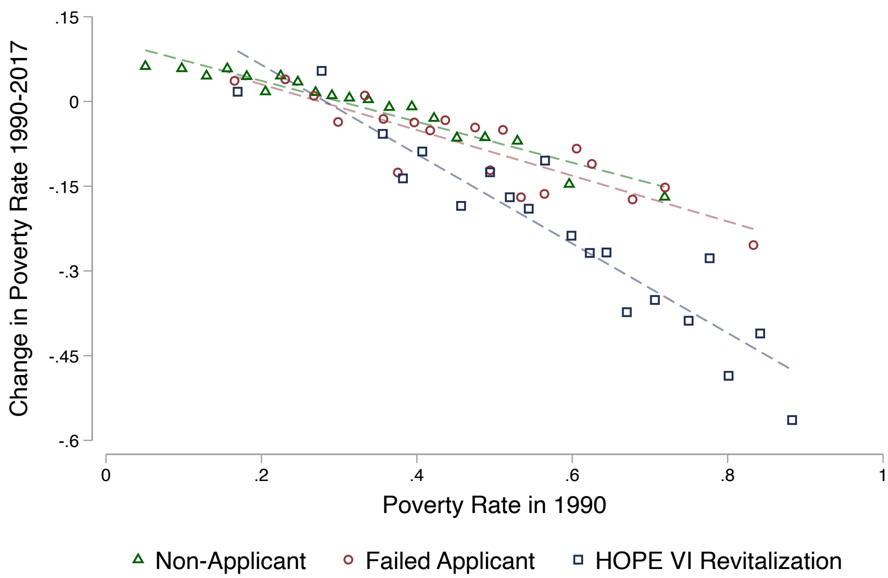
Notes: This figure plots estimates from the stacked difference-in-differences specification described in Equation 11. The two series present estimates from specifications that weight by the inverse propensity score and the baseline population of the neighborhood. The outcome is the share of current residents who are poor. The shaded regions denote the 95 percent confidence intervals.

Figure A.12: Changes in Neighborhood Poverty Rates in Public Housing Sites

(A) Before HOPE VI Revitalization Program

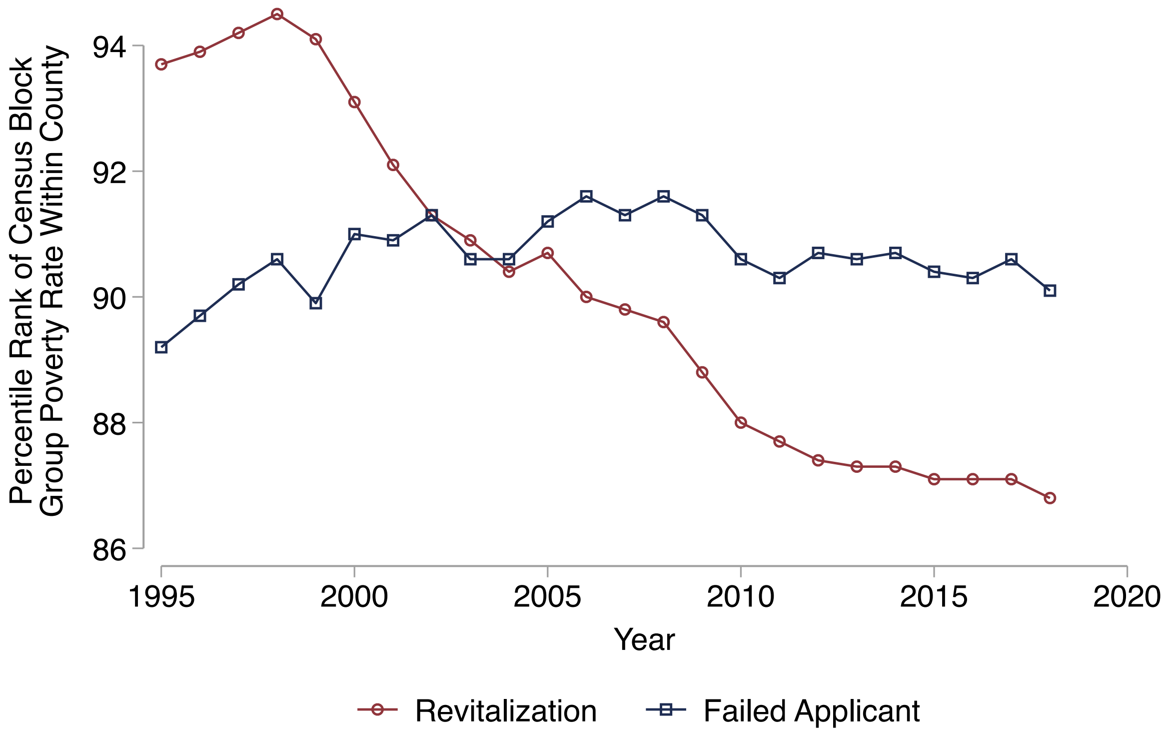


(B) After HOPE VI Revitalization Program



Notes: Each figure plots the change in neighborhood poverty rates against the baseline poverty rate. Within the non-applicant, failed applicant, and HOPE VI Revitalization samples, projects are grouped into population-weighted ventiles based on the baseline poverty rate. Each point represents a population-weighted average of observations in a given ventile. In Panel A the baseline year is 1980 and the change is measured between 1980 and 1990. In Panel B the baseline year is 1990 and the change is measured between 1990 and 2017.
Source: Authors' calculations based on summary files of Decennial Census survey and the ACS.

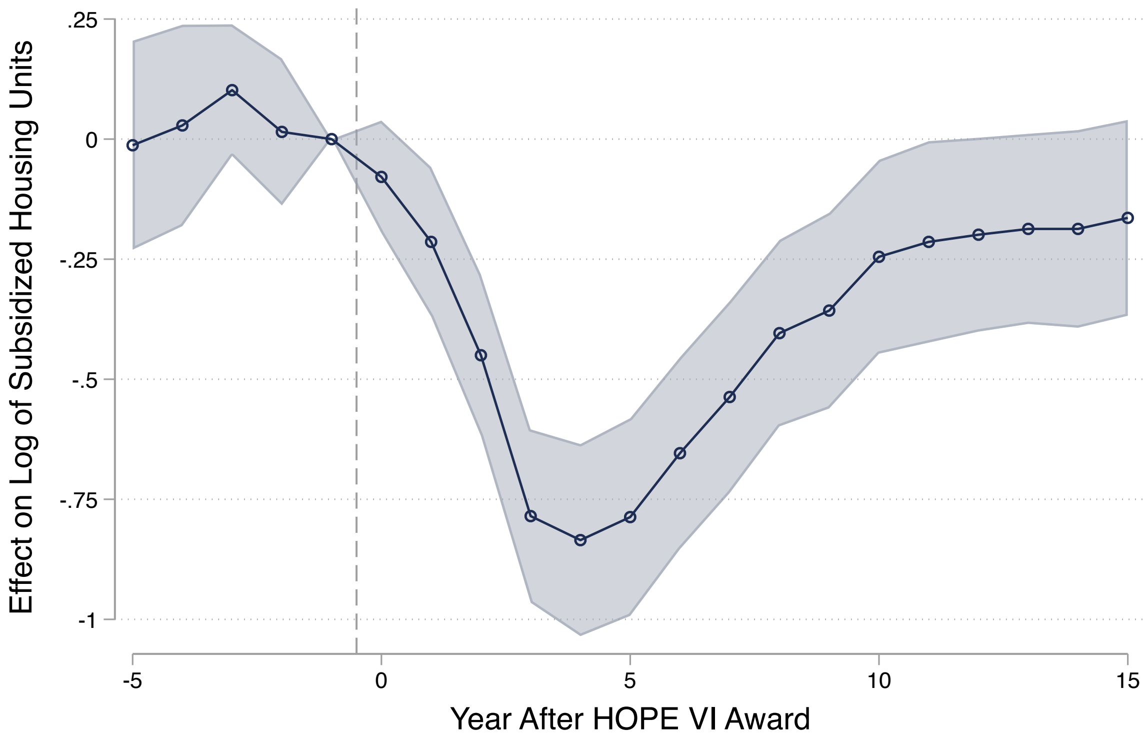
Figure A.13: Average Within-County Rank of Poverty Rate by Year and Treatment Status



Notes: For each census block group and year, we compute its percentile rank relative to all other block groups in the same county and year. Red circles show the average percentile rank for HOPE VI Revitalization neighborhoods; blue circles show the corresponding average for failed applicant neighborhoods.

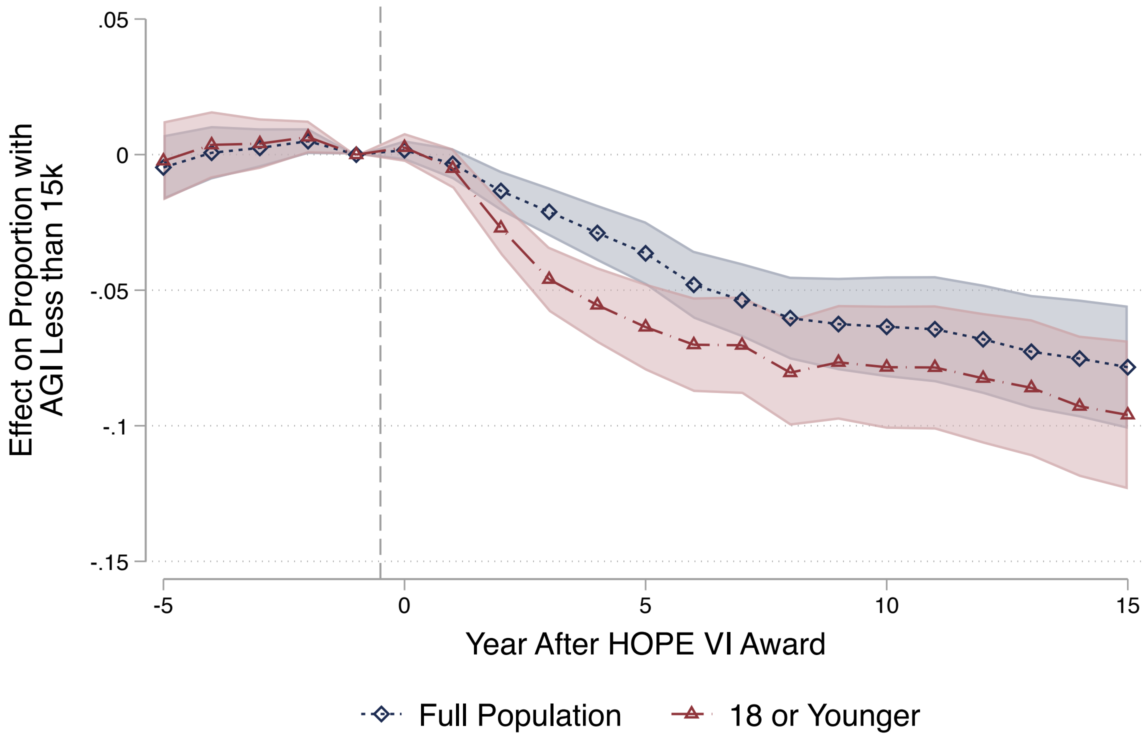
Source: Authors' calculations based on linked administrative data.

Figure A.14: Effect of HOPE VI Revitalization on Number of Subsidized Housing Units



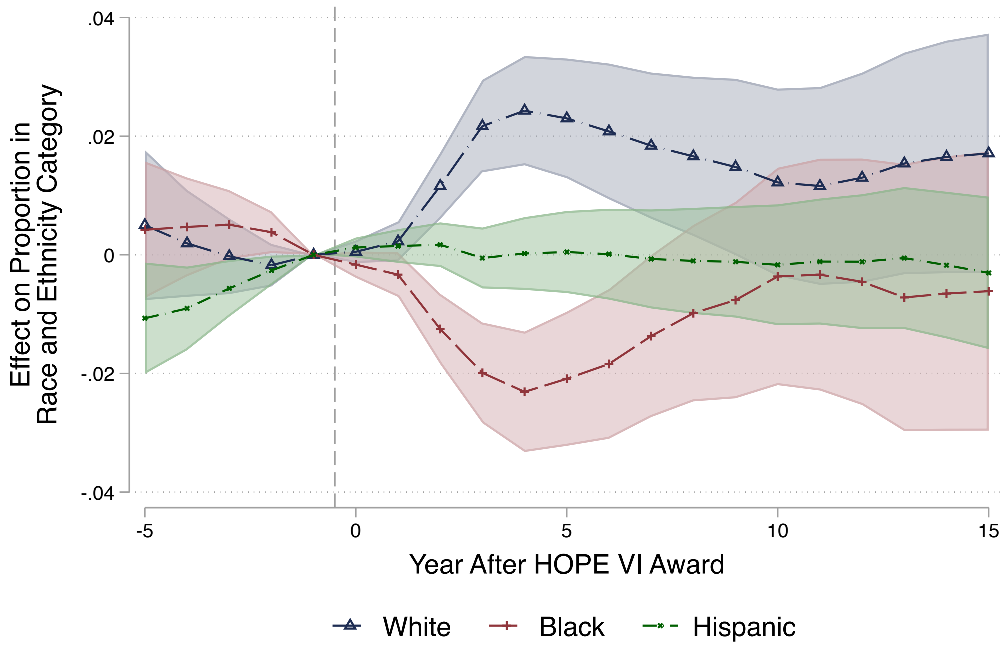
Notes: This figure plots estimates from the stacked difference-in-differences specification described in Equation 11. The outcome is the log of the total number of subsidized housing units in the neighborhood. The vertical bars denote the 95 percent confidence intervals.
Source: Authors' calculations based on linked administrative data.

Figure A.15: Effect of HOPE VI on Neighborhood Poverty, Among Families with Children



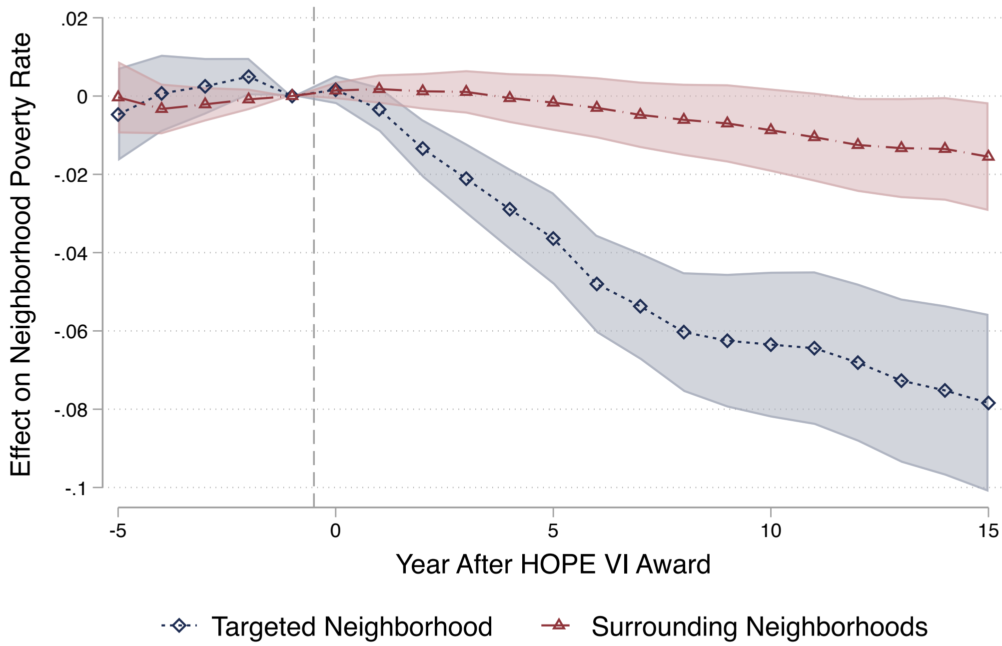
Notes: This figure plots estimates from the stacked difference-in-differences specification described in Equation 11. The outcome is the poverty rate for all individuals in the neighborhood (diamond markers) or the poverty rate for those 18 and younger (triangle markers). The shaded regions denote the 95 percent confidence intervals.
Source: Authors' calculations based on linked administrative data.

Figure A.16: Effect of HOPE VI on Racial and Ethnic Composition of Neighborhood



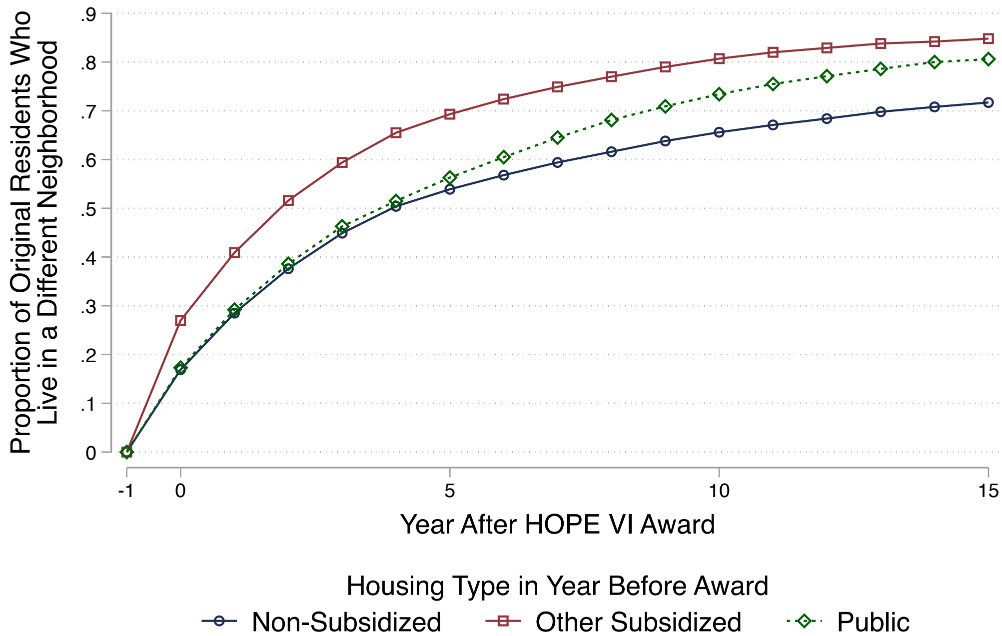
Notes: This figure plots estimates from the stacked difference-in-differences specification described in Equation 11. The outcome is the proportion of the population who is White non-Hispanic, Black non-Hispanic, or Hispanic. The shaded regions denote the 95 percent confidence intervals.
Source: Authors' calculations based on linked administrative data.

Figure A.17: Effect of HOPE VI on Poverty Rates in Surrounding Neighborhoods



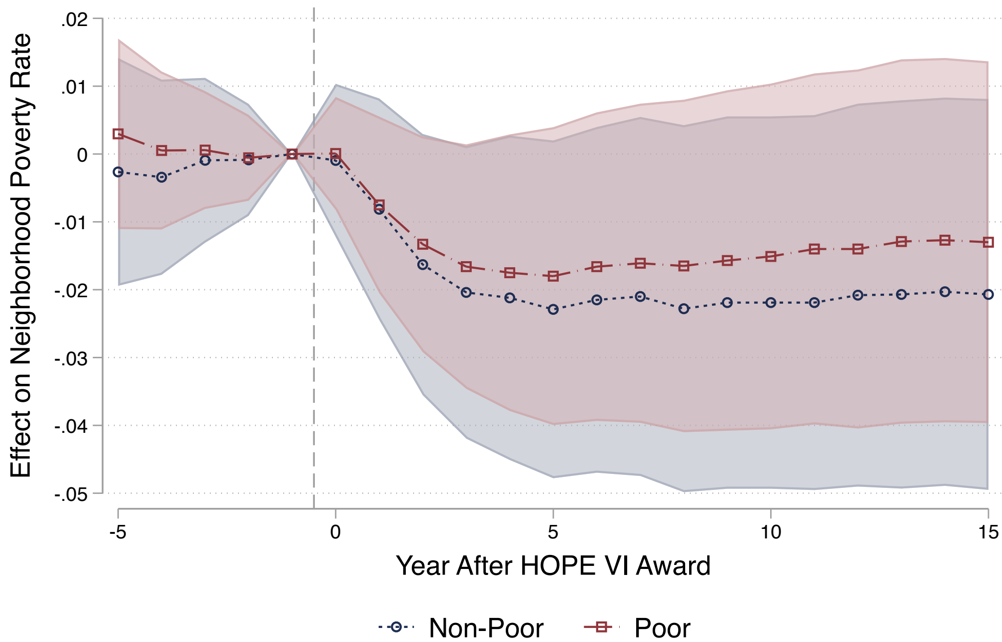
Notes: This figure plots estimates from the specification described in Equation 11. The outcome variable is the poverty rate in the target neighborhoods or the poverty rate in the contiguous Census block groups. The shaded region denotes the 95 percent confidence intervals.
Source: Authors' calculations based on linked administrative data.

Figure A.18: Residential Mobility of the Original Residents, Failed Applicants



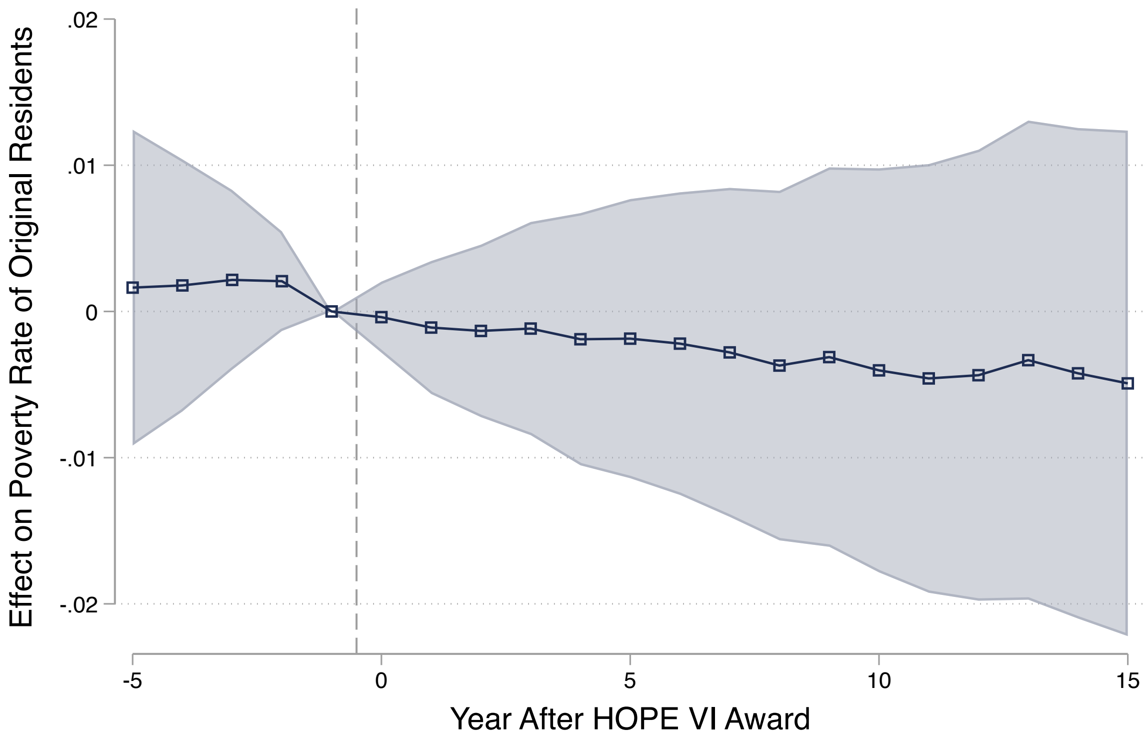
Notes: This figure plots the proportion of the original residents of the failed applicant neighborhoods who moved to a different Census block group by a given year after the award.
Source: Authors' calculations based on linked administrative data.

Figure A.19: Effect on Exposure to Poverty for Original Residents without Subsidized Housing, by Baseline Poverty Status



Notes: This figure plots estimates from the stacked difference-in-differences specification described in Equation 11. The outcome is the poverty rate of the current neighborhood, and the sample is the set of original residents without subsidized housing who are either poor or not poor in the year before the award. The shaded regions denote the 95 percent confidence intervals.
Source: Authors' calculations based on linked administrative data.

Figure A.20: Effect of HOPE VI Revitalization on Income of Original Residents

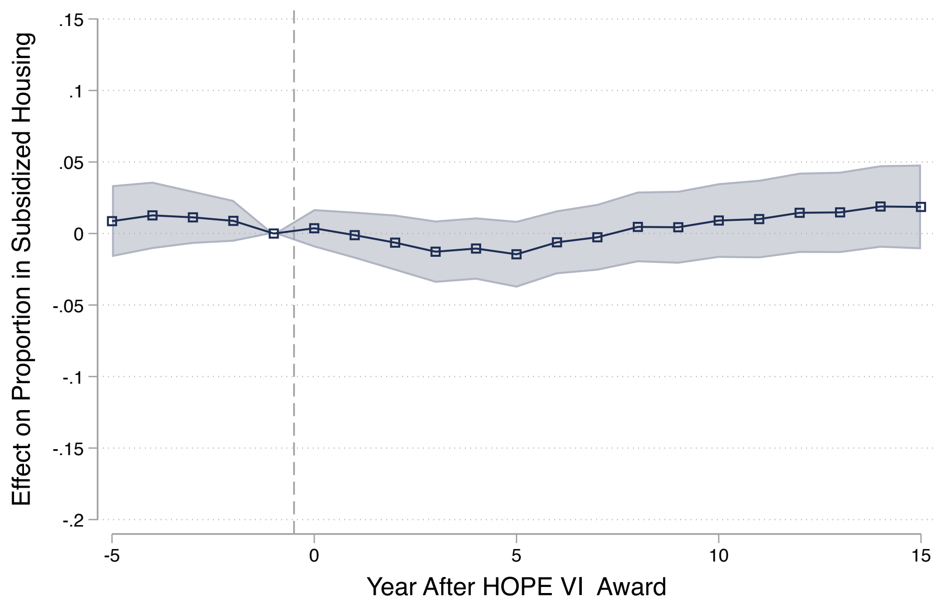


Notes: This figure plots estimates from the stacked difference-in-differences specification described in Equation 11. The outcome is the proportion of the original residents who are poor. The shaded regions denote the 95 percent confidence intervals.

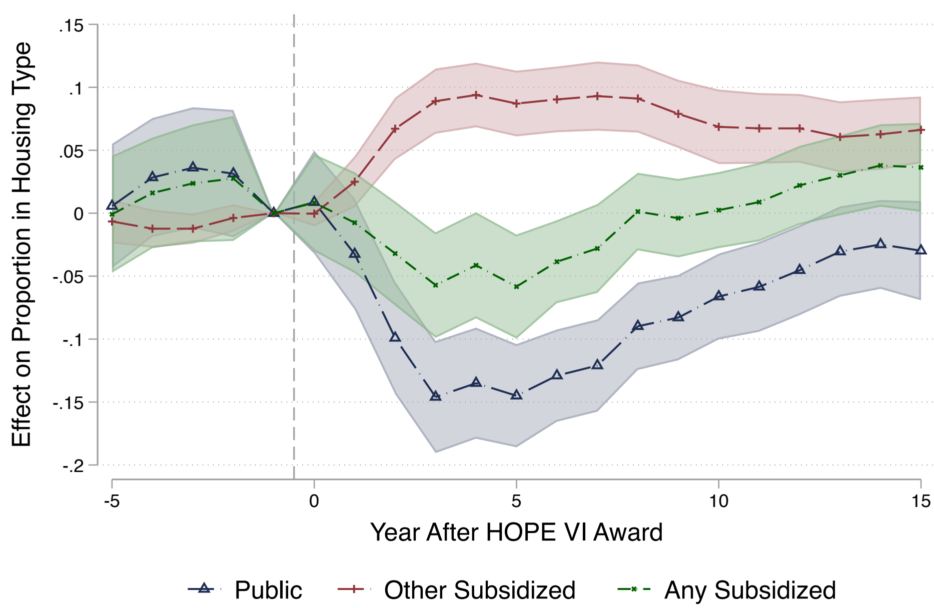
Source: Authors' calculations based on linked administrative data.

Figure A.21: Effect of HOPE VI Revitalization on Subsidy Status, Original Residents

(A) All Original Residents

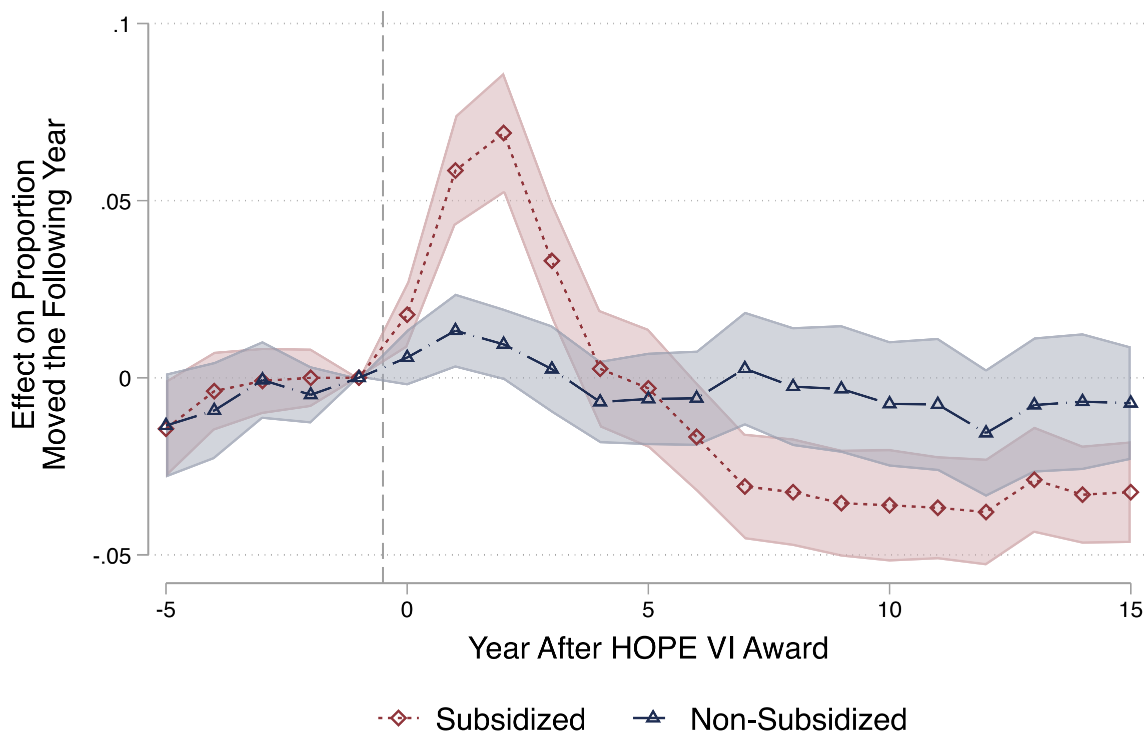


(B) Original Residents of Public Housing



Notes: This figure plots estimates from the stacked difference-in-differences specification described in Equation 11. The sample in Panel A includes all original residents and the outcome is the proportion of individuals in subsidized housing in a given year. The sample in Panel B includes original residents of the public housing projects and the outcome is the proportion of these residents who were living in public, other subsidized, or non-subsidized housing in a given year. The shaded regions denote the 95 percent confidence intervals.
Source: Authors' calculations based on linked administrative data.

Figure A.22: Effect of HOPE VI Revitalization on Out-Migration



Notes: This figure plots estimates from the specification described in Equation 11. The outcome variable is the proportion of current residents who move to a different neighborhood in the following year. The shaded region denotes the 95 percent confidence intervals.
Source: Authors' calculations based on linked administrative data.

Appendix B Proofs of Identification Results

B.1 Impact on Neighborhoods and Original Residents

We first prove the identification result for the first target estimand in Equation 1. Starting with the empirical comparison, we have:

$$\begin{aligned}
&\mathbb{E}_n^w[\bar{P}_{t''}(n) \mid D_{c(n)} = 1, T_n = 1] - \mathbb{E}_n^w[\bar{P}_{t''}(n) \mid D_{c(n)} = 0, T_n = 1] \\
&= \mathbb{E}_n^w[\bar{P}^1_{t''}(n) \mid D_{c(n)} = 1, T_n = 1] - \mathbb{E}_n^w[\bar{P}^0_{t''}(n) \mid D_{c(n)} = 0, T_n = 1] \\
&= \mathbb{E}_n^w[\bar{P}^1_{t''}(n) \mid D_{c(n)} = 1, T_n = 1] - \mathbb{E}_n^w[\bar{P}^0_{t''}(n) \mid D_{c(n)} = 1, T_n = 1] \\
&= \mathbb{E}_n[\bar{P}^1_{t''}(n) - \bar{P}^0_{t''}(n) \mid D_{c(n)} = 1, T_n = 1].
\end{aligned} \tag{B.1}$$

The second equality follows from Equation 7 when $j = \ell = t''$, $k = m = 0$, and $\mathcal{A} = \{N^0_{it''} = n\}$.

Next we prove the identification result for the second target estimand in Equation 4. Starting with the empirical comparison, we have:

$$\begin{aligned}
&\mathbb{E}_n^w[\mathbb{E}_i[\bar{P}_{t''}(N_{it''}) \mid N_{it'} = n, S_{it'} = 1] \mid D_{c(n)} = 1, T_n = 1] \\
&\quad - \mathbb{E}_n^w[\mathbb{E}_i[\bar{P}_{t''}(N_{it''}) \mid N_{it'} = n, S_{it'} = 1] \mid D_{c(n)} = 0, T_n = 1] \\
&=\mathbb{E}_n^w[\mathbb{E}_i[\bar{P}^1_{t''}(N^1_{it''}) \mid N_{it'} = n, S_{it'} = 1] \mid D_{c(n)} = 1, T_n = 1] \\
&\quad - \mathbb{E}_n^w[\mathbb{E}_i[\bar{P}^0_{t''}(N^0_{it''}) \mid N_{it'} = n, S_{it'} = 1] \mid D_{c(n)} = 0, T_n = 1] \\
&=\mathbb{E}_n^w[\mathbb{E}_i[\bar{P}^1_{t''}(N^1_{it''}) \mid N_{it'} = n, S_{it'} = 1] \mid D_{c(n)} = 1, T_n = 1] \\
&\quad - \mathbb{E}_n^w[\mathbb{E}_i[\bar{P}^0_{t''}(N^0_{it''}) \mid N_{it'} = n, S_{it'} = 1] \mid D_{c(n)} = 1, T_n = 1] \\
&= \mathbb{E}_n[\mathbb{E}_i[\bar{P}^1_{t''}(N^1_{it''}) - \bar{P}^0_{t''}(N^0_{it''}) \mid N_{it'} = n, S_{it'} = 1] \mid T_n = 1, D_{c(n)} = 1].
\end{aligned} \tag{B.2}$$

The second equality follows from Equation 7 when $j = \ell = t''$, $k = m = 0$, and $\mathcal{A} = \{N_{it'} = n, S_{it'} = 1\}$.

B.2 Impacts on Subgroups in Subsidized Housing After HOPE VI

B.2.1 Original Residents Who Received Subsidized Housing After ($\mathcal{G}_1$)

To prove Equation 26, we start with the estimator, and we have,

$$\begin{aligned}
&\mathbb{E}_{in}^w[\bar{P}_{t''}(N_{it''}) \mid D_{c(N_{it'})} = 1, T_{N_{it'}} = 1, S_{it''} = 1] \\
&\qquad - \mathbb{E}_{in}^w[\bar{P}_{t''}(N_{it''}) \mid D_{c(N_{it'})} = 0, T_{N_{it'}} = 1, S_{it''} = 1] \\
&=\mathbb{E}_{in}^w[\bar{P}^1_{t''}(N^1_{it''}) \mid D_{c(N_{it'})} = 1, T_{N_{it'}} = 1, S^1_{it''} = 1] \\
&\qquad - \mathbb{E}_{in}^w[\bar{P}^0_{t''}(N^0_{it''}) \mid D_{c(N_{it'})} = 0, T_{N_{it'}} = 1, S^1_{it''} = 1] \\
&=\mathbb{E}_{in}[\bar{P}^1_{t''}(N^1_{it''}) - \bar{P}^0_{t''}(N^0_{it''}) \mid i \in \mathcal{G}_1]
\end{aligned} \tag{B.3}$$

where the first equality holds by Assumption 2 and the second by Assumption 1 when $j = \ell = t''$, $k = m = 0$, and $\mathcal{A} = \{N_{it'} = n, S^1_{it''} = 1\}$.

B.2.2 New Residents of HOPE VI Neighborhoods ($\mathcal{G}_2$)

To prove Equation 29, we start by decomposing the target estimand into the effect on neighborhoods and the sorting effect,

$$\begin{aligned}\mathbb{E}_{in}[\bar{P}^1_{t''}(N^1_{it''}) - \bar{P}^0_{t''}(N^0_{it''}) \mid \mathcal{G}_2] =&\mathbb{E}_{in}[\bar{P}^1_{t''}(N^1_{it''}) - \bar{P}^0_{t''}(N^1_{it''}) \mid \mathcal{G}_2] \\ &+ \mathbb{E}_{in}[\bar{P}^0_{t''}(N^1_{it''}) - \bar{P}^0_{t''}(N^0_{it''}) \mid \mathcal{G}_2] \\ =&\mathbb{E}_{n}[\bar{P}^1_{t''}(n) - \bar{P}^0_{t''}(n) \mid D_{c(n)} = 1, T_n = 1] \\ &+ \mathbb{E}_{in}[\bar{P}^0_{t''}(N^1_{it''}) - \bar{P}^0_{t''}(N^0_{it''}) \mid \mathcal{G}_2].\end{aligned} \tag{B.4}$$

Assumption 3 allows us to rewrite the estimator as,[40]

$$\begin{aligned}&\mathbb{E}^w_{in}[\bar{P}_{t'}(N_{it'}) \mid D_{c(N_{it''})} = 1, T_{N_{it'}} = 0, T_{N_{it''}} = 1, S_{it''} = 1] \\ &-\mathbb{E}^w_{in}[\bar{P}_{t'}(N_{it'}) \mid D_{c(N_{it''})} = 0, T_{N_{it'}} = 0, T_{N_{it''}} = 1, S_{it''} = 1] \\ &\quad=\theta\big(\mathbb{E}^w_{in}[\bar{P}^0_{it''}(N^0_{it''}) \mid D_{c(N_{it''})} = 1, T_{N_{it'}} = 0, T_{N_{it''}} = 1, S_{it''} = 1] \\ &\quad\quad - \mathbb{E}^w_{in}[\bar{P}^0_{it''}(N^0_{it''}) \mid D_{c(N_{it''})} = 0, T_{N_{it'}} = 0, T_{N_{it''}} = 1, S_{it''} = 1]\big) \\ &\quad=\theta\big(\mathbb{E}^w_{in}[\bar{P}^0_{it''}(N^0_{it''}) \mid D_{c(N_{it''})} = 1, T_{N_{it'}} = 0, T_{N_{it''}} = 1, S_{it''} = 1] \\ &\quad\quad - \mathbb{E}^w_{n}[\bar{P}^0_{it''}(n) \mid D_{c(n)} = 0, T_n = 1]\big)\end{aligned} \tag{B.5}$$

When $j = \ell = t''$, $k = m = 0$, and $\mathcal{A} = \{N^0_{it''} = n\}$, Assumption 1 implies that, $\mathbb{E}^w_{in}[\bar{P}^0_{it''}(n) \mid D_{c(n)} = 0, T_n = 1] = \mathbb{E}^w_{in}[\bar{P}^0_{it''}(n) \mid D_{c(n)} = 1, T_n = 1]$, which yields,

$$\begin{aligned}&\theta\big(\mathbb{E}^w_{in}[\bar{P}^0_{it''}(N^0_{it''}) \mid D_{c(N_{it''})} = 1, T_{N_{it'}} = 0, T_{N_{it''}} = 1, S_{it''} = 1] \\ &\quad - \mathbb{E}^w_{in}[\bar{P}^0_{it''}(n) \mid D_{c(n)} = 0, T_n = 1]\big) \\ &=\theta\big(\mathbb{E}^w_{in}[\bar{P}^0_{it''}(N^0_{it''}) \mid D_{c(N_{it''})} = 1, T_{N_{it'}} = 0, T_{N_{it''}} = 1, S_{it''} = 1] \\ &\quad - \mathbb{E}^w_{in}[\bar{P}^0_{it''}(n) \mid D_{c(n)} = 1, T_n = 1]\big) \\ &=\theta\big(\mathbb{E}^w_{in}[\bar{P}^0_{it''}(N^0_{it''}) \mid D_{c(N_{it''})} = 1, T_{N_{it'}} = 0, T_{N_{it''}} = 1, S_{it''} = 1] \\ &\quad - \mathbb{E}^w_{in}[\bar{P}^0_{it''}(N^1_{it''}) \mid D_{c(N_{it''})} = 1, T_{N_{it'}} = 0, T_{N_{it''}} = 1, S_{it''} = 1]\big) \\ &=\theta\mathbb{E}_{in}[\bar{P}^0_{it''}(N^0_{it''}) - \bar{P}^0_{it''}(N^1_{it''}) \mid \mathcal{G}_2].\end{aligned} \tag{B.6}$$

B.2.3 Residents Who Would Have Moved in Absent HOPE VI ($\mathcal{G}_3$)

To prove Equation 32, we can rewrite the target estimand as,

$$\mathbb{E}_{in}[\bar{P}^1_{t''}(N^1_{it''}) - \bar{P}^0_{t''}(N^0_{it''}) \mid \mathcal{G}_3] = \mathbb{E}_{in}[\bar{P}^1_{t''}(N^1_{it''}) \mid \mathcal{G}_3] - \mathbb{E}_{in}[\bar{P}^0_{t''}(N^0_{it''}) \mid \mathcal{G}_3]. \tag{B.7}$$

[40] The assumption that ϵ_i is an idiosyncratic error term implies that $\mathbb{E}^w_{in}[\epsilon_i \mid D_{c(N_{it''})} = d, T_{N_{it'}} = 0, T_{N_{it''}} = 1, S_{it''} = 1] = 0$ for $d \in \{0, 1\}$.

For the first component, we have:

$$\begin{aligned}
\mathbb{E}_{in}[\bar{P}^1_{t''}(N^1_{it''}) \mid \mathcal{G}_3] =&\mathbb{E}_{in}[\bar{P}^0_{t''}(N_{it'}) \mid \mathcal{G}_3] \\
=&\mathbb{E}^w_{in}[\bar{P}^0_{t''}(N_{it'}) \mid D_{c(N^0_{it''})} = 1, S^1_{it''} = 1, T_{N_{it'}} = 0, T_{N^1_{it''}} = 0, T_{N^0_{it''}} = 1] \\
=&\mathbb{E}^w_{in}[\bar{P}^0_{t''}(N_{it'}) \mid D_{c(N^0_{it''})} = 0, S^1_{it''} = 1, T_{N_{it'}} = 0, T_{N^1_{it''}} = 0, T_{N^0_{it''}} = 1] \\
=&\mathbb{E}^w_{in}[\bar{P}^0_{t''}(N_{it'}) \mid D_{c(N_{it''})} = 0, T_{N_{it'}} = 0, T_{N_{it''}} = 1, S_{it''} = 1],
\end{aligned} \tag{B.8}$$

where the first equality follows from Equation 30 of Assumption 4; the second equality follows from the assumption that there is no migration out of the city, which implies that $D_{c(N^1_{it''})} = D_{c(N^0_{it''})}$; the third equality follows from Assumption 1 when $j = \ell = t''$, $k = 0$, and $\mathcal{A} = \{N^0_{it''} = n, T_{N_{it'}} = 0, T_{N^1_{it''}} = 0, S^1_{it''} = 1\}$; and the fourth equality follows from Equation 31 of Assumption 4.

For the second component, we have:

$$\begin{aligned}
\mathbb{E}_{in}[\bar{P}^0_{t''}(N^0_{it''}) \mid \mathcal{G}_3] =&\mathbb{E}^w_{in}[\bar{P}^0_{t''}(N^0_{it''}) \mid D_{c(N^1_{it''})} = 1, S^1_{it''} = 1, T_{N_{it'}} = 0, T_{N^1_{it''}} = 0, T_{N^0_{it''}} = 1] \\
=&\mathbb{E}^w_{n}[\bar{P}^0_{t''}(n) \mid D_{c(n)} = 1, T_n = 1] \\
=&\mathbb{E}^w_{n}[\bar{P}^0_{t''}(n) \mid D_{c(n)} = 0, T_n = 1] \\
=&\mathbb{E}^w_{in}[\bar{P}_{t''}(N_{it''}) \mid D_{c(N_{it''})} = 0, T_{N_{it'}} = 0, T_{N_{it''}} = 1, S_{it''} = 1],
\end{aligned} \tag{B.9}$$

where the third equality follows from Assumption 1 when $j = \ell = t''$, $k = m = 0$, and $\mathcal{A} = \{N^0_{it''} = n\}$.

Combining Equations B.8 and B.9 yields the identification result,

$$\begin{aligned}
&\mathbb{E}_{in}[\bar{P}^1_{t''}(N^1_{it''}) - \bar{P}^0_{t''}(N^0_{it''}) \mid \mathcal{G}_3] \\
&\quad = \mathbb{E}^w_{in}[\bar{P}_{t''}(N_{it'}) - \bar{P}_{t''}(N_{it''}) \mid D_{c(N_{it''})} = 0, T_{N_{it'}} = 0, T_{N_{it''}} = 1, S_{it''} = 1].
\end{aligned} \tag{B.10}$$

B.2.4 Residents of Other Neighborhoods ($\mathcal{G}_4$)

To prove Equation 37, note that for any neighborhood n such that $T_n = 0$, we have:

$$\begin{aligned}
\bar{P}^1_{t''}(n) - \bar{P}^0_{t''}(n) =& \lambda_n(\mathbb{E}[P^1_{it''} \mid N^1_{it''} = n, T_{N^0_{it''}} = 1] - \mathbb{E}[P^0_{it''} \mid N^0_{it''} = n, N^1_{it''} \neq n]) \\
\leq& \lambda_n \\
=& \frac{\sum_i \mathbb{1}\{N^1_{it''} = n, T_{N^0_{it''}} = 1\}}{\sum_i \mathbb{1}\{N^1_{it''} = n\}} \\
=& \frac{\sum_i \mathbb{1}\{N^1_{it''} = n, T_{N^0_{it''}} = 1, T_{N_{it'}} = 1\}}{\sum_i \mathbb{1}\{N^1_{it''} = n\}} \\
&+ \frac{\sum_i \mathbb{1}\{N^1_{it''} = n, T_{N^0_{it''}} = 1, T_{N_{it'}} = 0\}}{\sum_i \mathbb{1}\{N^1_{it''} = n\}} \\
=& \frac{\sum_i \mathbb{1}\{N^1_{it''} = n, T_{N^0_{it''}} = 1, T_{N_{it'}} = 1\}}{\sum_i \mathbb{1}\{N^1_{it''} = n\}} \\
&+ \frac{\sum_i \mathbb{1}\{N_{it'} = n, T_{N^0_{it''}} = 1, T_{N_{it'}} = 0\}}{\sum_i \mathbb{1}\{N^1_{it''} = n\}} \\
\leq& \frac{\sum_i \mathbb{1}\{N^1_{it''} = n, T_{N_{it'}} = 1\}}{\sum_i \mathbb{1}\{N^1_{it''} = n\}} + \frac{\sum_i \mathbb{1}\{N_{it'} = n, T_{N^0_{it''}} = 1\}}{\sum_i \mathbb{1}\{N^1_{it''} = n\}} \\
=& \frac{\sum_i \mathbb{1}\{N^1_{it''} = n, T_{N_{it'}} = 1\}}{\sum_i \mathbb{1}\{N^1_{it''} = n\}} + \frac{\sum_i \mathbb{1}\{N_{it'} = n, T_{N^0_{it''}} = 1\}}{\sum_i \mathbb{1}\{N^0_{it''} = n\}},
\end{aligned} \tag{B.11}$$

where $\lambda_n = \frac{\sum_i \mathbb{1}\{N^1_{it''}=n, T_{N^0_{it''}}=1\}}{\sum_i \mathbb{1}\{N^1_{it''}=n\}}$ is the share of neighborhood n's residents at time t'' who are displaced from treated neighborhoods. The first line follows from Equations 34, 35, and 36 of Assumption 5; the second line follows from the fact that poverty rates are bounded between 0 and 1; the third line follows from the definition of λ_n; the fourth line is simply rearranging terms; the fifth line follows from Equation 33 of Assumption 5; the sixth line drops some of the conditional statements; and the seventh line follows from Equation 34 of Assumption 5.

Appendix C Additional Empirical Results

To motivate the comparison between HOPE VI and failed applicant projects, Figure A.12(A) presents a binned scatter plot of the change in neighborhood poverty rates (measured from Census Surveys) between 1980 and 1990 against the average neighborhood poverty rate in 1980. Prior to the award, changes in neighborhood poverty rates were unrelated to treatment status. Figure A.12(B) repeats the analysis for changes in poverty rates between 1990 and 2017. Revitalization sites experienced substantially larger reductions in poverty than both failed applicants and non-applicants with similar baseline poverty rates. These patterns provide preliminary evidence that HOPE VI may have reduced poverty in targeted neighborhoods and highlight the importance of comparing treated sites to similar untreated projects.

Figure A.11 assesses the robustness to using alternative weights. We estimate Equation 11 using population weights and Figure A.11 compares these estimates to those from the baseline specification, which uses IPWs. The estimated impact on neighborhood poverty rates is very similar across the two specifications, which is a consequence of the fact that the pre-grant differences in observable characteristics are modest and the failed applicant neighborhoods were relatively stable throughout the study period while revitalized neighborhoods changed dramatically. Figure A.13 illustrates this by plotting the average within-county poverty rank for HOPE VI and failed applicant neighborhoods from 1995 to 2018.[41] In 1995—one year before the first HOPE VI grant—HOPE VI neighborhoods ranked in the 95th percentile of poverty within their counties. By 2018—eight years after the final grant—they had fallen to the 87th percentile. In contrast, failed applicant neighborhoods were consistently ranked between the 89th and 91st percentiles.

Table A.5 shows that HOPE VI substantially increased housing costs. We measure housing-related variables using responses to the ACS in the 10 to 15 years after the award.[42] Since we do not have ACS data in the pre-award period, we use Equation 8 to estimate the effect of the program on the outcomes from the ACS. Columns 1 to 9 of Table A.5 show the effect on rent, home values, and monthly mortgage payments. For each outcome we present results from a baseline model, which controls for pre-award neighborhood characteristics, a modified specification that also controls for observable housing unit characteristics (building type, year built, persons per room), which is our preferred specification, and a specification that limits the sample to units that were built prior to the award. Across all outcomes and specifications we find positive effects on housing prices. The robustness of the price effects to controlling for housing characteristics and limiting the sample to pre-existing housing units, suggests that the program made the neighborhoods more appealing places to live rather than just improving the physical characteristics of the housing stock. Columns 10 and 11 show that the program had no impact on home ownership rates.

[41] We use within-county ranks to net out city-wide fluctuations in poverty rates.

[42] When measuring housing prices, we exclude households who receive subsidized housing. Respondents to the ACS often report the rent they pay (the tenant payment) as opposed to their contract rent. Because HOPE VI reduces the share of units that are subsidized, using all responses to the ACS would lead us to overestimate the effect on prices. Indeed, when we use all respondents to the ACS, we find increases in rent that are 43 percent larger than the estimates in Table A.5.

Appendix D Residential Sorting Model

In this section we interpret the HOPE VI program through the lens of a model of residential sorting. Our model extends the framework of Schelling (1971, 2006) and Glaeser (2008) to include public housing and heterogeneous agents.

Consider a city that consists of two neighborhoods, the central city (C) and the suburbs (B), which both have a fixed, homogeneous housing stock of equal size. There are two periods (t and $t+1$). The population is fixed and equal to the size of the housing stock. There are two types of households: low-income (l) and high-income (h). All low-income and no high-income households have subsidized housing. The poverty rate of neighborhood j is P_{jt} (i.e., the share of households that are low-income) and it is defined in equilibrium, while the poverty rate of the city is fixed at P.[43]

The HOPE VI program sought to reduce the average neighborhood poverty rate for subsidized renters ($\bar{P}^l_t$), where $\bar{P}^l_t \equiv P_{Ct} \times \omega_{Ct} + P_{Bt} \times \omega_{Bt}$ and $\omega_{jt} = \frac{P_{jt}}{P_{Ct}+P_{Bt}}$, taking the city-level poverty rate as given. Figure D.1 depicts the relationship between segregation and exposure to poverty by plotting $\bar{P}^l_t$ against the difference between the neighborhood-level poverty rates, $P_{Ct} - P_{Bt}$. Distributing poor households evenly throughout the city minimizes exposure to neighborhood poverty for subsidized households.

Low-income households either live in public housing or receive housing vouchers and join high-income households in their search for housing in the private market.[44] In each neighborhood a share of units, H_{jt}, are reserved for public housing. Each household i in the private market chooses a neighborhood j to maximize their utility, given by:

$$\begin{aligned} u^h_{ijt} &= y^h - r_{jt} + A_{jt} - P_{jt} + \epsilon_{ijt} \\ u^l_{ijt} &= y^l - \phi r_{jt} + \delta A_{jt} - \alpha P_{jt} + \epsilon_{ijt}, \end{aligned} \tag{D.1}$$

where y is income; r_{jt} is the cost of housing; A_{jt} is the amenity value of neighborhood j; P_{jt} is the neighborhood poverty rate; and ϵ_{ijt} represents household-level heterogeneity in neighborhood valuations, which is assumed to be independently, identically distributed extreme value. $0 < \phi < 1$ captures the fact that low-income households who are not in public housing have housing vouchers and therefore pay less than full market rent. While housing subsidies typically require households to contribute a fixed percentage of their income to rent, a positive value of ϕ could capture other dimensions of the cost of living (e.g., more expensive groceries, etc.). $\delta > 0$ and $\alpha > 0$ allow low-income households to value neighborhood amenities and neighborhood poverty differently than high-income households. A spatial equilibrium is a vector of prices, $\{r^*_{Ct}, r^*_{Bt}\}$, and allocations, $\{P^*_{Ct}, P^*_{Bt}\}$, such that no household in the private market could be made better off by moving neighborhoods.

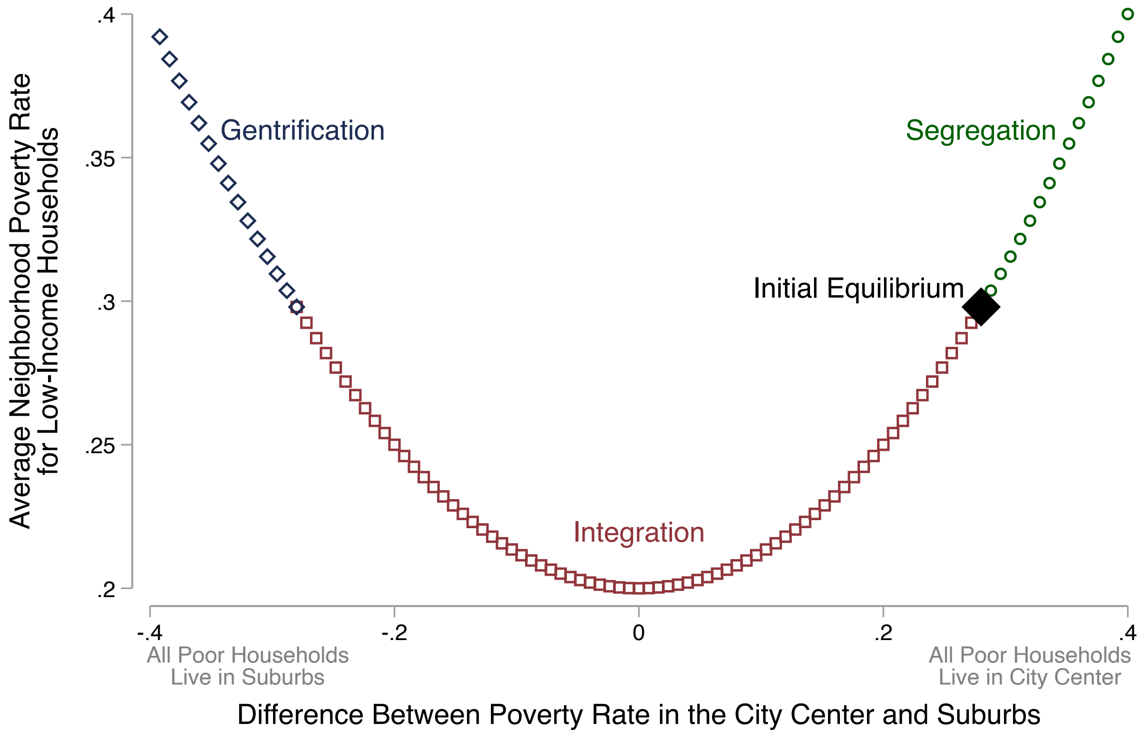
To model the city before the program we consider a unique, interior, stable equilibrium where there is no public housing in the suburbs and the city center has a higher poverty rate than the suburbs.[45] There are two cases in which such an equilibrium exists, *Case 1:* $A_{Ct} > A_{Bt}$ and $\delta > \phi$; and *Case 2:* $A_{Ct} < A_{Bt}$ and $\delta < \phi$. We model the HOPE VI program as increasing

[43]We drop the time subscript for variables that are time invariant.

[44]We assume that public housing is rationed and sufficiently appealing such that low-income households would always choose to accept public housing over searching for housing in the private market.

[45]Formally, the equilibrium satisfies $0 = H_B$, $0 < H_{Ct} < P^*_{Ct}$, and $0 < P^*_{Bt} < P^*_{Ct} < 2P$.

Figure D.1: Segregation and Exposure to Neighborhood Poverty



Notes: This figure plots $\bar{P}^l_t$ on the vertical axis against $P_{Ct} - P_{Bt}$ on the horizontal axis under the assumption that $P = 0.2$. The solid diamond depicts the initial equilibrium. The hollow diamond, square, and circle markers depict the three possible equilibria after the HOPE VI intervention.
Source: Simulated data.

neighborhood amenities ($A_{Ct} < A_{Ct+t)}$) and reducing the stock of public housing ($H_{Ct} > H_{Ct+1}$), keeping fixed the city-wide poverty rate. In equilibrium, neighborhood poverty rates are pinned down by: $(\delta - \phi)(A_B - A_{Ct}) = 2(\alpha - \phi)(P - P^*_{Ct}) + log(\frac{P^*_{Bt}}{P^*_{Ct}}) - \phi log(\frac{1-P^*_{Bt}}{1-P^*_{Ct}})$. Taking the derivative of P^*_{Ct} with respect to A_{Ct} yields, $\frac{dP^*_{Ct}}{dA_{Ct}} = \frac{\phi-\delta}{2}\left[\phi\left(\frac{(1-P^*_{Ct})(1-P^*_{Bt})-(1-P)}{(1-P^*_{Ct})(1-P^*_{Bt})}\right) - \left(\alpha + \frac{P}{P^*_{Ct}P^*_{Bt}}\right)\right]^{-1}$. The conditions of case 1 imply that $\frac{dP^*_{Ct}}{dA_{Ct}} > 0$ and the conditions of case 2 imply that $\frac{dP^*_{Ct}}{dA_{Ct}} < 0$. See Appendix D.1 for proofs.

The HOPE VI program produces three possible equilibria. First, case 1 implies that $\bar{P}^l_t < \bar{P}^l_{t+1}$. The program attracts low-income households to the city center because they value the amenities and housing vouchers prevent them from paying the full cost. We refer to this case as the *Segregation* equilibrium, which is depicted by the green circles in Figure D.1. Second, case 2 in addition to $A_B \geq A_{Ct+1}$ or $H_{Ct+1} \geq 2P - P^*_{Ct}$ implies that $\bar{P}^l_t > \bar{P}^l_{t+1}$. An increase in amenities attracts high-income households to the city center because they value the amenities more. We refer to this case as the *Integration* equilibrium, which is depicted by the red squares in Figure D.1. Third, case 2 in addition to $A_B < A_{Ct+1}$ and $H_{Ct+1} < 2P - P^*_{Ct}$ implies that $\bar{P}^l_t < \bar{P}^l_{t+1}$. In this case the poverty rate in the suburbs after the award is higher than the poverty rate in city center before the award ($P^*_{Bt+1} > P^*_{Ct}$) because the improvement in amenities attracts high-income households and the reduction in public housing allows the low-income households to be priced out. We refer to this scenario as the *Gentrification* equilibrium, which is depicted by the blue diamonds in Figure D.1. The program reduces poverty rates in the city center, but increases poverty rates in the suburbs enough to increase the average, city-wide neighborhood poverty rate for low-income households.

This model illustrates why the effect of the HOPE VI program on neighborhood poverty for subsidized renters is theoretically ambiguous. The program could increase exposure to neighborhood poverty by attracting poor households to the revitalized sites (segregation equilibrium) or relocating too many poor households to the suburbs (gentrification equilibrium). Whether these unintended consequences occur depends on how low-income households value neighborhood amenities (δ) and the extent to which they pay for them (ϕ). The program will be successful only if it attracts high-income households into the revitalized neighborhoods while maintaining a sufficient amount of affordable units for low-income households (integration equilibrium).

D.1 Proofs

The logit choice probability that household of type k selects neighborhood j is

$$\mathbb{P}^k_{jt} = \frac{exp\{\bar{u}^k_{ijt}\}}{exp\{\bar{u}^k_{iBt}\} + exp\{\bar{u}^k_{iCt}\}}, \tag{D.2}$$

where $\bar{u}^k_{ijt} \equiv u^k_{ijt} - \epsilon_{ijt}$. In equilibrium, the choice probabilities, $\mathbb{P}^k_{jt}$ must be consistent with the poverty rates that enter the utility function, P_{jt}. Without loss of generality assume the size of the population in each neighborhood is 1. Then a spatial equilibrium requires that,

$$P_{jt} = \mathbb{P}^l_{jt} 2P = 1 - \mathbb{P}^h_{jt} 2(1-P). \tag{D.3}$$

Combining the identity $(P_{Ct} + P_{Bt})/2 = P$ with equations D.2 and D.3 yields,

$$\begin{aligned} log(\mathbb{P}^l_{Bt}) - log(\mathbb{P}^l_{Ct}) &= -\phi(r_{Bt} - r_{Ct}) + \delta(A_{Bt} - A_{Ct}) - \alpha(P_{Bt} - P_{Ct}) \\ log(\frac{2P - P_{Ct}}{P_{Ct}}) &= -\phi(r_{Bt} - r_{Ct}) + \delta(A_{Bt} - A_{Ct}) - \alpha 2(P - P_{Ct}) \end{aligned} \tag{D.4}$$

and

$$\begin{aligned} log(\mathbb{P}^h_{Bt}) - log(\mathbb{P}^h_{Ct}) &= -(r_{Bt} - r_{Ct}) + (A_{Bt} - A_{Ct}) - (P_{Bt} - P_{Ct}) \\ log(\frac{2(1-P) - (1-P_{Ct})}{1 - P_{Ct}}) &= -(r_{Bt} - r_{Ct}) + (A_{Bt} - A_{Ct}) - 2(P - P_{Ct}). \end{aligned} \tag{D.5}$$

Rearranging terms in equations D.4 and D.5, we can write the price differential between neighborhoods B and C based on the difference in choice probabilities for household type k, $f^k(P_{Ct}) \equiv r_{Bt} - r_{Ct}$, as a function of the poverty rate in C,

$$\begin{aligned} f^l(P_{Ct}) &= \frac{\delta}{\phi}(A_B - A_{Ct}) - 2\frac{\alpha}{\phi}(P - P_{Ct}) - \frac{1}{\phi} log(\frac{2P - P_{Ct}}{P_{Ct}}) \\ f^h(P_{Ct}) &= (A_B - A_{Ct}) - 2(P - P_{Ct}) - log(\frac{2(1-P) - (1-P_{Ct})}{1 - P_{Ct}}) \end{aligned} \tag{D.6}$$

Equation D.6 is a system of two equations and two unknowns, which pins down the equilibrium neighborhood poverty rate, P_{Ct}, and relative prices, $r_{Bt} - r_{Ct}$.[46]

We consider a unique, interior, stable equilibrium in which there is public housing in C but not B and C has a higher poverty rate at time t. Let P^*_{jt} denote the equilibrium poverty rate in neighborhood j. Then the equilibrium satisfies the following conditions: $0 = H_B$, $0 < H_{Ct} < P^*_{Ct}$, and $0 < P^*_{Bt} < P^*_{Ct} < 2P$. Let $g(P_{Ct}) \equiv f^h(P_{Ct}) - f^l(P_{Ct})$, then the equilibrium level of price differentials, $r_{Bt} - r_{Ct}$, and the spatial distribution of poverty, P_{Ct} and P_{Bt}, is defined by the condition: $g(P^*_{Ct}) = 0$.

We prove the existence of a unique, interior, stable equilibrium by showing that three conditions are met: i) $\lim_{P_{Ct}\to 0} g(P_{Ct}) > 0$, ii) $\lim_{P_{Ct}\to 2P} g(P_{Ct}) < 0$, and iii) $dg(P_{Ct})/dP_{Ct} < 0$.

For i), note that $\lim_{P_{Ct}\to 0} \frac{1}{\phi} log(\frac{2P - P_{Ct}}{P_{Ct}}) = \infty$ for $\phi > 0$, which implies that $\lim_{P_{Ct}\to 0} g(P_{Ct}) > 0$. For ii), note that $\lim_{P_{Ct}\to 2P} \frac{1}{\phi} log(\frac{2P - P_{Ct}}{P_{Ct}}) = -\infty$ for $\phi > 0$, which implies that $\lim_{P_{Ct}\to 2P} g(P_{Ct}) < 0$. For iii), note that

$$\frac{dg(P_{Ct})}{dP_{Ct}} = 2\frac{\phi - \alpha}{\phi} - \frac{2(1-P)}{(1-P_{Bt})(1-P_{Ct})} - \frac{2P}{\phi P_{Bt} P_{Ct}} < 0. \tag{D.7}$$

Under the assumption that $\alpha > 0$ and $\phi > 0$, the inequality in Equation D.7 always holds because $\frac{\phi-\alpha}{\phi} \leq 1 < \frac{1}{1-P} \leq \frac{(1-P)}{(1-P_{Bt})(1-P_{Ct})}$ and $\frac{P}{\phi P_{Bt} P_{Ct}} > 0$.

To illustrate the intuition behind why these three conditions define an equilibrium, Figure D.2(a) plots $f^k(P_{Ct})$ and Figure D.2(b) plots $g(P_{Ct})$. If both conditions are met, then there is a unique poverty rate, P^*_{Ct} such that $g(P^*_{Ct}) = 0$. Furthermore, the fact that $g(P_{Ct})$ is downward sloping in Figure D.2(b) implies that the equilibrium is stable.

Under what conditions will $P^*_{Bt} < P^*_{Ct}$? We can rewrite $g(P^*_{Ct}) = 0$ as,

$$\frac{\delta - \phi}{\phi}(A_B - A_{Ct}) = -\frac{1}{\phi}\left[\alpha(P^*_{Ct} - P^*_{Bt}) + log(\frac{P^*_{Ct}}{P^*_{Bt}})\right] + \left[(1-P^*_{Bt}) - (1-P^*_{Ct}) - log(\frac{1-P^*_{Bt}}{1-P^*_{Ct}})\right]. \tag{D.8}$$

When $P^*_{Bt} < P^*_{Ct}$ the right hand side of Equation D.8 is always negative because $\alpha(P^*_{Ct} - P^*_{Bt}) + log(\frac{P^*_{Ct}}{P^*_{Bt}}) > 0$ and $(1-P^*_{Bt}) - (1-P^*_{Ct}) - log(\frac{1-P^*_{Bt}}{1-P^*_{Ct}}) < 0$. There are two conditions under which the left hand side is also negative: case i) $A_B < A_{Ct}$ and $\phi < \delta$ and case ii) $A_B > A_{Ct}$ and $\phi > \delta$. These are the two cases in which there exists an equilibrium that satisfies the above conditions (i.e., the equilibrium is a unique, interior, stable equilibrium in which there is public housing in C but not B and C has a higher poverty rate at time t).

Our main comparative static asks how poverty rates in the city will change if the amenity value increases. Using implicit differentiation with the condition $g(P^*_{Ct}) = 0$, yields,

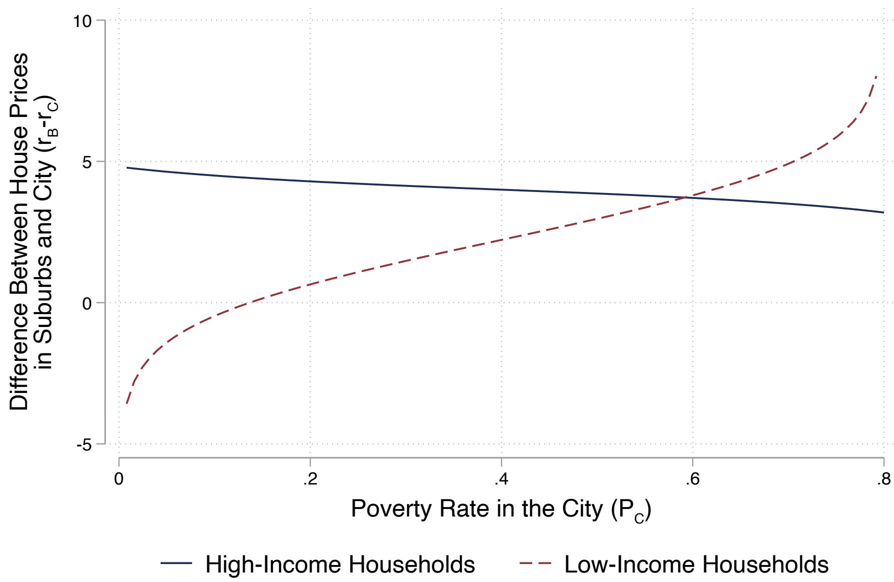
$$\frac{dP^*_{Ct}}{dA_{Ct}} = \frac{\phi - \delta}{2}\left[\phi(\frac{(1-P^*_{Ct})(1-P^*_{Bt}) - (1-P)}{(1-P^*_{Ct})(1-P^*_{Bt})}) - \left(\alpha + \frac{P}{P^*_{Ct}P^*_{Bt}}\right)\right]^{-1}. \tag{D.9}$$

Note that $\phi(\frac{(1-P^*_{Ct})(1-P^*_{Bt}) - (1-P)}{(1-P^*_{Ct})(1-P^*_{Bt})}) < 0$ because $(1-P^*_{Ct}) < (1-P) < (1-P^*_{Bt})$ and $\left(\alpha + \frac{P}{P^*_{Ct}P^*_{Bt}}\right. > 0$.

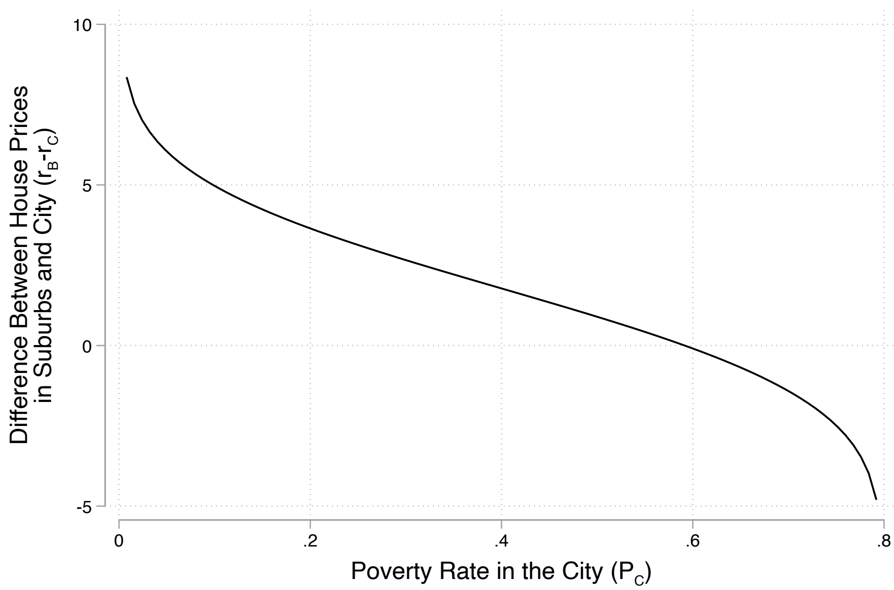
[46] Because we do not model housing supply, we can only pin down relative prices in equilibrium.

Figure D.2: Equilibrium

(A) High- and Low-Income Households



(B) Difference



Notes: Panel A plots $f^l(P_{Ct})$ and $f^h(P_{Ct})$, which describe the relationship between relative prices, $r_{Bt} - r_{Ct}$, and poverty rates in the city, P_{Ct}. Panel B plots $g(P_{Ct})$, which is defined as $g(P_{Ct}) \equiv f^h(P_{Ct}) - f^l(P_{Ct})$.
Source: The figure is based on simulated data under the assumptions that $P = 0.4$, $\phi = 0.9$, $\delta = 0.5$, $\alpha = 0.8$, and $(A_B - A_{Ct}) = 4$.

Thus, $\frac{dP^*_{Ct}}{dA_{Ct}} > 0$ if $\phi < \delta$ and $\frac{dP^*_{Ct}}{dA_{Ct}} < 0$ if $\phi > \delta$.[47]

References

Schelling, Thomas C. 1971. "Dynamic Models of Segregation." *Journal of Mathematical Sociology* 1(2): 143–186.

Schelling, Thomas C. 2006. *Micromotives and Macrobehavior*. New York: W. W. Norton & Company.

Glaeser, Edward L. 2008. *Cities, Agglomeration, and Spatial Equilibrium*. Oxford: Oxford University Press.

[47]If $\phi > \delta$ and the increase in A_{Ct} is sufficiently large such that $P^*_{Bt} > P^*_{Ct}$, then $\frac{dP^*_{Ct}}{dA_{Ct}} < 0$ still holds because $(1 - P^*_{Bt}) < (1 - P) < (1 - P^*_{Ct})$.

Appendix E Cost Benefit Calculation

Did the benefits of the program justify the costs? We use our estimates to quantify costs and benefits for the average PHA, which contained 9,021 subsidized units in 2018 and received two HOPE VI Revitalization grants totaling \$72.6 million in 2019 dollars to revitalize 17% of its subsidized units.

The aggregate effects in Section 5.4 imply that a PHA that revitalized 17% of its subsidized housing stock reduced neighborhood poverty rates among all subsidized renters by 1.5 pp, on average. Using experimental variation from MTO in housing prices to identify households' marginal utility of consumption, combined with structural estimates of household willingness to pay for neighborhood amenities, Galiani et al. (2015) estimate that a similarly low-income sample of subsidized renters has an annual willingness to pay of \$449 for a 1 pp decrease in the neighborhood poverty rate.[48] These estimates imply that subsidized renters in HOPE VI PHAs would be willing to pay \$674 more per year for housing in neighborhoods with poverty rates that are 1.5 pp lower ($674 = 1.5 \times 449$).

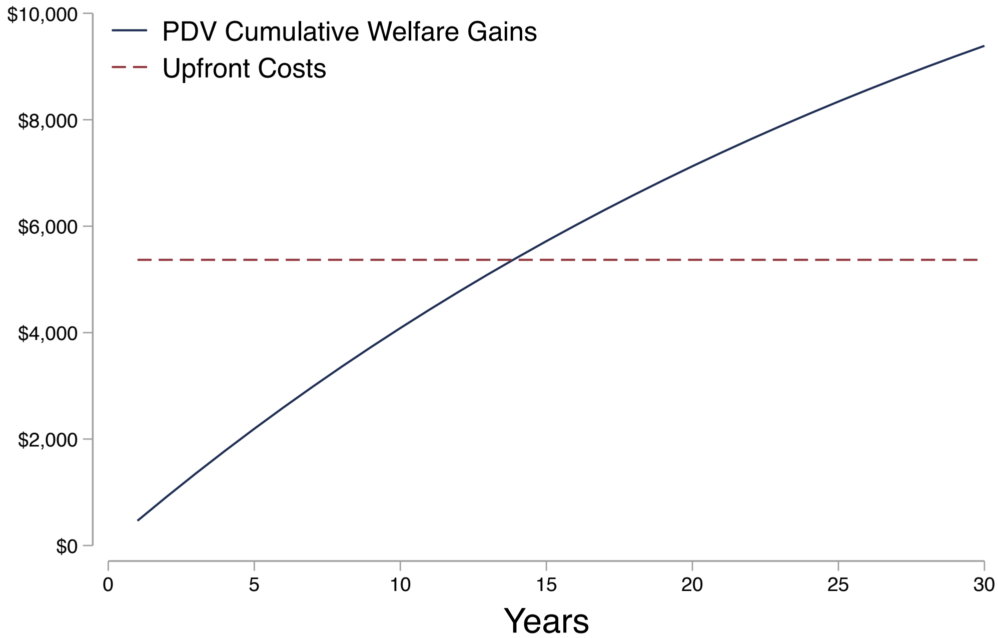
The upfront costs of the program are substantial: \$8,048 per subsidized unit in the PHA ($8{,}048 = 72{,}600{,}000/9{,}021$). However, the program produced durable reductions in neighborhood poverty for subsidized renters that lasted substantially longer than a single year: we observe poverty reductions that persist for at least 15 years after the intervention, the last year we consider due to data limitations. Figure E.1 plots the present discounted value (PDV) of cumulative annual welfare gains per subsidized unit as a function of the number of years over which benefits accrue, assuming a 3% annual discount rate. The figure shows that the PDV of welfare gains exceeds upfront costs if reductions in exposure to neighborhood poverty persist for at least 16 years.[49] Thus, when considered over a sufficiently long horizon, the HOPE VI program produced meaningful welfare gains for subsidized households. Moreover, these net gains in subsidized renter welfare would remain even if the full HOPE VI program costs had been passed through to subsidized households through rent increases.

Our estimates of the costs and benefits of the HOPE VI program are intentionally simple. In particular, they consider only financial costs and benefits from reduced exposure to neighborhood poverty for subsidized renters. While these are key inputs to any cost-benefit analysis, the program could have imposed other costs and produced other benefits that our analysis does not capture.

[48] These willingness-to-pay estimates correspond to the model prediction in Galiani et al. (2015) evaluated at the sample average of all observable characteristics that determine household willingness to pay for neighborhood amenities. To calculate willingness to pay, we use the formula $\frac{\beta^X_{i,k} I_i}{\beta^c}$ and combine estimates from Tables 1 and 3 of Galiani et al. (2015). We then convert the estimates from 1997 dollars to 2019 dollars.

[49] The PDV of 16 years of cumulative welfare gains from reduced exposure to neighborhood poverty is $8{,}466 = \sum_{t=1}^{16} 674/(1.03)^t$.

Figure E.1: Cumulative Welfare Gains vs Upfront Costs



Notes: This figure plots the present discounted value (PDV) of the cumulative welfare gains per subsidized unit as a function of the number of years over which these gains accrue. The red dashed line represents the upfront cost per subsidized units.